Polynomial calculus degree of the onto functional pigeonhole principle when the characteristic divides the surplus

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Abstract

The onto functional pigeonhole principle with $n+k$ pigeons and n holes has a polynomial calculus refutation of degree 1 over $\mathbb{F}_p$ when $p \nmid k$, and one of degree $p^{v_p(k)}$ in general, by a criterion of Riis. We study the lower bound when p divides k. In Lean 4 we prove that the degree is exactly p when p divides k exactly once ($n \geq 2p-3$), at least $p+1$ when p^2 divides k ($n \geq 2p-1$), and at least 5 for $p=2$, $k=8$ ($n \geq 7$). With written proofs that rest on named published theorems and have been checked by AI reviewers but not by people, we show that the degree is exactly $p^{v_p(k)}$ for $n \geq 2p^{v_p(k)}-1$, so that for surplus p^ℓ it grows as p^ℓ; we give two further mechanisms, one through Nullstellensatz designs (degree at least 2^ℓ with explicit thresholds) and one degree by degree (exact degree p^2 when $v_p(k)=2$); and we record five errata in two of the sources these proofs use, among them a gap in the published proof of a theorem we would otherwise have used. Every statement carries its evidence tier. All results were produced by AI research agents under the author's direction.

1 Introduction

1.1 The principle and the question

The onto functional pigeonhole principle onto-FPHP$_n^{n+k}$ states that $n+k$ pigeons can be mapped bijectively onto n holes. It is false for every $k \geq 1$. We use the *linear encoding*: Boolean variables x_{ij} (pigeon i sits in hole j) and the axioms

$$\textstyle\sum_j x_{ij} - 1, \qquad \sum_i x_{ij} - 1, \qquad x_{ij}x_{i'j}\ (i \neq i'), \qquad x_{ij}x_{ij'}\ (j \neq j'),$$

saying that every pigeon sits in a hole, every hole holds exactly one pigeon, no two pigeons share a hole and no pigeon sits in two holes. This is the system of Beame and Riis [2, Def. 7.2], also used in Razborov's survey [17]. All results of this paper are for this encoding. Translating the CNF clause by clause instead turns each pigeon clause into a polynomial of degree n [17, footnote 5], so degree bounds below n say nothing about that encoding.

We measure refutations in the polynomial calculus (PC) over a field of characteristic p: from the axioms and the Boolean axioms $x^2 - x$, derive 1 by linear combinations and multiplication by a variable. The *degree* of a refutation is the largest degree of a line, and the *PC degree* of a system is the least degree of a refutation. Section 2 fixes the conventions exactly.

If $p \nmid k$, summing the pigeon axioms and subtracting the hole axioms gives the constant $-k$, a refutation of degree 1. The question is what happens when p divides k. There the system has a refutation of degree p^v, where $v = v_p(k)$ is the exponent of p in k: Riis's criterion [19] (Theorem 8 in [17]) gives a refutation of degree d whenever $\binom{n+k}{d}$ and $\binom{n}{d}$ differ mod p, and by Lucas's theorem

$\binom{n+k}{p^v} - \binom{n}{p^v} \equiv k/p^v \not\equiv 0 \pmod{p}$. The criterion cannot give a smaller degree. So for fixed p and k the PC degree is bounded independently of n, and the questions are its exact value and how it grows with v.

1.2 What was known

Nullstellensatz. Beame and Riis [2, Thm. 8.1, Lemma 8.7] prove, for surplus $k = p^\ell$, that the Nullstellensatz degree is at least $2^\ell - 1$ for $n \geq ((p+2)^\ell - p^\ell)/2$ (Theorem 8.1) and at most $p^\ell - 1$ for $n \geq p^\ell$ (Lemma 8.7). Their degree measures the multipliers. PC line degree is at most Nullstellensatz *product* degree; for this encoding a certificate of multiplier degree d gives a PC refutation of degree at most $d + 2$. Nullstellensatz lower bounds alone therefore do not give PC lower bounds. Beame and Riis list non-trivial PC lower bounds for this family as open [2, §10].

Razborov's Question 7. Razborov's survey [17, §7, Question 7] asks, for fixed p, whether onto-FPHP$_n^{n+p^\ell}$ has no bounded-degree PC refutations in characteristic p, noting that the Nullstellensatz analogue is known. As printed, the question fixes p and ℓ and lets n grow. For fixed p and ℓ, Riis's upper bound p^ℓ above is independent of n. We also consider the distinct question of growth as ℓ varies. We state both readings and say which one each result addresses; none of our results settles the question in a sense we would call an answer, and we do not claim one.

Riis and Sitharam. Theorem 33 of the technical report of Riis and Sitharam [20] states a PC lower bound for the same family: degree at least $2^\ell - 1$ for surplus p^ℓ once $N \geq (21 + 3p^2)(2^\ell - 2) - p$. This is growth in ℓ. We cite it for comparison only: we have read its statement and the surrounding lemmas [20, pp. 36–38] but have not checked its proof, which delegates to an earlier tree construction.

Other work. General PC degree methods [15] and the cross-characteristic lower bounds for Tseitin and counting principles [3, 1, 4] concern regimes in which the characteristic does not divide the relevant modulus; Impagliazzo, Pudlák and Sgall [12] treat the injective principle without the onto and functional axioms, where the bound does not depend on p. None of them gives a low-degree reduction to onto-FPHP$_n^{n+k}$ with $p \mid k$. Section 8 gives the details and the search behind this paragraph.

1.3 Results

Table 1 lists the results with the evidence behind each. The tiers are defined in Section 1.4; in short, T-LEAN means kernel-checked in Lean 4, T-WRITTEN means a written proof checked by AI reviewers, and T-MEASURED means an exact finite computation. No person has reviewed any of the proofs.

The exact degree (E, with A and F). Result E says that Riis's upper bound is attained: for every k, the PC degree is exactly $p^{v_p(k)}$ once $n \geq 2p^{v_p(k)} - 1$. In particular, for surplus p^ℓ it is p^ℓ, so the degree grows with ℓ as fast as it can. E is the strongest statement in the paper and rests on the deepest inputs: Harman's stable equivalences, with the repairs of Section 7, and classical tilting theory (Appendix A). A shorter proof through the semisimplification functors of Hone et al. would give the threshold one lower, but it uses their Theorem 8.4.5, and while checking that theorem we found that its published proof is incomplete (Sections 5 and 7); we present that proof as conditional. The other results confirm parts of E on other inputs. At the Lean tier, A gives the

	Statement (characteristic p prime, surplus k)	Tier	Where
A	$k = pc$, $p \nmid c$, $n \geq 2p-3$: PC degree exactly p. Hence the same formula has PC degree p over $\mathbb{F}_p$ and 1 over $\mathbb{F}_q$ for every prime $q \nmid pc$.	T-LEAN	§3.1
B	$k = cp^2$, $c \geq 1$, $n \geq 2p-1$: no refutation of degree $\leq p$, so PC degree $\geq p+1$. At $p=2$, $k=4$, $n \geq 5$: exactly 4.	T-LEAN	§3.2
C	$p=2$, $k=8$, $n \geq 7$: no refutation of degree ≤ 4, so PC degree ≥ 5 (the upper bound is 8).	T-LEAN; certificate re-checked independently	§3.3
D	Growth. $k = p^\ell$, $\ell \geq 1$, $n \geq \frac{(p+2)^\ell - p^\ell}{2} + 2^{\ell+1} - 4$: PC degree $\geq 2^\ell$, over every field of characteristic p.	T-WRITTEN; inputs [5, 18, 2]	§4
E	Exact degree. For every $k \geq 1$, with $q = p^{v_p(k)}$: PC degree exactly q for $n \geq 2q-1$; for $n \geq 2q-2$ conditionally on [11, Thm. 8.4.5], whose published proof is incomplete (§7). Mechanism: for $q > D$ and $m \equiv m' \pmod q$ large, degree-$\leq D$ refutability of onto-FPHP$_n^m$ and onto-FPHP$_n^{m'}$ agree.	T-WRITTEN; inputs [10] and tilting theory (App. A)	§5
F	Bands. $q = p^{v_p(k)}$, $1 \leq D < \min(q, p^2)$, $n \geq 2D-1$: no refutation of degree $\leq D$. Hence, if $v_p(k) = 2$ and $n \geq 2p^2-3$: PC degree exactly p^2. Case $p=3$, $D=4$: $9 \mid k$, $n \geq 7$, no refutation of degree ≤ 4 (a second route through finite certificates).	T-WRITTEN; inputs [22, 24, 16]; $p=3$, $D=4$: Lean conditional (K100)	§6
G	Three points in Harman [10] and two in Hone et al. [11]; our proofs avoid each.	T-WRITTEN	§7

Table 1: Results. "Inputs" are published theorems the written proofs use and do not re-prove. Where the statement says "no refutation" the system is unsatisfiable and has a refutation of degree $p^{v_p(k)}$ by Riis's criterion. Throughout, lower-bound and exact-degree statements assume $n \geq 1$ and $k \geq 1$.

exact degree p when $v_p(k) = 1$, from $n = 2p-3$; B gives at least $p+1$ when $p^2 \mid k$, from $n = 2p-1$; C gives at least 5 for $p=2$, $k=8$, from $n=7$. At the written tier, F gives the exact degree p^2 when $v_p(k) = 2$, from $n = 2p^2-3$, using neither of E's inputs. For comparison, Theorem 33 of [20] gives $2^\ell - 1$ for $N \geq (21+3p^2)(2^\ell - 2) - p$.

Growth from Nullstellensatz designs (D). D gives the weaker bound 2^ℓ with explicit thresholds, by a transfer from Nullstellensatz designs to PC that uses neither E's inputs nor F's. Its degree bound is one more than the bound $2^\ell - 1$ of [20, Thm. 33], and its threshold in n is smaller for small ℓ and larger for large ℓ (for $p=3$: 12 against 93 at $\ell=2$, 1501 against 1437 at $\ell=5$). At $p=2$ its bound meets Riis's, so D gives the exact degree 2^ℓ over $\mathbb{F}_2$ independently of E, from $n \geq (4^\ell - 2^\ell)/2 + 2^{\ell+1} - 4$. None of D, E, F is formalised in Lean.

The bands (F). F is not a proof of growth: a bound that stops at p^2 cannot give unbounded growth at fixed p. It gives the fixed-valuation and fixed-degree statements above. Its F_3 case ($9 \mid k$, degree ≤ 4, all $n \geq 7$) has a Lean counterpart that is conditional on a finite-looking spanning statement for every board (Section 6); that condition is verified by exact computation on the smallest board at two of the three levels it needs.

Errata (G). Checking the sources of E led to five points in two recent sources: in Harman's paper, a statement that is false at $p = 2$, a misprinted modulus and an omitted justification; in Hone et al., a theorem that is false at the boundary of its hypotheses, and a lemma that is false as stated, on which the proof of their Theorem 8.4.5 depends. We sent the first four to the authors on 8 November 2026. Our proofs avoid each defect.

1.4 Evidence tiers

Every statement in this paper carries one of three tiers.

T-LEAN The stated formal theorem is kernel-checked in Lean 4, using only the axioms `propext`, `Classical.choice` and `Quot.sound`. Additional model re-proofs and independent certificate checks are described separately for each result (Section 9).

T-WRITTEN There is a written proof. At least one AI model other than its author reconstructed it, and a separate AI adjudicator (Astra, a Codex model that took no part in the run) checked it against the primary sources. Author and reconstructor are named per result; they are not always of different vendors. Published theorems that the proof uses are named as inputs and are not re-proved.

T-MEASURED An exact finite computation, with a reproducible script; a second, independently written implementation where we say so.

The tiers are not equally strong. A T-LEAN statement is as reliable as the Lean kernel and the match between the Lean definitions and the mathematics, which Section 2 states in full. A T-WRITTEN statement is as reliable as its model review and its inputs; we name, for each such result, the input we consider most exposed. No human expert has reviewed any proof in this paper.

The results were produced by AI research agents working in a harness that records every claim, review and computation; Section 9 describes this and what was checked. The Lean library and the checks are public with the four findings behind A, B, C and the $\mathbb{F}_3$ target (Zenodo DOIs 10.5281/zenodo.23996564, 10.5281/zenodo.23996585, 10.5281/zenodo.23996615 and 10.5281/zenodo.24056819).

1.5 Organisation

Section 2 defines the two formulations of PC used here and proves that they have the same refutations, and sets up the duality between refutations and designs. Section 3 gives prose proofs of A, B and C. Sections 4, 5 and 6 prove D, E and F. Section 7 records the errata, Section 8 the relation to the literature, and Section 9 how the results were produced and checked. Section 10 lists open problems.

2 Definitions and the two formulations of polynomial calculus

Lean names are given in footnotes, with namespaces abbreviated as in Section 3: `[0459]` stands for `Programme.«20261025-0459-proof-complexity»`, and so on; `[0635]` for `Programme.«20261102-0635-proof-complexity»`; `[maint]` stands for `Programme.«maintenance-20261108»`.

2.1 The formula

Fix $n, k \in \mathbb{N}$. There are $n+k$ pigeons $0 \le i < n+k$ and n holes $0 \le j < n$; the variable x_{ij} has index $in+j$, so the variables are $x_0, \dots, x_{N-1}$ with $N = (n+k)n$. The system ontoFPHP(n,k) is the list of integer polynomials

$$P_i = \textstyle\sum_{j<n} x_{ij} - 1, \quad H_j = \sum_{i<n+k} x_{ij} - 1, \quad x_{ij}x_{i'j}\ (i<i'), \quad x_{ij}x_{ij'}\ (j<j').$$

We call them pigeon, hole, collision and functional axioms.[1] They are the axioms of [2, Def. 7.2], with integer coefficients that are read mod p below. Lean represents a polynomial by a list of terms (an integer coefficient and a list of variables) and uses its raw syntactic degree, the maximum length of a term's variable list; rule (eq) below permits collecting coefficients and removing zero coefficients. For the displayed axioms, whose representations are normalised, this presents ordinary polynomial calculus over $\mathbb{F}_p$ with the same degree bound. No reduction by $x^2 = x$ is applied unless we say so.

2.2 Standard polynomial calculus

Definition 2.1 (standard PC)**.** Let $p \ge 2$, let A be a list of integer polynomials and $D \in \mathbb{N}$. $\mathrm{Der}_p(A, D)$ is the least set of integer polynomials such that

(ax) $P \in \mathrm{Der}_p(A,D)$ for every $P \in A$ with $\deg P \le D$;

(bool) $x^2 - x \in \mathrm{Der}_p(A,D)$ for every variable x, provided $D \ge 2$;

(lin) $uP + vQ \in \mathrm{Der}_p(A,D)$ for $P, Q \in \mathrm{Der}_p(A,D)$ and $u, v \in \mathbb{Z}$;

(mul) $xP \in \mathrm{Der}_p(A,D)$ for $P \in \mathrm{Der}_p(A,D)$ with $\deg P + 1 \le D$;

(eq) $Q \in \mathrm{Der}_p(A,D)$ whenever $P \in \mathrm{Der}_p(A,D)$, $\deg Q \le D$ and every monomial has congruent coefficients mod p in P and Q.

A *degree-D refutation* of A over $\mathbb{F}_p$ exists, written $\mathrm{Ref}_p(A,D)$, if $1 \in \mathrm{Der}_p(A,D)$.[2]

For prime p this is the usual polynomial calculus over $\mathbb{F}_p$ with Boolean axioms: rule (eq) identifies an integer polynomial with its image in $\mathbb{F}_p[x]$, and every line has degree at most D (by induction; (lin) does not raise degree). Two points of convention matter for comparisons. First, the degree is the degree of a *line*, measured syntactically, before any reduction by $x^2 = x$: rule (mul) is guarded by $\deg P + 1 \le D$ on the unreduced polynomial P. Second, Nullstellensatz degree in [2] is the degree of the multipliers; we quote their bounds in multiplier degree and state any conversion to product or PC line degree explicitly.

Lemma 2.2 (monotonicity)**.** *If $D \le D'$ then $\mathrm{Der}_p(A,D) \subseteq \mathrm{Der}_p(A,D')$. Hence $\neg\mathrm{Ref}_p(A,D')$ implies $\neg\mathrm{Ref}_p(A,D)$ for every $D \le D'$.*[3]

Proof. Every rule instance of degree D is a rule instance of degree D': each guard $\deg \le D$ or $\deg + 1 \le D$ (and $D \ge 2$) implies the same guard with D'. □

The parameter D is an upper bound on line degree. Monotonicity allows a lower bound at D to be restated for all smaller bounds; the library states those forms separately.[4]

[1] Lean: `[0459] OntoFPHP.ontoFPHP`.

[2] Lean: `[1156] StdPC.Der, StdPC.Refutes, StdPC.PEq, StdPC.boolAx`.

[3] Lean: `[maint] DegreeMono.Der.mono, Refutes.mono`.

[4] Lean: `[maint] DegreeMono.uniform_no_degree_le_p, uniform_no_standard_degree_le_p, no_deg_le4, deg_ge5`.

2.3 Semantic polynomial calculus on the cube

The Lean library proves its lower bounds for a formulation in which lines are functions on the Boolean cube. This removes the Boolean axioms and makes congruence a rule, which is convenient for the duality arguments of Section 2.6.

A *point* is a map $a\colon \mathbb{N} \to \{0,1\}$; a *line* is a function g from points to $\mathbb{Z}$. Each integer polynomial P defines a line $\hat{P}$ by evaluation. A line g has *cube degree at most* d if $\hat{Q}(a) \equiv g(a) \pmod p$ for all points a, for some integer polynomial Q with $\deg Q \le d$.[5]

Definition 2.3 (semantic PC)**.** The lines of a degree-D derivation over $\mathbb{Z}/p$ from A form the least set $\mathrm{PPC}_p(A,D)$ of lines containing (ax) $\hat{P}$ for each $P \in A$ with $\deg P \le D$; (lin) $uf + vg$ for $f, g \in \mathrm{PPC}_p(A,D)$ and $u, v \in \mathbb{Z}$; (mul) $x \cdot g$ for a variable x and $g \in \mathrm{PPC}_p(A,D)$ of cube degree at most d with $d+1 \le D$; (congr) every f with $f(a) \equiv g(a) \pmod p$ for all points a, for some $g \in \mathrm{PPC}_p(A,D)$. A *degree-$D$ refutation* is a derivation of the constant line 1; we write $\mathrm{PPCRef}_p(A,D)$.[6]

Here p is any integer ≥ 2, not necessarily prime; all our results take p prime. Degree monotonicity holds as in Lemma 2.2.[7]

Theorem 2.4. *For every $p \ge 2$, list A and D:* $\mathrm{Ref}_p(A,D) \iff \mathrm{PPCRef}_p(A,D)$.[8]

The two directions are Propositions 2.5 and 2.9. Consequently every lower bound in this paper holds verbatim for standard PC over $\mathbb{F}_p$ with Boolean axioms.

Proposition 2.5. *If $p \ge 2$ and $\mathrm{Ref}_p(A,D)$, then $\mathrm{PPCRef}_p(A,D)$.*[9]

Proof. We show by induction on $\mathrm{Der}_p(A,D)$ that for every $P \in \mathrm{Der}_p(A,D)$ either $\hat{P}(a) \equiv 0 \pmod p$ at every point a, or $\hat{P} \in \mathrm{PPC}_p(A,D)$.[10] The first alternative is needed because the Boolean axioms have no semantic counterpart: their value is 0, and the zero line need not be in $\mathrm{PPC}_p(A,D)$ (it is empty when A has no axiom of degree $\le D$). (ax) gives the second alternative and (bool) the first. (lin): if both $\hat{P}, \hat{Q}$ are derivable, so is $u\hat{P} + v\hat{Q}$; if one vanishes, $u\hat{P} + v\hat{Q}$ is congruent to a multiple of the other and is derivable by (lin) and (congr), or vanishes. (eq): polynomials with coefficients congruent mod p take congruent values, so (congr) applies. (mul): $\deg P + 1 \le D$ means $\hat{P}$ has cube degree at most $D-1$, so semantic (mul) applies to $\hat{P}$, and $\widehat{xP} = x \cdot \hat{P}$; if $\hat{P}$ vanishes, so does $x \cdot \hat{P}$.

Now let $1 \in \mathrm{Der}_p(A,D)$. The first case is impossible: $1 \not\equiv 0 \pmod p$ at any point since $p \ge 2$. So $\hat{1} = 1 \in \mathrm{PPC}_p(A,D)$. □

For the converse the only difficulty is (mul): a semantic line g of cube degree $\le D-1$ may be represented in the standard derivation by a polynomial Q of degree D, which may not be multiplied. We must derive a low-degree representative R of g instead. Two lemmas do this. For a monomial m let $\mathrm{ml}(m)$ be its multilinear form (the set of its variables), and extend ml linearly to polynomials.

Lemma 2.6 (multilinear reduction)**.** *If $\mathrm{Der}_p(A,D)$ contains some polynomial and m is a monomial with $|m| \le D$, then $m - \mathrm{ml}(m) \in \mathrm{Der}_p(A,D)$.*[11]

[5] Lean: `[1633] PCTseitin.CubeDeg`.
[6] Lean: `[0459] OntoFPHP.PPCDerivable, OntoFPHP.PPCRefutation`.
[7] Lean: `[maint] DegreeMono.PPCDerivable.mono, PPCRefutation.mono`.
[8] Lean: `[1156] StdPCConv.refutes_iff`.
[9] Lean: `[1156] StdPC.refutes_ppc`.
[10] Lean: `[1156] StdPC.der_sem`.
[11] Lean: `[1156] StdPCConv.der_dd`.

Proof. If m is multilinear, $m - \mathrm{ml}(m)$ has all coefficients 0: derive zero as $0 \cdot Q + 0 \cdot Q$ by (lin) from any derived line Q, and apply (eq). Otherwise some variable x occurs twice, $m = x^2 u$ with $|u| = |m| - 2$, and in particular $D \geq |m| \geq 2$. Then $m - xu = u \cdot (x^2 - x)$, obtained from the Boolean axiom by $|u|$ applications of (mul); the intermediate products have degree $\leq |m| \leq D$, so every guard holds. Since $\mathrm{ml}(xu) = \mathrm{ml}(m)$ and $|xu| < |m|$, induction on $|m|$ gives $xu - \mathrm{ml}(m)$, and (lin) adds the two. □

Lemma 2.7 (multilinear vanishing)**.** *If a multilinear integer polynomial $S = \sum_T c_T x_T$ satisfies $\hat{S}(a) \equiv 0 \pmod p$ at every point a, then $c_T \equiv 0 \pmod p$ for every T.*[12]

Proof. By induction on $|T|$. Evaluate at the indicator point a_T of T: $\hat{S}(a_T) = \sum_{T' \subseteq T} c_{T'}$. By induction $c_{T'} \equiv 0$ for $T' \subsetneq T$, so $c_T \equiv \hat{S}(a_T) \equiv 0$. □

Lemma 2.8. *Let $Q \in \mathrm{Der}_p(A, D)$, and let R be an integer polynomial with $\deg R \leq D$ and $\hat{R}(a) \equiv \hat{Q}(a) \pmod p$ at every point. Then $R \in \mathrm{Der}_p(A, D)$.*[13]

Proof. Put $S = R - Q$, a polynomial of degree $\leq D$ that vanishes mod p on the cube. Its multilinear form $\mathrm{ml}(S)$ takes the same values, so by Lemma 2.7 all its coefficients are $\equiv 0$; by (eq) applied to $0 \cdot Q$, $\mathrm{ml}(S) \in \mathrm{Der}_p(A, D)$. Adding the multiples $c\,(m - \mathrm{ml}(m))$ of Lemma 2.6 for the terms cm of S gives S. Then $R = S + Q$ by (lin). □

Proposition 2.9. *If* $\mathrm{PPCRef}_p(A, D)$*, then* $\mathrm{Ref}_p(A, D)$*.*[14]

Proof. By induction on $\mathrm{PPC}_p(A, D)$, every semantic line g has some $Q \in \mathrm{Der}_p(A, D)$ with $\hat{Q} \equiv g$ pointwise mod p. (ax), (lin) and (congr) are immediate. For (mul), let g have cube degree $\leq D - 1$, witnessed by R with $\deg R \leq D - 1$, and let Q be the derived representative of g. By Lemma 2.8, $R \in \mathrm{Der}_p(A, D)$, so $xR \in \mathrm{Der}_p(A, D)$ by (mul), and $\widehat{xR} \equiv x \cdot g$. Finally, if 1 is a semantic line, its representative Q agrees with the constant polynomial 1 on the cube, and Lemma 2.8 gives $1 \in \mathrm{Der}_p(A, D)$. □

2.4 The upper bound

Proposition 2.10 (Riis [19]; Theorem 8 of [17])**.** *Let p be prime. If $\binom{n+k}{d} \not\equiv \binom{n}{d} \pmod p$ for some $d \geq 1$, then* onto-FPHP$_n^{n+k}$ *has a PC refutation of degree d over $\mathbb{F}_p$. For $k = p^v c$ with $p \nmid c$, the choice $d = p^v$ applies by Lucas's theorem, giving degree p^v for every n.*

We use this only as an upper bound and do not re-prove it in general. In the case needed for Theorem A ($v = 1$) the library proves the degree-p refutation directly (Proposition 3.6). The criterion cannot give less than p^v: for $d < p^v$, Lucas's theorem gives $\binom{n+k}{d} \equiv \binom{n}{d}$.

2.5 Matchings, balance and extension

A *matching* is a set of variables no two of which share a pigeon or a hole; Lean represents it by its decreasing list (`CM`). For $f\colon \{\text{matchings}\} \to \mathbb{Z}$, a matching g and a pigeon i free in g, the *row sum* is $\mathrm{row}_f(g, i) = \sum_j f(g \cup \{x_{ij}\})$ over the holes j for which $g \cup \{x_{ij}\}$ is a matching; the *column sum* $\mathrm{col}_f(g, j)$ at a free hole j is defined likewise over pigeons. Write $|g|$ for the number of edges.

[12] Lean: `[1156] StdPCConv.cf_ml_dvd`.
[13] Lean: `[1156] StdPCConv.der_of_cube, der_of_zero, der_of_ml`.
[14] Lean: `[1156] StdPCConv.ppc_refutes, ppc_std`.

Definition 2.11. $\mathrm{Bal}_p(f, c, e)$ holds if for every matching g with $|g| = e$, every free row sum and every free column sum of f at g is $\equiv c(g) \pmod p$. For $p = 0$ this is exact equality. We call T a *window up to level* d if $\mathrm{Bal}_p(T, T, e)$ for all $e \le d$, and write $\mathcal{W}_d$ for the set of these.[15]

The parameters n, k are implicit in Bal; we write $\mathrm{Bal}_p^{(n,k)}$ when they vary.

Definition 2.12. $\mathrm{EXT}_p(n, k, d)$: for all f, c, if $d = 0$ or $\mathrm{Bal}_p(f, c, d-1)$, then there is y with $\mathrm{Bal}_p(y, f, d)$.[16]

For a set G of functions, $\mathrm{Span}_e(G)$ is the least set containing G and closed under sums, integer multiples, and replacement of f by any f' with $f'(g) \equiv f(g) \pmod p$ for all matchings g with $|g| = e$.[17]

Lemma 2.13. $\mathrm{Bal}_p(y, f, e)$ *is linear in the pair* (y, f) *and depends on* f *only through its residues on matchings of size* e*. Hence, if every* $f \in G$ *has some* y *with* $\mathrm{Bal}_p(y, f, e)$*, so does every* $f \in \mathrm{Span}_e(G)$*. In particular, if every* f *with* $\mathrm{Bal}_p(f, c, d-1)$ *(any* c*) lies in* $\mathrm{Span}_d(G)$*, then* $\mathrm{EXT}_p(n, k, d)$.[18]

Proof. Row and column sums are linear in f; the three closure rules of Span_e are matched by adding extensions, scaling them, and keeping the same y. □

Lemma 2.14 (flag lemma)**.** *If* $\mathrm{Bal}_p(f, c, e+1)$*, then at every matching* g *with* $|g| = e$*, every free row sum of* c *is congruent to every free column sum of* c*. Consequently there is* c' *with* $\mathrm{Bal}_p(c, c', e)$.[19]

Proof. Let i be a free pigeon and b a free hole of g. For a free hole $j \neq b$, $c(g+x_{ij}) \equiv \mathrm{col}_f(g+x_{ij}, b) = \sum_{i' \neq i} f(g + x_{ij} + x_{i'b})$, so $\mathrm{row}_c(g, i) \equiv c(g + x_{ib}) + \sum_{j \neq b} \sum_{i' \neq i} f(g + x_{ij} + x_{i'b})$. Expanding $\mathrm{col}_c(g, b)$ through the row sums of f at $g + x_{i'b}$ gives $c(g + x_{ib})$ plus the same double sum; the sums run over free j and free i'. Put $c'(g)$ equal to the column sum of c at a chosen free hole of g (0 if none). □

2.6 Designs and the extension criterion

Lower bounds are proved by exhibiting dual objects, *designs*, which vanish on every line of a low-degree derivation but not on 1.

Definition 2.15. Let $N = (n + k)n$. A *functional* E assigns an integer to each line, is linear mod p, respects congruence mod p, and reads its argument only at points whose variables of index $\ge N$ are 0. It is a *degree-*D *design* for A if $E(x_m \cdot P) \equiv 0 \pmod p$ for every $P \in A$ and every list m of variables with $|m| + \deg P \le D$, where $x_m = \prod_{v \in m} x_v$. E *extends* E' to degree d if $E(x_m) \equiv E'(x_m)$ for all $|m| \le d$.[20]

Lemma 2.16. *Let* $D \ge 1$*. Suppose every degree-*$(D-1)$ *design has an extension to degree* $D-1$ *that is a degree-*D *design, and some degree-*$(D-1)$ *design* E_0 *has* $E_0(1) \not\equiv 0$*. Then there is no degree-*D *refutation of* A.[21]

Proof. We show that every degree-D design E kills every line g of a degree-D derivation, by induction on the derivation. (ax) $E(1 \cdot P) \equiv 0$ since $\deg P \le D$. (lin), (congr) by linearity and congruence. (mul) Let g have cube degree $\le D-1$. The functional $h \mapsto E(x \cdot h)$ is a degree-$(D-1)$ design;[22] by hypothesis it has an extension E_3 to degree $D-1$ that is a degree-D design. By

[15] Lean: `[1402] ExtLemma.Bal, rowS, colS; [0607] Peel.Wd`.
[16] Lean: `[1402] ExtLemma.EXT`.
[17] Lean: `[1402] ExtLin.InSpan`.
[18] Lean: `[1402] ExtLin.Bal_add, Bal_smul, Bal_congr, ext_of_span, EXT_of_span`.
[19] Lean: `[0607] Flag.row_col_gen, Flag.flag; also [1434] RowColG.link_bal`.
[20] Lean: `[1627] DesignExt.Fn, IsDesign, Extends`.
[21] Lean: `[1627] DesignExt.ext_sound`.
[22] Lean: `[1627] DesignExt.xFn, xFn_design`.

induction $E_3(g) \equiv 0$; two functionals that agree on monomials of length $\leq D-1$ agree on lines of cube degree $\leq D-1$,[23] so $E(x \cdot g) \equiv E_3(g) \equiv 0$. Finally extend E_0 to a degree-D design E: then $E(1) \equiv E_0(1) \not\equiv 0$, so 1 is not a line. □

The hypothesis of Lemma 2.16 is stronger than the existence of one degree-D design with $E(1) \not\equiv 0$, which is the Nullstellensatz-type condition. The extra requirement, that *every* lower-degree design extend, is what makes the lemma sound for PC, where multiplication may follow linear combination.

Lemma 2.17 (extension step)**.** *Let $n > 0$, assume $\mathrm{EXT}_p(n,k,d)$, and let F be a degree-d design for $\mathrm{ontoFPHP}(n,k)$. Then some degree-$(d+1)$ design extends F to degree d.*[24]

Proof. Put $f(t) = F(x_t)$ for matchings t. (1) A degree-d design kills every monomial of length $\leq d$ that contains two distinct variables sharing a pigeon or a hole, since such a monomial is a multiple of a collision or functional axiom.[25] (2) $\mathrm{Bal}_p(f,f,d-1)$ when $d \geq 1$: for a matching g with $|g| = d-1$ and a free pigeon i, $F(x_g P_i) \equiv \sum_j F(x_g x_{ij}) - F(x_g)$, and the terms with $g \cup \{x_{ij}\}$ not a matching vanish by (1), so $F(x_g P_i) \equiv \mathrm{row}_f(g,i) - f(g)$; the design property gives $F(x_g P_i) \equiv 0$. Columns likewise.[26] (3) $\mathrm{EXT}_p(n,k,d)$ yields y with $\mathrm{Bal}_p(y,f,d)$. Let E be the moment functional whose value on x_S is $F(x_S)$ for $|S| \leq d$, $y(S)$ for matchings with $|S| = d+1$, and 0 for other sets with $|S| = d+1$.[27] It extends F to degree d.[28] (4) E is a degree-$(d+1)$ design. For $|m| + \deg P \leq d$ this follows from F being a design and the agreement.[29] For a collision or functional axiom with $|m| = d-1$, $x_m P$ is a monomial of length $\leq d+1$ with a conflicting pair: its value is 0 (non-matching at the top level) or killed by (1).[30] For a pigeon axiom with $|m| = d$ (holes are symmetric), let s be the set of m. If $|s| < d$, then $x_m = x_s$ on the cube and $x_s P_i$ has degree at most d, so $E(x_m P_i) \equiv F(x_s P_i) \equiv 0$; we may therefore assume $|s| = d$. If s is not a matching, every term of $E(x_s P_i)$ contains a conflict. If some $x_{ij} \in s$, all terms but one conflict and the remaining one cancels $E(x_s)$. Otherwise the line is free and $E(x_s P_i) \equiv \mathrm{row}_y(s,i) - f(s) \equiv 0$ by $\mathrm{Bal}_p(y,f,d)$.[31] □

Theorem 2.18 (EXT criterion)**.** *Let $p > 1$, $n_0 > 0$, $D \geq 1$, and suppose $\mathrm{EXT}_p(n_0,k,d)$ for every $d < D$. Then for every $n \geq n_0$ there is no degree-D refutation of $\mathrm{ontoFPHP}(n,k)$ over $\mathbb{Z}/p$. No hypothesis on k is needed.*[32]

Proof. At $n = n_0$: the functional with moment 1 at $\emptyset$ is a degree-0 design (vacuously, every axiom has degree ≥ 1) with value 1 at 1. Lemma 2.17 applied $D-1$ times gives a degree-$(D-1)$ design E_0 with $E_0(1) \equiv 1$, and once more shows that every degree-$(D-1)$ design extends.[33] Lemma 2.16 applies. For $n > n_0$ use restriction: substituting $x_{n+k,n} := 1$ and 0 for the other variables of pigeon $n+k$ and of hole n maps every axiom of $\mathrm{ontoFPHP}(n+1,k)$ to an axiom of $\mathrm{ontoFPHP}(n,k)$ or to a line derivable from them, so a degree-D refutation for $n+1$ yields one for n.[34] □

[23] Lean: `[1627] DesignExt.agree`.
[24] Lean: `[1402] ExtLemma.ext_step`.
[25] Lean: `[1402] ExtLemma.conf_ax, kill`.
[26] Lean: `[1402] ExtLemma.rowId, termsum, line_bal`.
[27] Lean: `[1402] ExtLemma.W, Ext; [1627] MomGen.sumWFn`.
[28] Lean: `[1402] ExtLemma.Ext_extends`.
[29] Lean: `[1627] DesignExt.low_kill`.
[30] Lean: `[1402] ExtLemma.Ext_kill`.
[31] Lean: `[1402] ExtLemma.Ext_line, line_conf, line_occ`.
[32] Lean: `[1402] ExtLemma.no_refutation_of_ext, no_refutation_of_ext_ge`.
[33] Lean: `[1402] ExtLemma.designs`.
[34] Lean: `[0459] OntoSubst.ontoFPHP_restrict, PPCDerivable.subst`.

The restriction step also shows that refutability is monotone in n for fixed k and D: lower bounds proved at one n_0 hold for all larger n.

Units. For prime p every $1 \le s \le p-1$ is invertible mod p (Bézout), hence so is $s!$ for $s \le p-1$.[35] Lean's `IsPrime p` is: $p \ge 2$ and every divisor of p is 1 or p.[36]

2.7 The matching algebra

Sections 4–6 work over an arbitrary field F and use a linear-algebra form of the same duality. We fix a board with a set P of $m = n+k$ pigeons and a set S of n holes, and write PC_F for Definition 2.1 with coefficients in F in place of integers read mod p.

Reduction. Let $\mathcal{B} = F[x]/(x_{ij}^2 - x_{ij},$ collision and functional axioms) be the *Boolean matching algebra*, and $\rho\colon F[x] \to \mathcal{B}$ the quotient map. The monomials $x_M = \prod_{(i,j)\in M} x_{ij}$, M a matching, form a basis of $\mathcal{B}$; ρ sends a monomial to x_M if its set of variables is a matching M and to 0 otherwise. Since ρ is a ring homomorphism, $\rho(xf) = \rho(x)\rho(f)$. Let $F_D \subseteq \mathcal{B}$ be the span of the x_M with $|M| \le D$; then $\rho(f) \in F_D$ whenever $\deg f \le D$.

The relation spaces. For a matching N and a pigeon i free in N, the *row relation* at (N,i) is $\rho(x_N(\sum_j x_{ij} - 1)) = \sum_j x_{N\cup\{(i,j)\}} - x_N$, the sum over holes j free in N; column relations at (N,j) are defined likewise. (If i is not free in N, the reduced product is 0.) Let

$$W_D = \mathrm{span}\{\text{row and column relations at bases } N \text{ with } |N| \le D-1\} \subseteq F_D,$$

so $W_0 = 0$ and $W_{D-1} \subseteq W_D \cap F_{D-1}$. We use two conditions:

$$\mathrm{U}_D:\ 1 \notin W_D, \qquad \mathrm{ND}_D:\ W_D \cap F_{D-1} = W_{D-1}.$$

Designs. A *unit degree-D design* on the board is a linear functional L on F_D with $L(W_D) = 0$ and $L(1) = 1$. In coordinates it is a function on matchings of size at most D with value 1 at $\emptyset$ such that, at every base N with $|N| < D$, every free row sum and every free column sum equals the value at N; this is a D-design in the sense of Beame and Riis [2, Def. 8.2], with values in F. By finite-dimensional duality, U_D holds if and only if a unit degree-D design exists, and ND_D holds if and only if every linear functional on F_{D-1} that vanishes on W_{D-1} extends to one on F_D that vanishes on W_D. (If every such functional extends, each of them vanishes on $W_D \cap F_{D-1}$, so $W_D \cap F_{D-1} \subseteq W_{D-1}^{\perp\perp} = W_{D-1}$; the converse is the same argument read backwards.)

U_D is the Nullstellensatz condition: reducing a Nullstellensatz certificate $\sum_r q_r A_r = 1$ of product degree at most D under ρ kills the Boolean, collision and functional terms and turns each row or column term into a combination of relations at bases of size at most $D-1$, so the certificate exists if and only if $1 \in W_D$. This is the product-degree convention; it differs from the multiplier degree of [2], and we convert between them only where stated.

Proposition 2.19 (from ND_D and U_D to PC)**.** *Let $D \ge 1$. If ND_D and U_D hold on a board, then* onto-FPHP *on that board has no PC_F refutation of degree at most D.*

[35] Lean: `[0607] Peel.units_of_prime, fact_unit`.
[36] Lean: `[0607] Peel.IsPrime`.

Proof. We show that $\rho(f) \in W_D$ for every line f of a degree-D derivation, by induction. Pigeon and hole axioms reduce to the relations at $N = \emptyset$, which lie in W_D since $D \geq 1$; Boolean, collision and functional axioms reduce to 0. Linear combinations, and replacement of coefficients by equal ones in F, preserve membership. For multiplication, the guard of rule (mul) gives $\deg f \leq D - 1$ for the unreduced line, so $\rho(f) \in W_D \cap F_{D-1} = W_{D-1}$ by ND_D. Each relation at a base N with $|N| \leq D - 2$, multiplied by a variable x and reduced, is 0 or a relation at the base $N \cup \{x\}$ (or at N itself, if $x \in N$), of size at most $D - 1$; so $\rho(xf) = \rho(x)\rho(f) \in W_D$. Since $1 \notin W_D$ by U_D, the line 1 is not derived. □

The induction is checked in Lean in abstract form: given any predicate on polynomials with the closure properties used above, no refutation exists.[37] Its instantiation with $\rho^{-1}(W_D)$ is the argument above and is not formalised. We use a representative of degree at most $D - 1$ before multiplying; the bounded normalisation argument of Lemma 2.8 justifies passing to such a representative. Checking only the reduced degree of the product would not justify an ordinary PC multiplication step.

Balanced arrays. For $e \geq 1$ let $H_e(P, S)$ be the space of functions f on the e-matchings of the board all of whose free row and column sums vanish: for every $(e-1)$-matching N and free pigeon i, $\sum_j f(N \cup \{(i,j)\}) = 0$, and likewise for free holes. If a functional L on F_t vanishes on W_t and on all matchings of size $< e$, with $1 \leq e \leq t$, then its layer $f(M) = L(x_M)$, $|M| = e$, lies in $H_e(P, S)$: the relation at a base of size $e - 1 \leq t - 1$ lies in W_t and says that the free row (or column) sum equals the value at the base, which is 0.

3 Results proved in Lean

This section reconstructs the programme's Lean proofs of A, B and C in mathematical prose, with the unreconstructed dependencies marked. The proofs are Lean 4 developments in the programme library (directory `programmes/proof-complexity/library`). The prose follows the Lean proofs lemma by lemma; the footnote at each statement names the Lean declaration that proves it. Namespaces are abbreviated by the run they come from: `[0459]` stands for

`Programme.«20261025-0459-proof-complexity»`,

and likewise `[1627]`, `[1402]`, `[0607]`, `[1434]`, `[0841]`, `[1156]`, `[0634]`, `[1249]` and `[1633]` for the runs 20261025-1627, 20261026-1402, 20261027-0607, 20261027-1434, 20261028-0841, 20261029-1156, 20261030-0634, 20261102-1249 and 20261024-1633. Names under `[1249]` live in the sub-namespace `UniformC3` (and `UniformC3.Block`).

Where a Lean step is a finite computation evaluated by the kernel (`decide`, `decide +kernel`) or a case split closed by automation (`simp`, `omega`, `grind`), we say so and do not supply a conceptual argument in its place. Lemmas from library files that we cite without reconstructing their proofs are marked "(stated, not reconstructed: …)" with their Lean names.

Framework. The formula ontoFPHP(n, k), the two formulations of PC and their equivalence (Theorem 2.4), degree monotonicity (Lemma 2.2), balance and extension (Definitions 2.11, 2.12), designs and the EXT criterion (Theorem 2.18) and the flag lemma (Lemma 2.14) are in Section 2. All three proofs below go through the EXT criterion or the peeling of Section 3.1.

[37] Lean: `[0635] StepSix.Invariant, StepSix.lines_mem, StepSix.no_refutation`.

3.1 A. Exact degree p when p divides the surplus exactly once

Theorem 3.1 (A)**.** *Let p be prime, $c \in \mathbb{N}$ with $p \nmid c$, and $n \geq 2p-3$. For every $D \in \mathbb{N}$,* ontoFPHP(n, pc) *has a degree-D semantic PC refutation over $\mathbb{Z}/p$ if and only if $p \leq D$.*[38]

Here $2p-3$ is natural-number subtraction (for $p=2$ the threshold is $n \geq 1$). By Theorem 2.4 the same holds verbatim for standard PC.[39] Theorem A combines an upper bound (Proposition 3.6, Lemma 3.7) with the lower bound:

Theorem 3.2 (A, lower bound)**.** *Let p be prime, $p \mid k$ and $n \geq 2p-3$. Then* ontoFPHP(n,k) *has no degree-$(p-1)$ refutation over $\mathbb{Z}/p$.*[40]

Corollary 3.3 (separation)**.** *Let p, q be primes, $p \nmid c$, $q \nmid pc$, $n \geq 2p-3$. Then* $\mathrm{PPCRef}_p(\text{ontoFPHP}(n, pc), D) \iff p \leq D$ *for all D, and* $\mathrm{PPCRef}_q(\text{ontoFPHP}(n, pc), 1)$.[41]

Proof. The first part is Theorem A. For the second, $pc > 0$ and there is u with $upc \equiv 1 \pmod q$ (Bézout). The degree-1 line $-u\sum_i P_i + u\sum_j H_j = -u(S-(n+pc)) + u(S-n) = upc$, where $S = \sum_{i,j} x_{ij}$, is $\equiv 1$.[42] □

Proof of Theorem A. If $D < p$ and a degree-D refutation existed, it would be a degree-$(p-1)$ refutation (Lemma 2.2), contradicting Theorem 3.2 (note $p \mid pc$). If $D \geq p$, Proposition 3.6 with $t = p$ and Lemma 3.7 give a degree-p refutation, which is one of degree D. □

A.1 The upper bound

Let $r_i = \sum_{j<n} x_{ij}$ (pigeon i) and $c_j = \sum_{i<n+k} x_{ij}$ (hole j), and let $e_t(f_0, \ldots, f_{s-1})$ be the elementary symmetric polynomial.

Lemma 3.4. *Let $D \geq 1$ and assume the zero line is derivable in degree D. For $s \leq n+k$ and $t \leq D$, the line $e_t(r_0, \ldots, r_{s-1}) - \binom{s}{t}$ is derivable in degree D from the pigeon axioms; likewise $e_t(c_0, \ldots, c_{s-1}) - \binom{s}{t}$ for $s \leq n$ from the hole axioms.*[43]

Proof. For $s = 0$ or $t = 0$ the line is zero. Otherwise induction on s, using

$$e_t^{(s+1)} - \tbinom{s+1}{t} = (e_t^{(s)} - \tbinom{s}{t}) + r_s(e_{t-1}^{(s)} - \tbinom{s}{t-1}) + \tbinom{s}{t-1}(r_s - 1).$$

The middle term is obtained by multiplying the line $e_{t-1}^{(s)} - \binom{s}{t-1}$, of cube degree $t-1 \leq D-1$, by each variable x_{sj} (rule (mul)) and summing; the last term is a multiple of the axiom P_s. □

Lemma 3.5. *Let every quadratic monomial axiom $x_u x_w$ of A have $u \neq w$, and call a Boolean point* good *if it satisfies all of them. If the zero line is derivable in degree D, then every line of cube degree $\leq D$ that vanishes mod p at all good points is derivable in degree D.*[44]

[38] Lean: `[0607] Lucas.pc_degree_exact`.
[39] Lean: `[1156] StdPCExact.std_pc_degree_exact`.
[40] Lean: `[0607] Peel.pc_degree_ge_p`.
[41] Lean: `[0607] Lucas.separation`.
[42] Lean: `[0459] OntoFPHP.ontoFPHP_deg1; [0607] Lucas.unit_of_ndvd`.
[43] Lean: `[0459] OntoLucas.esym_der`.
[44] Lean: `[0459] MatchQuot.quot`.

Proof (as in the source). Write $Q = Q|_{x_v=0} + x_v\,\partial_v Q$ variable by variable, carrying a monomial prefix S of variables already set to 1. When no variable is left, the line is x_S times a constant. If S contains a forbidden pair, x_S is a monomial axiom times $|S|-2$ variables; otherwise the indicator point of S is good, so the constant is $\equiv 0$. In detail, for a list S of variables write $x_S = \prod_{s\in S} x_s$ (a line), and for a polynomial Q and a variable v let $Q|_{v=0}$ consist of the terms of Q not containing v, and $\partial_v Q$ of the terms containing v with every occurrence of v deleted. On the cube $\hat{Q} = \widehat{Q|_{v=0}} + x_v\,\widehat{\partial_v Q}$; moreover $\deg Q|_{v=0} \le \deg Q$, $\deg \partial_v Q \le d$ whenever $\deg Q \le d+1$, and $\widehat{Q|_{v=0}}(a) = \hat{Q}(a[v := 0])$.[45] The Lean proof establishes, by induction on a duplicate-free list V of variables:

Claim. Let Q have all its variables in V, let S be a list of variables disjoint from V, and let d satisfy $\deg Q \le d$ and $|S| + d \le D$. If $\hat{Q}(a) \equiv 0 \pmod p$ at every good point a with $a_s = 1$ for all $s \in S$, then $x_S \cdot \hat{Q}$ is derivable in degree D.[46]

The lemma is the case $S = \emptyset$, $d = D$, $V = \{0, 1, \ldots, m\}$ (m the largest variable of Q), where Q, $\deg Q \le D$, witnesses the cube degree of the given line g; Q vanishes mod p at good points because g does, and (congr) replaces $\hat{Q}$ by g.[47] The only applications of (mul) are made in the following auxiliary fact: *for an axiom $P \in A$ and a list m of variables with $\deg P + |m| \le D$, the line $x_m\hat{P}$ is derivable.*[48] Indeed, for $m = \emptyset$ this is (ax), as $\deg P \le D$; and $x_{x::m}\hat{P} = x \cdot (x_m\hat{P})$, where $x_m\hat{P}$ has cube degree at most $\deg P + |m|$ (witness: P with m prepended to each monomial) and $\deg P + |m| + 1 \le D$, so (mul) applies to the line given by induction. Every intermediate line $x_{m'}\hat{P}$ (m' a suffix of m) thus has cube degree at most $\deg P + |m'| < D$ when it is multiplied.

Base case $V = \emptyset$. Then $\hat{Q}$ is a constant c. (i) If S contains u, w with $x_u x_w \in A$, then $u \ne w$ by hypothesis; removing one occurrence of each from S leaves S' with $|S'| = |S| - 2$, and $x_S = x_{S'}\,x_u x_w$ on the cube. The auxiliary fact with $P = x_u x_w$, $m = S'$ applies because $2 + |S'| = |S| \le |S| + d \le D$; then $c \cdot x_{S'} x_u x_w + 0 \cdot 0$ (rule (lin), using the zero line) is $x_S\hat{Q}$. (ii) Otherwise the indicator point of S is good and is 1 on S, so $p \mid c$; hence $x_S\hat{Q} \equiv 0$ pointwise, and (congr) applied to the zero line gives it.

Step $V = v :: V'$. Here $v \notin V'$ and $v \notin S$; $Q|_{v=0}$ and $\partial_v Q$ have their variables in V'.

- $x_S \cdot \widehat{Q|_{v=0}}$: the induction hypothesis with the same S and d. Its vanishing hypothesis holds because $\widehat{Q|_{v=0}}(a) = \hat{Q}(a[v := 0])$ and setting x_v to 0 keeps a point good and keeps it 1 on S ($v \notin S$).
- $x_{v::S} \cdot \widehat{\partial_v Q}$: if $d = 0$, all terms of Q are constants, $\widehat{\partial_v Q} = 0$, and the line is the zero line. If $d = d' + 1$, use the induction hypothesis with prefix $v :: S$ and bound d': $\deg \partial_v Q \le d'$ and $|v :: S| + d' = |S| + d \le D$. For its vanishing hypothesis let a be good with $a_v = 1$ and $a = 1$ on S; then $\hat{Q}(a) = \hat{Q}(a[v := 0]) + \widehat{\partial_v Q}(a)$ and both values of $\hat{Q}$ are $\equiv 0$, so $\widehat{\partial_v Q}(a) \equiv 0$.

Rule (lin) with coefficients $1, 1$ gives $x_S\widehat{Q|_{v=0}} + x_v x_S\widehat{\partial_v Q} = x_S\hat{Q}$.

Degree bookkeeping. The invariant $|S| + d \le D$ is preserved: passing to $\partial_v Q$ moves one unit from d to $|S|$, and passing to $Q|_{v=0}$ changes neither. The prefix x_S is never multiplied onto a derived line: the recursion only combines lines by (lin) and (congr), which have no degree guard, and every product with a variable is formed inside the auxiliary fact, building $x_{S'}x_u x_w$ upward from the axiom $x_u x_w$ with all intermediate lines of cube degree $< |S| \le D$. The zero line is used in case (ii), in the case $d = 0$, and as the second argument of (lin) in case (i).

[45] Lean: `[0459] MatchQuot.pz, pd, eval_split, deg_pz, deg_pd, eval_pz`.
[46] Lean: `[0459] MatchQuot.quot_aux`.
[47] Lean: `[0459] MatchQuot.quot`.
[48] Lean: `[0459] MatchQuot.mulMono`.

□

Proposition 3.6. *Let $t \geq 2$ and $n + k > 0$. If $\binom{n+k}{t} - \binom{n}{t}$ is a unit mod p, then* ontoFPHP(n, k) *has a degree-t refutation over $\mathbb{Z}/p$.*[49]

Proof. The zero line is $0 \cdot P_0$. At a good point (collision and functional axioms hold) every r_i and every c_j is 0 or 1 and $\sum_i r_i = \sum_j c_j = N$, so $e_t(r) = \binom{N}{t} = e_t(c)$.[50] By Lemma 3.5 (here $t \geq 2$ makes the quadratic axioms available) $e_t(r) - e_t(c)$ is derivable in degree t. With Lemma 3.4, $(e_t(r) - \binom{n+k}{t}) - (e_t(c) - \binom{n}{t}) - (e_t(r) - e_t(c)) = \binom{n}{t} - \binom{n+k}{t}$ is derivable; multiply by $-u$, where u inverts $\binom{n+k}{t} - \binom{n}{t}$ mod p, and apply (congr). □

Lemma 3.7. *Let p be prime. Then $p \mid \binom{p}{r}$ for $0 < r < p$; $\binom{m+p}{r} \equiv \binom{m}{r}$ for $r < p$; $\binom{m+p}{p} \equiv \binom{m}{p} + 1$; hence $\binom{n+pc}{p} - \binom{n}{p} \equiv c \pmod{p}$, which is a unit when $p \nmid c$.*[51]

Proof. $p \mid \binom{p}{r}$: from the absorption identity $r\binom{p}{r} = p\binom{p-1}{r-1}$ and Euclid's lemma, since $p \nmid r$. The two shift congruences follow by induction on m from Pascal's rule, the base case $m = 0$ being the first statement (and $\binom{p}{p} = 1$). Telescoping c times gives the congruence; Bézout gives the inverse of c. □

Remark 3.8. This congruence is what Lucas's theorem gives here, but the Lean proof does not invoke Lucas's theorem (the file is called `Lucas.lean`); it proves the shift congruences directly as above.

A.2 The lower bound: peeling

Theorem 3.2 follows from Theorem 2.18 with $D = p - 1$ and $n_0 = 2p - 3$ once $\mathrm{EXT}_p(2p - 3, k, d)$ holds for all $d \leq p - 2$. That is the case $t = p - 1$ of:

Proposition 3.9. *Let $p \mid k$, $2d + 1 \leq n$, and suppose s is invertible mod p for $1 \leq s \leq d + 1$. Then* $\mathrm{EXT}_p(n, k, d)$.[52]

For $d \leq p - 2$ we have $2d + 1 \leq 2p - 3$ and $d + 1 \leq p - 1$, so the hypotheses hold for prime p; this proves Theorem 3.2.[53] The rest of A.2 proves Proposition 3.9.

Rectangle products. Let $j \geq 0$, let $P = ((a_s, a'_s))_{s<j}$ be pigeon pairs and $Q = ((b_s, b'_s))_{s<j}$ hole pairs, with the $2j$ pigeons distinct and the $2j$ holes distinct. Put $\rho_s(x_{ib}) = ([i = a_s] - [i = a'_s])([b = b_s] - [b = b'_s])$, the signed indicator of the rectangle $\{a_s, a'_s\} \times \{b_s, b'_s\}$. For a matching g with $|g| = j$, $\mathrm{RP}_j(P, Q)(g) = \sum_\sigma \prod_{s<j} \rho_s(\sigma(s))$, the sum over the bijections σ from slots to the edges of g (a permanent; $\mathrm{RP}_0 = 1$).[54]

Lemma 3.10 (S_j)**.** *Let $2(j + 1) \leq n$ and let $(j + 1)!$ be invertible mod p. If* $\mathrm{Bal}_p(f, 0, j)$ *("f has zero links at level j"), then f lies in* Span_{j+1} *of the rectangle products* $\mathrm{RP}_{j+1}(P, Q)$.[55]

[49] Lean: `[0459] OntoLucas.ontoFPHP_upper`.
[50] Lean: `[0459] OntoLucas.good_eq`.
[51] Lean: `[0607] Lucas.dvd_bin, shift, shiftp, telescope, unit_c`.
[52] Lean: `[0607] Peel.ext_d`.
[53] Lean: `[0607] Peel.no_refutation_units`.
[54] Lean: `[0607] SJ.RP, SJ.jp, LemK.VP`.
[55] Lean: `[0607] SJ.SJ`.

Proof outline (from the Lean structure). For an injective pigeon tuple u and hole tuple w of length $j+1$, put $F(u,w) = f(\{x_{u_s w_s}\})$. Zero links translate into vanishing slot sums: the pigeon-slot sums of F are column sums of f and the hole-slot sums are row sums. *Lemma K* (characteristic-free): a function on injective j-tuples from a ground set with at least $2j$ points whose slot sums all vanish is, on valid tuples, a finite sum $\sum_P \mathrm{DP}_P(u)\, c_P(w)$ with $\mathrm{DP}_P(u) = \prod_s([u_s = P_s^1] - [u_s = P_s^2])$ and each c_P an integer combination of the functions $F(u', \cdot)$.[56] Applying it over pigeons and then over holes gives $F \equiv \sum d\, \mathrm{DP}_P(u)\mathrm{DP}_Q(w)$.[57] Summing over the $(j+1)!$ orderings of a matching g turns $\mathrm{DP}_P(u)\mathrm{DP}_Q(w)$ into $\mathrm{RP}_{j+1}(P,Q)(g)$, so $(j+1)!\, f \equiv \sum d\, \mathrm{RP}_{j+1}(P,Q)$ on matchings of size $j+1$;[58] multiply by the inverse of $(j+1)!$.

It remains to prove Lemma K, which we restate in the form proved in Lean. Let X be a finite set (Lean: a Boolean predicate read below a bound M) and W any set of parameters. A *valid* j-tuple is $u = (u_0, \ldots, u_{j-1})$ of distinct points of X. For $F\colon \{\text{valid } j\text{-tuples}\} \times W \to \mathbb{Z}$, the *slot-$t$ sum* at a valid u is $\sigma_t^X F(u,w) = \sum_a F(u[t := a], w)$ over the $a \in X$ different from all u_s, $s \neq t$. Write $\mathrm{KC}_m(X, j, F)$ if $m \mid \sigma_t^X F(u,w)$ for all valid u, $t < j$, w ($m \in \mathbb{Z}$ arbitrary, $m = 0$ meaning exactly). A pair list $P = ((P_s^1, P_s^2))_{s<j}$ is *valid* if its $2j$ points are distinct points of X, and $\mathrm{DP}_P(u) = \prod_{s<j} \delta_{P_s}(u_s)$ with $\delta_q(x) = [x = q^1] - [x = q^2]$. Finally $\mathrm{LC}_X(F)$ is the set of functions $c(w) = \sum_\ell a_\ell F(u^{(\ell)}, w)$, a finite integer combination with valid $u^{(\ell)}$.[59]

Lemma K. If $|X| \geq 2j$ and $\mathrm{KC}_m(X, j, F)$, there is a finite list of pairs (P, c_P), P valid and $c_P \in \mathrm{LC}_X(F)$, with $F(u,w) \equiv \sum_P \mathrm{DP}_P(u)\, c_P(w) \pmod m$ for all valid u and all w.[60]

Three facts are used. (K1) *Difference products have zero slot sums*: if the pairs of P are pairwise disjoint and the two points of P_t are distinct and lie in Y, then $\sigma_t^Y \mathrm{DP}_P(u) = 0$ exactly. For $\mathrm{DP}_P(u[t := a]) = \delta_{P_t}(a)\, R$ with R the product of the other factors, so the sum is $R\,([P_t^1 \text{ free}] - [P_t^2 \text{ free}])$; if P_t^1 (say) is not free, it equals some u_s, $s \neq t$, and $\delta_{P_s}(u_s) = 0$ by disjointness, so $R = 0$.[61] (K2) *Vanishing off an anchor*: if $z \in X$, $\mathrm{KC}_m(X, j+1, H)$, and $H \equiv 0$ at all valid tuples not containing z, then $H \equiv 0$ at all valid tuples. If $u_t = z$, the term $a = z$ of $\sigma_t^X H(u)$ is $H(u)$ itself and every other term is at a tuple without z, so $H(u) \equiv \sigma_t^X H(u) \equiv 0$.[62] (K3) LC is closed under sums and integer multiples, and is transitive: a combination of functions each in $\mathrm{LC}_X(F)$ lies in $\mathrm{LC}_X(F)$.[63]

Proof of Lemma K. Induction on j. For $j = 0$ the only valid tuple is empty; take the single pair $(\emptyset, F(\emptyset, \cdot))$, with $\mathrm{DP}_\emptyset = 1$.[64] For the step let $|X| \geq 2j+2$ and $\mathrm{KC}_m(X, j+1, F)$.[65]

1. *Anchor.* Fix $z \in X$ and put $X' = X \setminus \{z\}$, so $|X'| \geq 2j+1$.
2. *Slot defects.* For $t \leq j$ and a valid j-tuple v over X' let $G_t(v,w) = -F(\mathrm{ins}_t(z,v), w)$, where $\mathrm{ins}_t(z,v)$ inserts z at slot t (a valid $(j+1)$-tuple over X). A point can fill slot t' of v over X' if and only if it can fill the corresponding slot (t' if $t' < t$, else $t'+1$) of $\mathrm{ins}_t(z,v)$ over X, so the slot sums of G_t are minus slot sums of F and $\mathrm{KC}_m(X', j, G_t)$.[66] The induction hypothesis ($|X'| \geq 2j$) gives a list B_t of pairs (Q, c), Q valid over X', with $G_t(v,w) \equiv \sum_{B_t} \mathrm{DP}_Q(v) c(w)$; by (K3) each c lies in $\mathrm{LC}_X(F)$ (at length $j+1$).

[56] Lean: `[0607] LemK.lemK (file LemKMain.lean)`.
[57] Lean: `[0607] SJ.SJ_tuple`.
[58] Lean: `[0607] SJ.SJ_avg`.
[59] Lean: `[0607] LemK.Valid, sig, KC, DP, VP, LC`.
[60] Lean: `[0607] LemK.LK, lemK (LemKMain.lean)`.
[61] Lean: `[0607] LemK.sig_DP`.
[62] Lean: `[0607] LemK.zero_of_del`.
[63] Lean: `[0607] LemK.LC_add, LC_smul, LC_tr`.
[64] Lean: `[0607] LemK.lemK_zero`.
[65] Lean: `[0607] LemK.lemK_step`.
[66] Lean: `[0607] LemK.free_ins, ins_upd, valid_ins`.

3. *Correction.* For each such Q choose $w_Q \in X'$ outside the $2j$ points of Q, possible as $|X'| > 2j$.[67] Let $\varepsilon(u, w)$ be the sum, over $t \le j$ and $(Q, c) \in B_t$, of $\mathrm{DP}_P(u)c(w)$ with $P = \mathrm{ins}_t((w_Q, z), Q)$ (the pair (w_Q, z) inserted at slot t; P is valid over X), and put $\psi = F - \varepsilon$.[68]

4. *ψ has zero slot sums over X'.* Let u be valid over X' and $s \le j$. Since z is not in u, it is free at slot s, and $u[s := z] = \mathrm{ins}_s(z, \mathrm{skip}_s u)$ (skip_s deletes slot s); hence $\sigma_s^{X'} F(u) = \sigma_s^X F(u) + G_s(\mathrm{skip}_s u)$. In $\sigma_s^{X'}\varepsilon(u)$ the terms with $t \ne s$ vanish by (K1), as the slot-s pair of P is a pair of Q, inside X'. For $t = s$, $\mathrm{DP}_P(u[s := a]) = \delta_{(w_Q, z)}(a)\,\mathrm{DP}_Q(\mathrm{skip}_s u)$; summing over free $a \in X'$ ($z \notin X'$) gives $\mathrm{DP}_Q(\mathrm{skip}_s u)$ if w_Q is free, and otherwise $0 = \mathrm{DP}_Q(\mathrm{skip}_s u)$, because then w_Q is an entry of $\mathrm{skip}_s u$ lying off the pairs of Q.[69] So

$$\sigma_s^{X'}\psi(u) = \sigma_s^X F(u) + \Big(G_s(\mathrm{skip}_s u) - \sum_{B_s} \mathrm{DP}_Q(\mathrm{skip}_s u)\, c\Big) \equiv 0 + 0.$$

5. *Slices.* For $a \in X'$ let $\psi_a(v) = \psi((a, v))$ on valid j-tuples v over $X'' = X' \setminus \{a\}$, where (a, v) prepends a. Slot s of v is slot $s+1$ of (a, v), so $\mathrm{KC}_m(X'', j, \psi_a)$; and $|X''| = |X| - 2 \ge 2j$. The induction hypothesis gives B_a, with coefficients in $\mathrm{LC}_X(F)$ by (K3) ($\psi_a(v)$ is $F((a, v))$ minus a combination of the coefficients of ε). Let B_ψ consist of the pairs $(\mathrm{ins}_0((a, z), Q), c)$ for $a \in X'$, $(Q, c) \in B_a$. For u valid over X', $u_0 \ne z$ gives $\mathrm{DP}_{\mathrm{ins}_0((a,z),Q)}(u) = [u_0 = a]\,\mathrm{DP}_Q(\mathrm{skip}_0 u)$, so $\sum_{B_\psi} \mathrm{DP}\, c = \sum_{B_{u_0}} \mathrm{DP}_Q(\mathrm{skip}_0 u)\, c \equiv \psi_{u_0}(\mathrm{skip}_0 u) = \psi(u)$.[70]

6. *Assembly.* Take the pairs of ε followed by B_ψ, and let $H = F - \sum \mathrm{DP}_P\, c_P$ over this list. $\mathrm{KC}_m(X, j+1, H)$ holds since F satisfies it and each DP_P (P valid over X) has zero slot sums over X by (K1). At tuples avoiding z, $H = \psi - \sum_{B_\psi} \mathrm{DP}\, c \equiv 0$ by step 5. By (K2), $H \equiv 0$ at every valid tuple.

The size hypothesis is used four times: the anchor exists ($|X| > 0$), the induction hypothesis applies on X' and on X'' ($\ge 2j$ points), and the fresh points w_Q exist ($|X'| > 2j$). No division occurs, which is why Lemma K holds for every modulus m.

□

Windows. The next lemma realises every rectangle product, up to a factorial, as a window.

Lemma 3.11 (windows for rectangle data)**.** *Let $p \mid k$, $j \le d$, $2d + 1 \le n$, and P, Q as above with j pairs each. There is $W \in \mathcal{W}_d$ with $W(g) = (d - j + 1)!\,\mathrm{RP}_j(P, Q)(g)$ for every matching g with $|g| = j$.*[71]

Proof. Split pigeons and holes into blocks: the j rectangle blocks $\{a_s, a'_s\} \times \{b_s, b'_s\}$; $T = n - j - d - 1$ blocks $\{x_t\} \times \{h_t\}$ on fresh pigeons and holes; and a rest block with $m = n - 2j - T = d - j + 1$ holes and $n + k - 2j - T \equiv m \pmod p$ pigeons (here $p \mid k$ is used).[72] W is the block product: on a matching with an edge joining two different blocks it is 0; otherwise it is the product of local factors evaluated on the parts of the matching inside the blocks.[73] The local factors are: on a rectangle block, A_s with $A_s(\emptyset) = 0$, $A_s = \rho_s$ on single edges, and $A_s = x_{a_s b_s} x_{a'_s b'_s} - x_{a_s b'_s} x_{a'_s b_s}$ (the determinant) on its two 2-matchings;[74] the constant 1 on each 1×1 block; and on the rest block the symmetric design of

[67] Lean: `[0607] LemK.fresh`.
[68] Lean: `[0607] LemK.insP, VP_insP`.
[69] Lean: `[0607] LemK.DP_insP, DP_off`.
[70] Lean: `[0607] LemK.valid_cns, free_cns, cns_skip`.
[71] Lean: `[0607] WinMain.window`.
[72] Lean: `[0607] WinMain.good_of, Blocks.fresh_fn`.
[73] Lean: `[0607] WinJ.Wn, BProd.PB`.
[74] Lean: `[1402] Win.Aw`.

degree $D = d-j+1$, whose value at a local s-matching is $\mathrm{cv}(s) = (m-s)(m-s-1)\cdots(m-D+1)$, so $\mathrm{cv}(s) = (m-s)!$ as $m = D$.[75]

Balance for $e < D+j = d+1$.[76] A free line at a matching lies in one block, and its sum factors as the local line sum times the other factors.[77] Each local factor satisfies the local window identity wherever a free line exists: for the symmetric design at a local s-matching with $s < D$ the free row sum is $(m-s)\,\mathrm{cv}(s+1) = \mathrm{cv}(s)$ exactly and the free column sum is $(\#\text{pigeons}-s)\,\mathrm{cv}(s+1) \equiv \mathrm{cv}(s) \bmod p$;[78] the constant block is trivial;[79] and for A_s the identity can be checked by hand: at $\emptyset$ the row sum at a_s is $\rho_s(x_{a_sb_s}) + \rho_s(x_{a_sb'_s}) = 1-1 = 0 = A_s(\emptyset)$ (other rows and the columns alike), and at the edge $x_{a_sb_s}$ the only free cell is $x_{a'_sb'_s}$, with $A_s(\{x_{a_sb_s}, x_{a'_sb'_s}\}) = 1 = A_s(x_{a_sb_s})$ (the other three edges alike, with signs).[80] A rest line at a matching in which every rectangle block is occupied sees at most $e-j < D$ rest edges;[81] otherwise some rectangle block is empty and its factor $A_s(\emptyset) = 0$ makes both sides vanish.

Value. At a matching g with $|g| = j$ and $W(g) \neq 0$ every rectangle block carries exactly one edge, so the rest block is empty and contributes $\mathrm{cv}(0) = D!$; the permanent of functions with pairwise disjoint supports is the product of their sums,[82] giving $W(g) = (d-j+1)!\,\mathrm{RP}_j(P,Q)(g)$. □

Corollary 3.12. *If moreover $(d-j+1)!$ is invertible mod p, every $\mathrm{RP}_j(P,Q)$ lies in $\mathrm{Span}_j(\mathcal{W}_d)$.*[83]

Proof. $\mathrm{RP}_j \equiv u\,W$ on j-matchings, u the inverse of $(d-j+1)!$. □

Peeling. Call f *admissible at level* 0 always, and *at level* $\ell+1$ if $\mathrm{Bal}_p(f,c,\ell)$ for some c admissible at level ℓ. By the flag lemma (Lemma 2.14) iterated, $\mathrm{Bal}_p(f,c,\ell)$ already implies that f is admissible at level $\ell+1$.[84] $\mathcal{W}_d$ is closed under sums and multiples, so every member of $\mathrm{Span}_\ell(\mathcal{W}_d)$ agrees mod p on ℓ-matchings with a single $T \in \mathcal{W}_d$.[85]

Proposition 3.13 (peeling)**.** *Under the hypotheses of Proposition 3.9, for every $\ell \le d$ every f admissible at level ℓ lies in $\mathrm{Span}_\ell(\mathcal{W}_d)$.*[86]

Proof. Induction on ℓ. $\ell = 0$: Lemma 3.11 with $j = 0$ gives a window W with $W(\emptyset) = (d+1)!$ (the *type*-0 *window*: $n-d-1$ unit blocks and a rest block with $d+1$ holes); $f \equiv f(\emptyset)\,u\,W$ at $\emptyset$, u the inverse of $(d+1)!$. $\ell \to \ell+1$: let $\mathrm{Bal}_p(f,c,\ell)$ with c admissible. By induction $c \equiv T$ on ℓ-matchings for some $T \in \mathcal{W}_d$. Since $\mathrm{Bal}_p(T,T,\ell)$, the residual $f-T$ satisfies $\mathrm{Bal}_p(f-T,\, c-T,\, \ell)$, i.e. it has zero links at level ℓ. By Lemma 3.10 (with $(\ell+1)!$ invertible and $2(\ell+1) \le n$) it lies in the span of the $\mathrm{RP}_{\ell+1}$, which by Corollary 3.12 lies in $\mathrm{Span}_{\ell+1}(\mathcal{W}_d)$. Add T back. □

Proof of Proposition 3.9. Each $T \in \mathcal{W}_d$ has the extension T itself ($\mathrm{Bal}_p(T,T,d)$). An f with $\mathrm{Bal}_p(f,c,d-1)$ is admissible at level d, so lies in $\mathrm{Span}_d(\mathcal{W}_d)$ by Proposition 3.13; Lemma 2.13 concludes. All factorials used are at most $(d+1)!$. □

[75] Lean: `[1402] Win.cvf, SymDesign.cv`.
[76] Lean: `[0607] WinJ.win_bal`.
[77] Lean: `[0607] BProd.line_PB, bal_winB`.
[78] Lean: `[1402] SymDesign.cv_step; [0607] LocWin.sym_rowD, sym_colD`.
[79] Lean: `[0607] LocWin.unit_row, unit_col`.
[80] Lean: `[0607] LocWin.rect_row, rect_col via [1402] Win.blk0_row, blk0_col`.
[81] Lean: `[0607] Blocks.cnt_filter`.
[82] Lean: `[0607] Blocks.jp_prod; WinJ.win_val`.
[83] Lean: `[0607] Peel.rp_span`.
[84] Lean: `[0607] Peel.Adm, adm_of_bal`.
[85] Lean: `[0607] Peel.span_rep`.
[86] Lean: `[0607] Peel.claim`.

Mechanism of A. The upper bound is Riis's symmetric-function refutation at $t = p$, made a unit by the Lucas-type congruence $\binom{n+pc}{p} - \binom{n}{p} \equiv c$. The lower bound is *peeling*: a balanced function at level d is written, level by level, as a window plus a zero-link residual; zero-link residuals are spans of rectangle products (S_j, via Lemma K), and rectangle products are windows up to the factorial $(d-j+1)!$. The only arithmetic input is that $1!, \ldots, (d+1)!$ are units, which for $d = p-2$ is exactly why the bound stops at $p-1$: at $d = p-1$ the type-0 window has value $p! \equiv 0$. The board must hold the blocks, which gives $n \geq 2d+1 = 2p-3$.

3.2 B. Degree at least $p+1$ when p^2 divides the surplus

Theorem 3.14 (B)**.** *Let p be prime and $c, n \in \mathbb{N}$ with $n \geq 2p-1$. Then* ontoFPHP(n, cp^2) *has no degree-p semantic PC refutation over $\mathbb{Z}/p$, and no degree-p standard PC refutation over $\mathbb{F}_p$.*[87]

The Lean statement allows $c = 0$ (then the formula is satisfiable and the claim is trivial) and states the bound $D = p$; by monotonicity (Lemma 2.2) there is no refutation of degree at most p (for $c \geq 1$),[88] so the PC degree is at least $p+1$. The surplus is written $(c \cdot p) \cdot p$. The proof produces, as an intermediate result:

Theorem 3.15 (unit design)**.** *For prime p and every c there is T with* $\mathrm{Bal}_p^{(2p-1,\, cp^2)}(T, T, e)$ *for all $e \leq p-1$ and $T(\emptyset) \equiv 1 \pmod p$.*[89]

The proof has four steps: a criterion turning such a T into the lower bound (B.1); the construction of T from a *rectangular lift* (B.2); the rectangular lift from a *square lift* on the $(2p-1) \times (2p-1)$ board (B.3); and an explicit square lift for every prime (B.4). In Lean, B.2–B.3 are proved first under the hypothesis that a square lift exists (`SquareLiftAssumption`),[90] and B.4 discharges it.[91]

B.1 The design-to-lower-bound criterion

Theorem 3.16. *Let p be prime, $p \mid k$, $n_0 \geq 2p-1$, and let T_0 satisfy* $\mathrm{Bal}_p^{(n_0,k)}(T_0, T_0, e)$ *for all $e \leq p-1$ and $T_0(\emptyset) \equiv 1$. Then for every $n \geq n_0$,* ontoFPHP(n,k) *has no degree-p refutation over $\mathbb{Z}/p$.*[92]

Proof. Apply Theorem 2.18 with $D = p$; we need $\mathrm{EXT}_p(n_0, k, d)$ for $d \leq p-1$. For $d \leq p-2$ this is Proposition 3.9 ($2d+1 \leq 2p-3 \leq n_0$, $d+1 \leq p-1$). For $d = p-1$ run the peeling of Proposition 3.13 with T_0 in place of the type-0 window: at level 0, $f \equiv f(\emptyset) T_0$ at $\emptyset$ because $T_0(\emptyset) \equiv 1$; at levels $\ell + 1 \geq 1$ the argument is unchanged and uses only the units $(\ell+1)!$ with $\ell + 1 \leq p-1$ (Lemma 3.10) and $(d-j+1)!$ with $j \geq 1$, i.e. at most $(p-1)!$ (Corollary 3.12).[93] The board condition is $2d+1 = 2p-1 \leq n_0$. The type-0 window would have needed $(d+1)! = p!$, which is 0 mod p; T_0 is exactly what replaces it. □

B.2 From a rectangular lift to a unit design

Definition 3.17. A *rectangular lift* for (p, k) is a function H on matchings of the board with $2p-1$ holes and $2p-1+k$ pigeons such that $H(g) \equiv 1 \pmod p$ for every matching with $|g| = p$, and at

[87] Lean: `[1249] uniform_no_degree_p_unconditional, uniform_no_standard_degree_p_unconditional`.
[88] Lean: `[maint] DegreeMono.uniform_no_degree_le_p, uniform_no_standard_degree_le_p`.
[89] Lean: `[1249] uniform_unit_design_unconditional`.
[90] Lean: `[1249] Main.lean: uniform_unit_design, uniform_no_degree_p`.
[91] Lean: `[1249] squareLiftAssumption (Unconditional.lean)`.
[92] Lean: `[0607] VUnit.no_refutation_of_unit`.
[93] Lean: `[0607] VUnit.claimU, rp_span', ext_unit`.

every matching with $|g| = p-1$ every free row sum and every free column sum of H is exactly 0 (as an integer).[94]

Lemma 3.18 (top layer)**.** *Let p be prime and H a rectangular lift for (p, cp^2). Put $U(g) = (H(g)-1)/p$ for $|g| = p$. Then* $\mathrm{Bal}_p^{(2p-1,cp^2)}(U, -1, p-1)$.[95]

Proof. Let $|g| = p-1$. At a free pigeon there are $(2p-1)-(p-1) = p$ free holes, so $0 = \mathrm{row}_H(g,i) = p + p\,\mathrm{row}_U(g,i)$ and $\mathrm{row}_U(g,i) = -1$. At a free hole there are $(2p-1+k)-(p-1) = p+k$ free pigeons, so $0 = (p+k) + p\,\mathrm{col}_U(g,j)$, i.e. $\mathrm{col}_U(g,j) = -1 - k/p = -1 - cp \equiv -1$. This is the step that uses $p^2 \mid k$.[96] □

Lemma 3.19 (symmetric boundary)**.** *Let $n > 0$, $p \mid k$, and* $\mathrm{Bal}_p^{(n,k)}(U, -1, d)$. *Define $T(g) = -\mathrm{cv}_{n,d}(|g|) = -(n-|g|)(n-|g|-1)\cdots(n-d+1)$ for $|g| \le d$ and $T(g) = U(g)$ for $|g| = d+1$. Then $T(\emptyset) = -n(n-1)\cdots(n-d+1)$ and* $\mathrm{Bal}_p(T, T, e)$ *for every $e \le d$.*[97]

Proof. For $e < d$ the free row sum is $-(n-e)\mathrm{cv}(e+1) = -\mathrm{cv}(e) = T(g)$ and the free column sum is $-(n+k-e)\mathrm{cv}(e+1) \equiv -\mathrm{cv}(e) = T(g)$, using $p \mid k$. For $e = d$ the line sums are sums of U, which are $\equiv -1 = T(g)$ since $\mathrm{cv}(d) = 1$. □

Proposition 3.20. *If p is prime and a rectangular lift for (p, cp^2) exists, then the conclusion of Theorem 3.15 holds.*[98]

Proof. Lemma 3.18 and Lemma 3.19 with $n = 2p-1$, $d = p-1$ give T with $T(\emptyset) = -(2p-1)(2p-2)\cdots(p+1)$. Each factor $p+r$ $(1 \le r \le p-1)$ is $\equiv r$, a unit, so $T(\emptyset)$ is a unit;[99] scale T by its inverse. □

Theorem B follows from Proposition 3.20 and Theorem 3.16 with $n_0 = 2p-1$ ($p \mid cp^2$), and the standard form from Theorem 2.4.[100]

B.3 From the square to the rectangle

Definition 3.21. A *square lift* (integral lift) for p is a function H on matchings of the square board, $2p-1$ pigeons and $2p-1$ holes ($k = 0$), with $H(g) \equiv 1$ for $|g| = p$ and all free row and column sums exactly 0 at matchings with $|g| = p-1$.[101]

Proposition 3.22. *For prime p, a square lift for p and any c yield a rectangular lift for (p, cp^2).*[102]

Proof. Let $k = cp^2$. For each set A of $2p-1$ pigeons among $2p-1+k$, let H_A be H transported to the pigeons A by a relabelling and extended by 0 to every matching that uses a pigeon outside A.[103] Put $\widetilde{H} = \sum_A H_A$.[104]

Exact harmonicity. Padding by zeros keeps all free line sums at $(p-1)$-matchings exactly 0: a row of an outside pigeon sums zeros, a line of an inside pigeon or of a hole only gains zero terms,

[94] Lean: `[1249] RectangularLift (Boundary.lean)`.
[95] Lean: `[1249] RectangularLift.top, top_bal`.
[96] Lean: `[1249] RectangularLift.line_eq; [1402] SymDesign.rowS_const, colS_const`.
[97] Lean: `[0634] FactorBoundary.design_of_top, lower_bal, lower_top; TopSplice.splice_bal`.
[98] Lean: `[1249] unit_design_of_rectangular_lift`.
[99] Lean: `[0634] FactorUnit.ff_unit, empty_unit`.
[100] Lean: `[1249] no_refutation_of_rectangular_lift; uniform_no_standard_degree_p`.
[101] Lean: `[1249] IntegralLift (Lift.lean)`.
[102] Lean: `[1249] rectangleFromSquare (Transfer.lean)`.
[103] Lean: `[1249] Selection.embed, Selection.perm, Pad.pad`.
[104] Lean: `[1249] transferred`.

and at a matching that already uses an outside pigeon every term is 0. Relabelling preserves Bal, and a sum of exactly harmonic functions is exactly harmonic.[105]

Unit. For $|g| = p$, $H_A(g) \equiv$ [the pigeons of g lie in A]. The p pigeons of g are distinct, so the number of such A is $\binom{2p-1+k-p}{(2p-1)-p} = \binom{k+p-1}{p-1}$.[106] Finally $\binom{cp^2+p-1}{p-1} \equiv \binom{p-1}{p-1} = 1$, by iterating $\binom{a+p}{r} \equiv \binom{a}{r}$ ($r < p$, Lemma 3.7) cp times.[107] (This step only needs $p \mid k$.) □

B.4 An explicit square lift

Split the rows and the columns of the square board into the blocks $B_0 = \{0, \dots, p-2\}$, $B_1 = \{p-1, \dots, 2p-3\}$, $B_2 = \{2p-2\}$ of sizes $a = (p-1, p-1, 1)$.[108] For a matching g let $M(g) \in \mathbb{N}^{3\times 3}$ count its edges from row block I to column block J, with row sums rc_I and column sums cc_J. M is *feasible* if $\mathrm{rc}_I \le a_I$ and $\mathrm{cc}_J \le a_J$; its *total* is the number of edges. E_{IJ} is the unit matrix at (I, J).[109]

Lemma 3.23 (reduction to count matrices)**.** *Let $V\colon \mathbb{N}^{3\times 3} \to \mathbb{Z}$ satisfy, for feasible M: (unit) $V(M) \equiv 1$ if the total is p; and, if the total is $p-1$,*

$$\textstyle\sum_J (a_J - \mathrm{cc}_J(M))\, V(M + E_{IJ}) = 0 \text{ for every } I,$$
$$\textstyle\sum_I (a_I - \mathrm{rc}_I(M))\, V(M + E_{IJ}) = 0 \text{ for every } J.$$

Then $H(g) = V(M(g))$ is a square lift.[110]

Proof. At a matching g, a free pigeon in row block I has $a_J - \mathrm{cc}_J(M(g))$ free holes in column block J, and each extension has count matrix $M(g) + E_{IJ}$. □

Define, for feasible M of total p,

$$V_0(M) = -(-1)^{M_{01}+M_{10}}\,(p-2)!\,N(M) \cdot (p-1-\mathrm{rc}_0)!\,(p-1-\mathrm{rc}_1)!\,(p-1-\mathrm{cc}_0)!\,(p-1-\mathrm{cc}_1)!$$

and $V_0(M) = 0$ otherwise, where the numerator $N(M)$ is, for feasible M (cases read top to bottom; feasibility allows at most one edge in block row 2 and one in block column 2),

- $N(M) = p-1$ if $M_{22} = 1$;
- $N(M) = M_{00}M_{11} - M_{01}M_{10}$ if $\mathrm{rc}_2 = \mathrm{cc}_2 = 0$;
- if $\mathrm{rc}_2 = 0$ and $\mathrm{cc}_2 = 1$: $N(M) = -(M_{10} + M_{11})$ when $M_{02} = 1$, and $-(M_{00} + M_{01})$ when $M_{12} = 1$;
- if $\mathrm{rc}_2 = 1$ and $\mathrm{cc}_2 = 0$: $N(M) = -(M_{01} + M_{11})$ when $M_{20} = 1$, and $-(M_{00} + M_{10})$ when $M_{21} = 1$;
- if $\mathrm{rc}_2 = \mathrm{cc}_2 = 1$ and $M_{22} = 0$: $N(M) = -1$ when $M_{02} = M_{20}$, and 1 otherwise.[111]

[105] Lean: `[1249] Pad.pad_bal, embedded_bal, embed_harmonic, bal_sum; [1402] Relabel.bal_relab`.
[106] Lean: `[1249] embed_unit, selection_count, SubsetCount.coverCount_eq, transferred_congr_count`.
[107] Lean: `[1249] Multiplicity.subset_multiplicity_congr, bin_shift_multiple`.
[108] Lean: `[1249] Block.squarePartition, capacity, blockOf`.
[109] Lean: `[1249] Block.counts, Feasible, inc`.
[110] Lean: `[1249] Block.Certificate, Certificate.toIntegralLift, row_formula, col_formula, feasible_counts`.
[111] Lean: `[1249] Block.numerator, explicitValue (ExplicitFormula.lean)`.

(A comment in the source says the formula was obtained from a generating-function coefficient; that derivation is not part of the Lean proof and is not used.)

Lemma 3.24. *Let* $w(M) = \prod_I (a_I - \mathrm{rc}_I)! \prod_J (a_J - \mathrm{cc}_J)!$ *and* $\varphi(M) = (-1)^{M_{01}+M_{10}} N(M)$ *when* $\mathrm{rc}_2, \mathrm{cc}_2 \le 1$, *else* 0. *For feasible* M *of total* p, $V_0(M) = -(p-2)!\, w(M)\, \varphi(M)$; *and* $(a_I - \mathrm{rc}_I)(a_J - \mathrm{cc}_J)\, w(M + E_{IJ}) = w(M)$ *when* $\mathrm{rc}_I < a_I$, $\mathrm{cc}_J < a_J$.[112]

Lemma 3.25. *Let* M *be feasible of total* $p-1$ *and* $\mathrm{rc}_I < a_I$. *Then* $\sum_J \varphi(M + E_{IJ}) = 0$. *Also* $\varphi(M) = 0$ *whenever* M *has total* p *and some* $\mathrm{cc}_J > a_J$; *and* $\varphi(M^{\mathsf{T}}) = \varphi(M)$.[113]

Proof. Feasibility leaves ten patterns for the entries in block row and column 2 (at most one edge in row block 2 and one in column block 2).[114] The Lean proof checks the identity in each of the 10×3 cases (pattern, row block I) by rewriting and linear arithmetic (`simp`, `omega`, `grind`). One case, for illustration: no edges in block row or column 2 and $I = 0$. Writing $\sigma = (-1)^{M_{01}+M_{10}}$, the three terms are $\sigma((M_{00}+1)M_{11} - M_{01}M_{10})$, $-\sigma(M_{00}M_{11} - (M_{01}+1)M_{10})$ and $-\sigma(M_{10} + M_{11})$, which sum to 0. □

Proposition 3.26 (exact harmonicity)**.** *For feasible* M *of total* $p-1$: $\sum_J (a_J - \mathrm{cc}_J) V_0(M + E_{IJ}) = 0$ *for every* I *and* $\sum_I (a_I - \mathrm{rc}_I) V_0(M + E_{IJ}) = 0$ *for every* J.[115]

Proof. If $\mathrm{rc}_I = a_I$, every $M + E_{IJ}$ is infeasible and all terms vanish. Otherwise multiply the row sum by $a_I - \mathrm{rc}_I \neq 0$. By Lemma 3.24 each term becomes $-(p-2)!\, w(M)\, \varphi(M + E_{IJ})$ (when $\mathrm{cc}_J = a_J$ the coefficient is 0 and $\varphi(M + E_{IJ}) = 0$ by Lemma 3.25), and the sum vanishes by Lemma 3.25. Columns follow from $V_0(M^{\mathsf{T}}) = V_0(M)$.[116] □

Lemma 3.27 (factorial reflection)**.** *For every* $p \ge 1$ *and* $r \le p-1$: $(-1)^r\, r!\, (p-1-r)! \equiv (p-1)! \pmod p$. *For* $p \ge 2$: $(p-2)! \equiv -(p-1)!$.[117]

Proof. Induction on r: passing from r to $r+1$ multiplies by $-(r+1)$ and divides by $p-1-r \equiv -(r+1)$. The difference of consecutive terms is an explicit multiple of p, so no division is needed. For the second claim, $(p-1)! = (p-1)(p-2)! \equiv -(p-2)!$. □

Proposition 3.28 (common residue)**.** *For feasible* M *of total* p, $V_0(M) \equiv -\big((p-1)!\big)^3 \pmod p$.[118]

Proof. The Lean proof treats the ten boundary patterns separately. In each it rewrites the four factorial arguments using the total p, reduces $V_0(M) \bmod p$ to $(p-2)!\, (-1)^{r+t}\, r!\, s!\, t!\, u!$ with $r+s = t+u = p-1$, and applies Lemma 3.27 twice: $(p-2)!\, (p-1)!^2 \equiv -(p-1)!^3$. The first pattern (no edges in block row or column 2) runs as follows. Then $\mathrm{rc}_0 + \mathrm{rc}_1 = p$ with $\mathrm{rc}_I \le p-1$, so both are ≥ 1, and similarly for columns; the factorial arguments are $p - 1 - \mathrm{rc}_0 = \mathrm{rc}_1 - 1$, etc. Since $M_{00}M_{11} - M_{01}M_{10} = pM_{00} - \mathrm{rc}_0\, \mathrm{cc}_0$, and $(-1)^{\mathrm{rc}_0 + \mathrm{cc}_0} = (-1)^{M_{01}+M_{10}}$,

$$V_0(M) \equiv (p-2)!\, (-1)^{\mathrm{rc}_0}\, \mathrm{rc}_0!\, (\mathrm{rc}_1 - 1)! \cdot (-1)^{\mathrm{cc}_0}\, \mathrm{cc}_0!\, (\mathrm{cc}_1 - 1)! \pmod p,$$

which has the stated form with $r = \mathrm{rc}_0$, $s = \mathrm{rc}_1 - 1$, $t = \mathrm{cc}_0$, $u = \mathrm{cc}_1 - 1$. The other nine patterns are verified in Lean by the same rewriting followed by `simp`/`grind`; we have not written them out. □

[112] Lean: `[1249] Block.weight, explicit_eq, weight_inc; ExplicitAlgebra.phi, jCore`.
[113] Lean: `[1249] Block.phi_row, phi_bad_col, phi_transpose`.
[114] Lean: `[1249] Block.boundary_cases`.
[115] Lean: `[1249] Block.explicit_row, explicit_col, weighted_term`.
[116] Lean: `[1249] Block.explicit_transpose`.
[117] Lean: `[1249] Block.factorial_reflection, factorial_tail`.
[118] Lean: `[1249] Block.explicit_common_congruence, commonValue, main_pair`.

Theorem 3.29. *For every prime p a square lift exists.*[119]

Proof. $(p-1)!$ is a unit (units), so $-\big((p-1)!\big)^3$ has an inverse u.[120] Then $V = u\,V_0$ satisfies the hypotheses of Lemma 3.23 by Propositions 3.28 and 3.26. □

Remark 3.30. For $p = 2$ and $p = 3$ the library also contains explicit square lifts given as data, whose defining properties are verified by kernel computation (`decide`); they give the statements for $p = 2$, $n \geq 3$ and $p = 3$, $n \geq 5$ independently of B.4.[121]

Mechanism of B. An explicit integer function on p-matchings of the $(2p-1) \times (2p-1)$ board, depending only on the 3×3 block-count matrix, has all free line sums at $(p-1)$-matchings exactly 0 and is $\equiv$ a unit on every p-matching (the *square lift*). Summing its copies over all choices of $2p-1$ pigeons among $2p-1+k$ keeps exact harmonicity and multiplies the residue by $\binom{k+p-1}{p-1} \equiv 1$. Because the line sums are exact, the top layer $(H-1)/p$ is balanced against -1; this is where $p^2 \mid k$ enters, since the column count $p+k$ must leave $k/p \equiv 0$. Spliced under the symmetric falling-factorial levels it is a *balanced unit design* of degree p on $2p-1$ holes. The *design-to-lower-bound criterion* (B.1) is the peeling of A run at level $p-1$, with this design replacing the type-0 window whose value $p!$ vanishes mod p.

3.3 C. No degree-4 refutation over $\mathbb{F}_2$ at surplus 8

Theorem 3.31 (C)**.** *For every $n \geq 7$,* ontoFPHP$(n, 8)$ *has no degree-4 semantic PC refutation over $\mathbb{Z}/2$, and no degree-4 standard PC refutation over $\mathbb{F}_2$.*[122]

By monotonicity (Lemma 2.2) there is no refutation of degree at most 4, so the PC degree is at least 5.[123] The proof is Theorem 2.18 with $p = 2$, $k = 8$, $n_0 = 7$, $D = 4$: it suffices to prove $\mathrm{EXT}_2(7, 8, d)$ for $d = 0, 1, 2, 3$.[124] Level 0 is explicit and levels 1, 2 use small kernel-checked certificates; level 3 is reduced to a single finite statement, Proposition 3.42, which is established by a kernel-checked certificate. Throughout, $n = 7$, $k = 8$: 15 pigeons, 7 holes, 105 variables.

Relabellings. Permutations σ of the pigeons and τ of the holes act on functions of matchings, and $\mathrm{Bal}_p(y, f, e)$ implies $\mathrm{Bal}_p(y^{\sigma\tau}, f^{\sigma\tau}, e)$.[125] So every relabelling of a function with an extension has one.

Lemma 3.32 (level 0)**.** *If $p \mid k$ and $n > 0$, then* $\mathrm{EXT}_p(n, k, 0)$.[126]

Proof. Given $f(\emptyset)$, put $y(x_{ii}) = f(\emptyset)$ for $i < n$, $y(x_{i0}) = f(\emptyset)$ for $i \geq n$, and $y = 0$ elsewhere. Every row sums to $f(\emptyset)$, columns $j \neq 0$ sum to $f(\emptyset)$, and column 0 sums to $(k+1)f(\emptyset) \equiv f(\emptyset)$. □

Lemma 3.33 (level 1)**.** $\mathrm{EXT}_2(7, 8, 1)$.[127]

[119] Lean: `[1249] squareLiftAssumption, Block.squareLiftAssumption`.
[120] Lean: `[1249] Block.commonValue_unit`.
[121] Lean: `[1249] integralLift2, integralLift3, no_standard_degree_two, no_standard_degree_three`.
[122] Lean: `[0841] L3.Main.no_deg4; [1156] StdPCCor.std_no_deg4_k8`.
[123] Lean: `[maint] DegreeMono.no_deg_le4, deg_ge5`.
[124] Lean: `[1434] Ext278b.hard_of_ext3`.
[125] Lean: `[1402] Relabel.relab, bal_relab`.
[126] Lean: `[1402] Ext0.ext0`.
[127] Lean: `[1434] Ext278a.ext1`.

Proof. Let H be the degree-3 design of the library: $H(\emptyset) = 1$ and $\mathrm{Bal}_2(H, H, e)$ for $e \le 2$.[128] It is a two-block product: on pigeons $< 12 \times$ holes < 4 it is the library design of ontoFPHP$(4, 8)$ reindexed, on pigeons $12, 13, 14 \times$ holes $4, 5, 6$ the indicator of the diagonal, and 0 on cross edges. Every function c on edges whose row and column sums at $\emptyset$ are all $\equiv \kappa$ lies in Span_1 of the relabellings of H: this is a kernel-checked peeling certificate (85 elimination steps by sums of relabellings in echelon order, then 21 line propagations), sound by the general checker theorem.[129] Each relabelling of H extends itself at level 1; Lemma 2.13 concludes. □

Lemma 3.34 (level 2)**.** $\mathrm{EXT}_2(7, 8, 2)$.[130]

Proof. Let $\mathrm{Bal}_2(f, c, 1)$. By the flag lemma at $g = \emptyset$ the link c is level-0 balanced,[131] so by the level-1 certificate $c \in \mathrm{Span}_1$(relabellings of H). Since each generator extends itself, there is u in Span_2 of the same generators with $\mathrm{Bal}_2(u, c, 1)$.[132] Then $f - u$ has zero links at level 1, and a kernel-checked level-2 certificate (files `L2C0`–`L2C8`) puts it in Span_2 of the relabellings of R_0, the product of the rectangles on $\{0, 1\} \times \{0, 1\}$ and $\{2, 3\} \times \{2, 3\}$, and of T, the indicator of 2-matchings inside the 3×3 block on pigeons and holes $\{0, 1, 2\}$.[133] Each generator has a level-2 extension: R_0 by a three-block product with block extensions (uses $n \ge 7$, $p \mid k$), T by an explicit two-block product, H by itself.[134] □

Lemma 3.35 (lifting to level 3)**.** *Let $1 < n$ and let G, G_z be sets of functions such that (i) each $\gamma \in G$ has L with $\mathrm{Bal}_p(L, \gamma, 2)$ and a level-3 extension of L; (ii) every c with $\mathrm{Bal}_p(c, h, 1)$ for some h lies in $\mathrm{Span}_2(G)$; (iii) each $z \in G_z$ has a level-3 extension; (iv) every z with $\mathrm{Bal}_p(z, 0, 2)$ lies in $\mathrm{Span}_3(G_z)$. Then $\mathrm{EXT}_p(n, k, 3)$.*[135]

Proof. Let $\mathrm{Bal}_p(f, c, 2)$. The link c is level-1 balanced (flag lemma), so $c \in \mathrm{Span}_2(G)$ by (ii). Lifts are linear, so by (i) there is L with $\mathrm{Bal}_p(L, c, 2)$ and an extension y of L. Then $f - L$ has zero links at level 2, so by (iv), (iii) and Lemma 2.13 it has an extension y'. $y + y'$ extends f. □

Let $\mathcal{S}_{23}$ be the set of g with $\mathrm{Bal}_2(g, g, 2)$ and $\mathrm{Bal}_2(g, g, 3)$, and $\mathcal{S}_3$ those with $\mathrm{Bal}_2(g, g, 3)$.[136] For $G = \mathcal{S}_{23}$ take $L = \gamma$ in (i); for $G_z = \mathcal{S}_3$ take the extension z itself in (iii).

Lemma 3.36 (condition (ii))**.** *Every c with $\mathrm{Bal}_2(c, h, 1)$ lies in $\mathrm{Span}_2(\mathcal{S}_{23})$.*[137]

Proof. The link h is level-0 balanced. A second kernel-checked level-1 certificate, over the library's degree-4 unit design D (with $D(\emptyset) = 1$ and $\mathrm{Bal}_2(D, D, e)$ for $e \le 3$) and a rectangle tower R_{Bd}, puts h in Span_1 of functions self-balanced at levels $1, 2, 3$, which lifts to $u \in \mathrm{Span}_2(\mathcal{S}_{23})$ with $\mathrm{Bal}_2(u, h, 1)$.[138] Then $c - u$ has zero links at level 1, so the level-2 certificate of Lemma 3.34 puts it in the span of relabellings of R_0 and T. By kernel evaluation on all pairs of variables, R_0 agrees mod 2 on 2-matchings with a self-balanced tower R_{RM}, and T with a relabelling of a tower T_B; both are in $\mathcal{S}_{23}$.[139] □

[128] Lean: `[1434] HDes.H, HDes.h_design`.
[129] Lean: `[1434] Ext278a.span1; L1Data.cert, L1Data.run_ok; Peel.span_of_run`.
[130] Lean: `[1434] Ext278b.ext2, span2, g_ext`.
[131] Lean: `[1434] Ext278a.link_bal; [1402] RowCol.row_col`.
[132] Lean: `[1434] Ext278a.lift1`.
[133] Lean: `[1434] L2Base.R0; [1402] TExt.T; [1434] Ext278b.bases2, cov, p0`.
[134] Lean: `[1402] RectExt.rect_ext, TImage.t_image_ext, TExt.t_ext; [1434] Ext278a.gh_ext`.
[135] Lean: `[1434] Lift.EXT_lift (with d = 1), Lift.lift_of_span`.
[136] Lean: `[1434] Ext278c.SB23, SB3`.
[137] Lean: `[1434] Ext278c.span2b`.
[138] Lean: `[1434] Ext278c.span1b, lift1b, bases1b; L1bData; K5Bal; Towers.RBd`.
[139] Lean: `[1434] Ext278c.zl2, gz_span; Cong.r0_cong, Cong.t_cong; Towers.rrm_bal`.

Generator lemmas for levels 1–3. We prove the properties of the generators used in Lemmas 3.33, 3.34 and 3.36. Throughout $n = 7$, $k = 8$ unless stated otherwise; pigeon i and hole j give the variable x_{ij} of index $7i + j$.

Block products. Block maps β_P on pigeons and β_H on holes, with values $< m$, put the edge x_{ij} in block b if $\beta_P(i) = \beta_H(j) = b$, and make it a *cross edge* otherwise. For factors φ_b on local matchings whose values depend only on the set of edges, the block product is $\Pi(g) = 0$ if g has a cross edge and $\Pi(g) = \prod_b \varphi_b(g_b)$ otherwise, g_b the edges of g in block b. For a local matching σ of block b and a pigeon i of block b free in σ, the *local row sum* $\mathrm{lrow}_{\varphi_b}(\sigma, i)$ is $\sum \varphi_b(\sigma + x_{it})$ over the holes t of block b free in σ; local column sums are defined likewise. We use:

(P1) (stated, not reconstructed: `[1402] Win.bal_win`; its hypotheses are read off the call in `HDes.lean`.) If at every matching g with $|g| = e$ (under a further hypothesis on g that the proof of Lemma 3.37 does not use) every local row sum $\mathrm{lrow}_{\varphi_b}(g_b, i)$, $b = \beta_P(i)$, i free, is $\equiv \varphi_b(g_b)$, and likewise every local column sum, then $\mathrm{Bal}_p(\Pi, \Pi, e)$ $(m = 3)$.

(P2) The same with the weaker hypothesis that, for g without cross edges, $(\mathrm{lrow}_{\varphi_b}(g_b, i) - \varphi_b(g_b)) \prod_{c \neq b} \varphi_c(g_c) \equiv 0$, and likewise for columns.[140] Proof: if g has a cross edge, Π vanishes at g and at every one-edge extension of it; otherwise the free row sum of Π at g is $\mathrm{lrow}_{\varphi_b}(g_b, i) \prod_{c \neq b} \varphi_c(g_c)$, since an edge from i to a hole of another block is a cross edge (stated, not reconstructed: `[1402] ProdExt.line_Prod`), while $\Pi(g) = \varphi_b(g_b) \prod_{c \neq b} \varphi_c(g_c)$.

(P3) (stated, not reconstructed: `[1434] ProdM.bal_prodM`, in the form described in the paragraph on the generators below.) Say φ_b is *locally balanced up to* E_b if the local line identities hold at every local matching of block b with at most E_b edges.[141] If each φ_b is locally balanced up to E_b, vanishes on local matchings with fewer than s_b edges, and $e \leq E_b + \sum_{c \neq b} s_c$ for every b, then $\mathrm{Bal}_p(\Pi, \Pi, e)$.

A *length-only* factor is $\varphi(\sigma) = G(|\sigma|)$. On a block with P pigeons and Q holes its local row sum at σ is exactly $(Q - |\sigma|)\, G(|\sigma| + 1)$ and its local column sum $(P - |\sigma|)\, G(|\sigma| + 1)$, by counting free cells.[142] Local balance of a length-only factor is packaged as (stated, not reconstructed: `[1434] LocLf.lf_locbal`); in every application below its numerical hypotheses are closed by `omega` and `decide`.

Lemma 3.37 (the design H)**.** *$H(\emptyset) = 1$ and $\mathrm{Bal}_2(H, H, e)$ for $e \leq 2$.*[143]

Proof. H is the block product with $\beta_P(i) = [i \geq 12]$, $\beta_H(j) = [j \geq 4]$ and factors $\varphi_0 = B_7$, $\varphi_1 = \mathrm{Dg}$ (the third slot of the Lean product is unused). Here $B_7(\sigma) = B(\rho(\sigma))$, where B is the library design of ontoFPHP$(4, 8)$ and $\rho(x_{ij}) = 4i + j$ renames an edge of block 0 to the variable x_{ij} of the 4-hole board; and $\mathrm{Dg}(\sigma) = 1$ if every edge of σ is one of $(12, 4)$, $(13, 5)$, $(14, 6)$, else 0.[144] $H(\emptyset) = 1$ is evaluated by `decide`. For the balance we apply (P1) and check the local identities at a matching g with $|g| = e \leq 2$.

Block 0. ρ is injective on block 0, preserves pigeons, holes and conflicts, and maps local matchings to matchings of ontoFPHP$(4, 8)$ of the same size. So for a pigeon $i < 12$ free in g the local row sum of B_7 at g_0 over the holes $t < 4$ is the row sum $\mathrm{row}_B^{(4,8)}(\rho(g_0), i)$, with i free in $\rho(g_0)$; likewise

[140] Lean: `[1402] TExt.bal_win'`.
[141] Lean: `[1434] ProdM.LocBal (definition read off its uses in Towers.lean)`.
[142] Lean: `[1402] TExt.gsym_row, gsym_col`.
[143] Lean: `[1434] HDes.h_design`.
[144] Lean: `[1434] HDes.H, B7, Dg, rho`.

for columns $j < 4$.[145] Since $|\rho(g_0)| \le 2$, the library fact $\mathrm{Bal}_2^{(4,8)}(B, B, e')$ for $e' \le 2$ (stated, not reconstructed: `[1402] BDesMain.b_design`) gives $\equiv B(\rho(g_0)) = B_7(g_0)$.

Block 1. For a pigeon $i \in \{12, 13, 14\}$ free in g, among the holes $4, 5, 6$ only $t = i - 8$ can give a nonzero term $\mathrm{Dg}(g_1 + x_{it})$. If $\mathrm{Dg}(g_1) = 0$ all terms are 0; if $\mathrm{Dg}(g_1) = 1$, then g_1 is diagonal, the hole $i - 8$ is free (its diagonal pigeon i is), and the term is 1. So the local row sum equals $\mathrm{Dg}(g_1)$ exactly; columns $j \in \{4, 5, 6\}$ likewise, with the single pigeon $j + 8$.[146] □

Lemma 3.38 (the design D)**.** *$D(\emptyset) = 1$ and $\mathrm{Bal}_2(D, D, e)$ for $e \le 3$.*[147]

Proof. $D = \mathrm{Wt}(sup)$ is the $\mathbb{Z}_{15} \times \mathbb{Z}_7$-invariant weight of the library design of run 1627 (stated, not reconstructed: `[1627] K5Defs.Wt` with its properties `Wt_congr`, `Wt_nm`); Wt depends only on the set of a list and vanishes on lists that are not matchings. $D(\emptyset) = 1$ is the kernel-checked fact `chk_W0`. Let $|g| \le 3$. By the two properties of Wt, each term of the row sum of D at g and pigeon i equals $\mathrm{Wt}(x_{ij} :: g)$, so $\mathrm{row}_D(g, i) - D(g)$ is the pigeon defect $\mathrm{pigS}(sup, i, g)$ of the run-1627 development, and likewise columns.[148] The library lemma `[1627] K5Lem.lin_all` (stated, not reconstructed), fed with the kernel-checked certificates `chk_all` and `chk_empty` (`[1627] K5Chk`), states that both defects are even for every g of size ≤ 3; the freeness of i is not used.[149] □

Lemma 3.39 (towers)**.** *The four block products of Table 2 satisfy $\mathrm{Bal}_2(\Phi, \Phi, e)$ for every $e \le 3$.*[150]

tower	block (pigeons × holes)	factor	E_b	s_b
R_{Bd}	$\{0, \dots, 11\} \times \{0, \dots, 3\}$	B_7	2	0
	$\{12, 13\} \times \{4, 5\}$	$[s \ge 1]$	∞	1
	$\{14\} \times \{6\}$	1	∞	0
T_B	$\{0, \dots, 11\} \times \{0, \dots, 3\}$	B_7	2	0
	$\{12, 13, 14\} \times \{4, 5, 6\}$	$[s \ge 2]$	∞	2
R_{RM}	$\{0, 1\}^2$, $\{2, 3\}^2$	$[s \ge 1]$ each	∞	1
	$\{4, \dots, 14\} \times \{4, 5, 6\}$	M	1	0
R_{RRRM}	$\{0, 1\}^2$, $\{2, 3\}^2$, $\{4, 5\}^2$	$[s \ge 1]$ each	∞	1
	$\{6, \dots, 14\} \times \{6\}$	1	0	0

Table 2: The towers of `Towers.lean` (s = local size; ∞ is 100 in Lean).

Proof. Each tower is a block product, and we apply (P3) with the E_b and s_b of Table 2.[151] *Vanishing below s_b*: $[s \ge 1]$ at $s = 0$ and $[s \ge 2]$ at $s \le 1$ are 0 (`simp`). *Truncation*: $e \le E_b + \sum_{c \ne b} s_c$ for $e \le 3$ and all b is checked by `decide`; the binding cases are $3 \le 2 + 1 + 0$ for B_7 in R_{Bd}, $3 \le 2 + 2$ for B_7 in T_B, $3 \le 1 + 1 + 1$ for M, and $3 \le 0 + 1 + 1 + 1$ for the all-ones block of R_{RRRM}. *Local balance*:

- Length-only factors, by `LocLf.lf_locbal` with `decide`. With the local sums $(Q - s)G(s + 1)$ and $(P - s)G(s + 1)$ the identities read, mod 2: for $[s \ge 1]$ on 2×2, $2 \cdot 1 \equiv 0$ at $s = 0$ and $1 \cdot 1 = 1$ at $s = 1$; for $[s \ge 2]$ on 3×3, $3 \cdot 0 = 0$, $2 \cdot 1 \equiv 0$, $1 \cdot 1 = 1$ at $s = 0, 1, 2$; for the pin, $1 \cdot 1 = 1$ at $s = 0$; for the all-ones 9×1 block, $1 \cdot 1 = 1$ (rows) and $9 \cdot 1 \equiv 1$ (columns) at $s = 0$. At larger s there is no free line in the block, except for rows of the 9×1 block at $s = 1$, where the identity would read $0 \equiv 1$; this is why $E_b = 0$ there.

[145] Lean: `[1434] HDes.row0, col0, red_CM, red_len, red_pfree, red_hfree`.
[146] Lean: `[1434] HDes.row1, col1`.
[147] Lean: `[1434] K5Bal.d_design`.
[148] Lean: `[1434] K5Bal.term, row_eq, col_eq`.
[149] Lean: `[1434] K5Bal.d_bal`.
[150] Lean: `[1434] Towers.towers_bal, rbd_bal, tb_bal, rrm_bal, rrrm_bal`.
[151] Lean: `[1434] Towers.RBd, TB, RRM, RRRM`.

- B_7 (on pigeons < 12, holes < 4) up to $E = 2$: by the reindexing argument of Lemma 3.37 and `BDesMain.b_design`.[152]

- M up to $E = 1$: $M(\emptyset) = 1$, $M(\{v\}) = [v \in S_1]$, $M(\{v, w\}) = [\{v, w\} \in S_2]$ for explicit lists S_1 (11 edges) and S_2, and $M = 0$ on larger sets. The local identities at $\emptyset$ and at every single edge of the block are verified by kernel computation (`decide +kernel`).[153] □

Lemma 3.40 (rectangle pairs extend)**.** *Let $n \geq 7$, $p \mid k$, and let $f = \rho \otimes \rho'$ be the product (`tprod`, defined in `RectBlock`) of the signed rectangles ρ on $\{a, a'\} \times \{b, b'\}$ and ρ' on $\{c, c'\} \times \{d, d'\}$, with a, a', c, c' distinct and b, b', d, d' distinct. Then there is Y with* $\mathrm{Bal}_p(Y, f, 2)$.[154]

Proof. Choose fresh pigeons $x \neq y$ and fresh holes $h \neq h'$ (possible as $n \geq 7$).[155] Blocks: $0 = \{a, a', x\} \times \{b, b', h\}$, $1 = \{c, c', y\} \times \{d, d', h'\}$, and 2 the rest, with $n + k - 6$ pigeons and $n - 6 \geq 1$ holes.[156]

Block 2. There is μ on the block-2 cells with every row sum and every column sum $\equiv 1 \pmod p$. This holds for any L_1 pigeons and $L_2 \geq 1$ holes with $L_2 \leq L_1$ and $p \mid L_1 - L_2$ (here $L_1 - L_2 = k$), by induction on L_2: for $L_2 = 1$ take $\mu \equiv 1$ (the column sum is $L_1 \equiv L_2 = 1$); otherwise put $\mu = 1$ at one cell (s, q), $\mu = 0$ on the rest of row s and column q, and take μ on the remaining $(L_1 - 1) \times (L_2 - 1)$ board by induction.[157] Let M_2 be μ on single edges and 0 elsewhere; its local line sums at $\emptyset$ are $\equiv 1$.

The extension. The library combines block data by (stated, not reconstructed: `[1402] ProdExt.bal_prod3`), applied with local degrees $1, 1, 0$. Its data are, on block 0, the function A_0 equal to ρ on single edges and 0 elsewhere, and the degree-2 function $Y_0 = \mathrm{Yb}(a, a', x; b, b', h)$, a signed sum of three products of edge functions (stated, not reconstructed: `[1402] RectBlock.Yb`); on block 1 the same for ρ'; on block 2 the indicator A_2 of $\emptyset$ and $Y_2 = M_2$. The hypotheses supplied are: the local line sums of A_0 at $\emptyset$ are 0 (in row a, $\rho(x_{ab}) + \rho(x_{ab'}) + \rho(x_{ah}) = 1 - 1 + 0$, and likewise elsewhere); at each block-0 edge e, every free local line sum of Y_0 at $\{e\}$ equals $\rho(e)$ exactly (stated, not reconstructed: `[1402] RectBlock.r_row, r_col, Yb_row, Yb_col`); the same on block 1; and the line sums of M_2 at $\emptyset$ are $\equiv 1 = A_2(\emptyset)$. The result is Y with $\mathrm{Bal}_p(Y, \Pi_A, 2)$, where Π_A is the block product of A_0, A_1, A_2.

Agreement. On 2-matchings $\Pi_A = \rho \otimes \rho'$: ρ and ρ' vanish off blocks 0 and 1, and the remaining case split on the blocks of the two edges is closed by `simp` and `grind`.[158] Since $\mathrm{Bal}_p(Y, \cdot, 2)$ depends only on residues on 2-matchings (Lemma 2.13), $\mathrm{Bal}_p(Y, f, 2)$. □

Lemma 3.41 (the generator T extends)**.** *Let $n \geq 4$ and $2 \mid k$, and let T be the indicator of the 2-matchings inside the block $\{0, 1, 2\} \times \{0, 1, 2\}$. Then* $\mathrm{Bal}_2(Y, T, 2)$ *for the block product Y below.*[159]

Proof. Blocks: $0 = \{0, 1, 2\} \times \{0, 1, 2\}$ and 2 the remaining pigeons and holes (the middle slot of the Lean product is empty, with factor 1). Factors: $\varphi_0(\sigma) = [|\sigma| \geq 2]$; $\varphi_2(\emptyset) = 1$, $\varphi_2(\{x_{ij}\}) = 1$ if $i = j$ or ($i \geq n$ and $j = 3$), else 0, and $\varphi_2 = 0$ on two or more edges.[160] We show $\mathrm{Bal}_2(Y, Y, 2)$ by (P2); let g have $|g| = 2$ and no cross edge.[161]

[152] Lean: `[1434] Towers.b7_loc`.
[153] Lean: `[1434] Towers.M, MS1, MS2, M_chk0, M_chk1r, M_chk1c, M_loc`.
[154] Lean: `[1402] RectExt.rect_ext, rect_ext_lab`.
[155] Lean: `[1402] RectExt.fresh`.
[156] Lean: `[1402] RectExt.bpf, count2`.
[157] Lean: `[1402] RectExt.mu_exists`.
[158] Lean: `[1402] RectExt.rect_zero, tprod_eq_prod`.
[159] Lean: `[1402] TExt.t_ext`.
[160] Lean: `[1402] TExt.bt, G0, dd, dl, Y, T`.
[161] Lean: `[1402] TExt.Y_bal`.

Lines in block 0. The bracket of (P2) is $(3-s)\varphi_0(s+1)-\varphi_0(s)$ with $s=|g_0|\le 2$ (three holes, resp. three pigeons, in the block), i.e. 0, 2, 0 for $s=0,1,2$: even, by `decide`.

Lines in block 2. If $g_2=\emptyset$: a pigeon $i\ge 3$ has exactly one block-2 hole with $\varphi_2=1$, namely i if $i<n$ and 3 if $i\ge n$ (here $n\ge 4$), so its local row sum is $1=\varphi_2(\emptyset)$;[162] a hole $j\ge 3$ has local column sum $1+[j=3]\,k\equiv 1$, using $2\mid k$.[163] If $g_2\neq\emptyset$: the local line sums of φ_2 at g_2 are 0 (φ_2 vanishes on two edges), and the other factor $\varphi_0(g_0)=0$ because $|g_0|\le 1$, so the bracket times the other factor is 0.[164]

Agreement. On 2-matchings $Y=T$: if both edges lie in block 0, $Y=1\cdot 1=T$; otherwise either there is a cross edge, or an edge in block 2 forces $|g_0|\le 1$ and $\varphi_0(g_0)=0$, and in both cases $Y=0=T$.[165] Hence $\mathrm{Bal}_2(Y,T,2)$ (Lemma 2.13). □

Lemma 3.37 is the design H of Lemma 3.33; Lemmas 3.40 and 3.41 are the extensions of R_0 and T in Lemma 3.34; and Lemmas 3.38 and 3.39 give the self-balanced functions D, R_{Bd}, R_{RM} and T_B of Lemma 3.36. The tower R_{RRRM} (an RRR-type product) is not cited in Lemma 3.36.

It remains to verify condition (iv) for $G_z=\mathcal{S}_3$; this is the level-3 certificate.

Proposition 3.42 (level-3 zero-link span)**.** *Every z on matchings of* ontoFPHP$(7,8)$ *with* $\mathrm{Bal}_2(z,0,2)$ *lies in* $\mathrm{Span}_3(\mathcal{G}_3)$*, where $\mathcal{G}_3$ is the set of valid RRR and $T\otimes R$ generators defined below; $\mathcal{G}_3\subseteq\mathcal{S}_3$.*[166]

Given the proposition, Lemma 3.35 gives $\mathrm{EXT}_2(7,8,3)$,[167] and with Lemmas 3.32–3.34 and Theorem 2.18 this proves Theorem C.[168]

The generators. Both families are block products: matchings with an edge between different blocks get 0, and otherwise the value is a product of local factors. A *length-only* factor on a block with P pigeons and Q holes has a value $G(s)$ depending only on the size s of the local matching; it is locally balanced up to length L when, for every $s\le L$, $(Q-s)G(s+1)\equiv G(s) \pmod p$ if $s<P$ (a free pigeon exists) and $(P-s)G(s+1)\equiv G(s) \pmod p$ if $s<Q$ (a free hole exists).[169] A product of locally balanced factors is balanced against itself at level e, with a truncation rule: a factor need only be balanced up to some length E_b provided it vanishes below some length s_b and $e\le E_b+\sum_{c\neq b}s_c$ for every block b.[170]

- RRR(π): three 2×2 rectangle blocks $\{p_c,q_c\}\times\{j_c,h_c\}$ ($c<3$) and the all-ones block on the remaining 9 pigeons $\times$ 1 hole.[171]
- $T\otimes R(\pi)$: the T tower on a 3×3 block, a rectangle block, the all-ones block on 9 pigeons $\times$ the remaining hole, and a 1×1 pin.[172]

For valid parameters (a decidable condition) both are self-balanced at every level $e\le 3$; in the Lean proof the local conditions and the truncation inequality are discharged by `decide`.[173] They

[162] Lean: `[1402] TExt.row_out`.
[163] Lean: `[1402] TExt.col_out`.
[164] Lean: `[1402] TExt.loc_big, G0_small`.
[165] Lean: `[1402] TExt.Y_eq`.
[166] Lean: `[1434] Ext278d.z3_of_run, g3_sb3; [0841] L3.Main.run_all, cov_all`.
[167] Lean: `[1434] Ext278c.ext3_of_z3, hard_of_z3`.
[168] Lean: `[1434] Ext278d.hard_of_run`.
[169] Lean: `[1434] LocLf.lf_locbal`.
[170] Lean: `[1434] ProdM.bal_prodM`.
[171] Lean: `[1434] Gen3.GRR`.
[172] Lean: `[1434] Gen3T.GTR`.
[173] Lean: `[1434] Gen3.grr_bal, Gen3T.gtr_bal`.

vanish on matchings of size < 3, so they have zero links at level 2.[174] On a matching with 3 edges each value is the number of occurrences of the matching's code in an explicit list: the 64 products of one edge from each rectangle for RRR, and the 72 products of a 2-matching inside the T block with a rectangle edge for $T \otimes R$.[175]

The certificate and what it asserts. Code the matchings with three edges, i.e. decreasing lists $[a, b, c]$ of variables < 105, by $r_3(a, b, c) = \binom{a}{3} + \binom{b}{2} + c < 187460$ (injective).[176] A *state* for z is a pair (r, P) of a residual function r and a set P of *finalised* codes such that (S1) $\mathrm{Bal}_2(r, 0, 2)$; (S2) $z - r \in \mathrm{Span}_3(\mathcal{G}_3)$; (S3) r is even on every finalised matching.[177] Initially $r = z$ and $P = \emptyset$.[178] Each step names a pivot $[a, b, c]$, which must be a matching with three edges, and adds it to P:

- `k1`$(gs; a, b, c)$: valid generators gs such that every code in their joint support list is the pivot or not finalised, and the pivot occurs an odd number of times. Replace r by $r - r(\text{pivot}) \sum gs$. The new residual is even at the pivot and unchanged mod 2 at finalised codes (the generators are 0 there); (S1) and (S2) are preserved because the generators have zero links and lie in $\mathcal{G}_3$.[179]

- `kx`: the same, with the parity condition checked through the XOR mask of the support list, which allows cancellations among the generators.[180]

- `pr`$(u, w, \mathit{line}; a, b, c)$: a free row or column at the 2-matching $[u, w]$ all of whose codes other than the pivot are finalised, the pivot occurring an odd number of times. By (S1) the line sum of r is even, and by (S3) all other terms are even, so $r(\text{pivot})$ is even; r is unchanged.[181]

If after the run every matching with three edges is finalised, then $r \equiv 0$ on all of them and $z \equiv z - r$, so $z \in \mathrm{Span}_3(\mathcal{G}_3)$ by (S2).[182]

The Lean proof of Proposition 3.42 consists of the soundness theorem just described, which is a conventional proof, and two finite facts, each *verified by kernel computation* (`decide +kernel`):

1. the checker accepts the step list: $\mathrm{run}_3(\emptyset, ss)$ returns a final mask P_f. The list is stored in files `L3/F0`–`F89`; their headers record 95,550 steps in 2,154 pieces, one kernel check per piece, and the files contain 35,811 `k1`, 1,394 `kx` and 58,345 `pr` steps. The pieces are chained by kernel-checked equalities of the intermediate masks;[183]

2. P_f covers every matching with three edges: for every $a < 105$, every $b < a$, $c < b$ with $[a, b, c]$ a matching, $r_3(a, b, c) \in P_\mathrm{f}$ (60 kernel checks).[184]

We do not reproduce the certificate. Its correctness rests on the kernel's evaluation of these two facts; no part of the argument above depends on how the step list was found. (The finding page reports that a second agent re-checked the certificate with an independent program; that check is outside the Lean proof.)

[174] Lean: `[1434] Peel3.GS.zero2, GS.zl`.
[175] Lean: `[1434] Gen3.grr_count, Gen3T.gtr_count; Peel3.GS.count`.
[176] Lean: `[1434] Code3.r3, r3_inj`.
[177] Lean: `[1434] Peel3.State3, Inv3`.
[178] Lean: `[1434] Peel3.state_init`.
[179] Lean: `[1434] Peel3.chk1, kill_sound`.
[180] Lean: `[1434] Peel3.chkX; [0841] L3.Big.Part (splitting of large masks)`.
[181] Lean: `[1434] Peel3.chkP, line_par, step_sound`.
[182] Lean: `[1434] Peel3.run3_sound, span_of_cov, covA`.
[183] Lean: `[0841] L3.F`i`.r`j`, L3.Main.lk0-lk89, run_all; [1434] Peel3.run3_append`.
[184] Lean: `[0841] L3.Cov.cov, L3.Main.cov_all`.

Mechanism of C. The same EXT criterion as in A and B, at $(p, k, n_0) = (2, 8, 7)$. Levels 0–2 follow the peeling pattern with fixed designs and small certificates. Level 3 reduces, by the lifting lemma, to one finite linear-algebra statement over $\mathbb{F}_2$: the zero-link space on matchings with three edges is spanned by the self-balanced RRR and $T \otimes R$ products. A Gaussian-elimination-style certificate with 95,550 steps proves it.

4 Growth in ℓ: the padding transfer

This section proves result D. The mechanism is a transfer from Nullstellensatz designs to PC lower bounds that costs $2D - 2$ extra holes. It is applied to the explicit designs of Beame and Riis.

Theorem 4.1 (padding transfer)**.** *Let F be a field, $D \geq 1$, $n_0 \geq 1$ and $k \geq 0$. If the board with $n_0 + k$ pigeons and n_0 holes has a unit degree-D design over F, then for every*

$$n \;\geq\; n_0 + 2D - 2$$

the board with $n + k$ pigeons and n holes satisfies ND_D *and* U_D*. In particular* onto-FPHP_n^{n+k} *has no* PC_F *refutation of degree at most D.*

Corollary 4.2 (D)**.** *Let p be prime, $\ell \geq 1$ and F a field of characteristic p. For*

$$n \;\geq\; \frac{(p+2)^\ell - p^\ell}{2} + 2^{\ell+1} - 4,$$

every PC_F *refutation of* onto-$\mathrm{FPHP}_n^{n+p^\ell}$ *has degree at least 2^ℓ.*

Tier and inputs. T-WRITTEN. The proof uses three published results that we do not re-prove: Corollary B of Church and Ellenberg [5], Theorem 3.12 of Rhoades [18], and the design construction of Beame and Riis [2, Def. 8.2, Lemma 8.4, proof of Thm. 8.1]. It uses no representation theory beyond these, and no result of Harman or of Hone et al. The input we consider most exposed is the conversion of Rhoades's theorem into filtered completeness on the square (Lemma 4.5); it is short, and we give it in full. Section 4.7 records who wrote and checked each step.

4.1 External inputs

FI-modules. An *FI-module* assigns an abelian group $V(T)$ to each finite set T and a map $V(T) \to V(T')$ to each injection $T \hookrightarrow T'$, functorially. It is *generated in degree* $\leq d$ if every $V(T)$ is spanned by the images of the $V(T_0)$ with $|T_0| \leq d$. For a module W that assigns a group to each finite set and an action of bijections only, the *induced* ("free") FI-module is

$$M(W)(T) = \bigoplus_{T_0 \subseteq T} W(T_0),$$

with injections acting by relabelling [5, §2.1, eq. (2)]. "Free" here does not mean projective, and no restriction on characteristic is involved.

Theorem 4.3 (Church–Ellenberg [5, Cor. B])**.** *Let M be an induced FI-module generated in degree $\leq a$ and V an FI-module generated in degree $\leq b$. Then the kernel of every FI-module map $V \to M$ is generated in degree $\leq a + b + 1$.*

The theorem is stated for abelian groups. We apply it to F-vector spaces by forgetting the scalar structure; the images of smaller pieces are F-subspaces, so generation as groups gives spanning over F.

The square. On an $s \times s$ board let $I_s \subseteq F[x]$ be generated by all row sums $\sum_j x_{ij}$, all column sums $\sum_i x_{ij}$, and all products $x_{ij}x_{ij'}$, $x_{ij}x_{i'j}$ of two variables in one row or one column, squares included [18, Def. 1.1]. Let $I(\mathcal{P}_s)$ be the ideal of polynomials vanishing on the $s!$ permutation matrices, and $\operatorname{gr} I(\mathcal{P}_s)$ the ideal of top-degree homogeneous parts of its elements.

Theorem 4.4 (Rhoades [18, Thm. 3.12])**.** *Over every field,* $I_s = \operatorname{gr} I(\mathcal{P}_s)$.

Rhoades's later results on the module structure assume characteristic 0 or $p > s$; we use only Theorem 4.4, which has no such assumption.

Beame–Riis designs. For $\ell \geq 0$ put $N_\ell = ((p+2)^\ell - p^\ell)/2$. Beame and Riis construct, by induction on ℓ, a $(2^\ell - 1)$-design over $\mathbb{Z}_p$ on the board with $N_\ell + p^\ell$ pigeons and N_ℓ holes [2, proof of Thm. 8.1]. The step is their Lemma 8.4: a d-design on a $D_0 \times R_0$ board gives a $(2d+1)$-design on a board with $(p+1)|D_0| + |R_0|$ pigeons and $|D_0| + (p+1)|R_0|$ holes. Their d-designs (Definition 8.2) are exactly our unit degree-d designs with values in $\mathbb{F}_p$. We use the construction, not Theorem 8.1 itself; this avoids converting between their multiplier degree and our product degree.

4.2 Square filtered completeness

Lemma 4.5. *On an $s \times s$ board, for every t, W_t is the set of elements of F_t that vanish at every permutation matrix. Consequently every linear functional on F_t that vanishes on W_t is the restriction of a linear combination, with coefficients in F, of evaluations at permutation matrices.*

Proof. Elements of W_t vanish on permutation matrices, since each row or column relation does. Conversely let $g \in F_t$ vanish on permutation matrices; lift it to a multilinear polynomial f of degree $\leq t$ supported on matchings. Then $f \in I(\mathcal{P}_s)$. By Theorem 4.4 its top homogeneous part, of degree $t' \leq t$, is a combination $\sum_r h_r\gamma_r$ of the generators γ_r of I_s with homogeneous h_r and $\deg(h_r\gamma_r) = t'$. Let $\tilde{\gamma}_r$ be the corresponding axiom: $\gamma_r - 1$ (a pigeon or hole axiom) for a row or column sum, $x^2 - x$ (a Boolean axiom) for a square x^2, and γ_r itself (a collision or functional axiom) for a product of two distinct variables in one line. Each $\tilde{\gamma}_r$ has top part γ_r. Then $f - \sum_r h_r\tilde{\gamma}_r$ has degree $< t'$ and still vanishes on permutation matrices. Repeating, f is a combination of the axioms with multipliers of degree at most $t-1$ for the linear axioms and $t-2$ for the quadratic ones. Applying ρ, the Boolean, collision and functional terms vanish and each linear term becomes a combination of row and column relations at bases of size $\leq t-1$, so $g = \rho(f) \in W_t$.

For the second statement: a functional on F_t vanishing on W_t factors through the evaluation map $F_t \to F^{\mathcal{P}_s}$, whose kernel is W_t. Extend the induced functional from the image to all of $F^{\mathcal{P}_s}$; every functional on $F^{\mathcal{P}_s}$ is a combination of point evaluations. □

The functional obtained in this way need not be invariant, nonnegative or unital. Every evaluation at a permutation matrix w, viewed as the function $M \mapsto [M \subseteq w]$ on matchings of every size, satisfies all row and column relations at every base: if $M \subseteq w$ and i is free in M, then $w(i)$ is free in M, so exactly one term of the free row sum is 1. So do their linear combinations.

4.3 Balanced arrays have small support

Lemma 4.6. *Let $e \geq 1$. Every $f \in H_e(P,S)$ is a sum of arrays $f_\alpha \in H_e(P,S)$, each supported on matchings inside $P_\alpha \times S_\alpha$ with $|P_\alpha| \leq 2e$ and $|S_\alpha| \leq 2e$.*

Proof. Fix the pigeon set P and regard everything as a functor of the hole set. Let $C_e(T)$ be the space of arrays on e-matchings $P \leftrightarrow T$ all of whose free column sums vanish. Each column

condition involves a single e-element hole set holes$(N)\cup\{j\}$, so $C_e(T)=\bigoplus_{T_0\subseteq T,\,|T_0|=e}U(T_0)$, where $U(T_0)$ is the space of column-balanced arrays using exactly the holes T_0. With injections acting by relabelling and extension by zero, C_e is the induced module $M(U)$, generated in degree e. Let $B_{e-1}(T)$ have a basis of pairs (N,i), N an $(e-1)$-matching $P\leftrightarrow T$ and i a pigeon free in N; it is induced, generated in degree $e-1$, since (N,i) uses exactly the holes of N. The row-sum map $(Rf)(N,i)=\sum_j f(N\cup\{(i,j)\})$ is a map of FI-modules $C_e\to B_{e-1}$: after extension by zero, the new terms vanish, and coordinates (N,i) with N outside the old hole set get the value 0 on both sides. Its kernel is $H_e(P,\cdot)$. By Theorem 4.3 with $a=e-1$, $b=e$, the kernel is generated in degree $\le 2e$: every $f\in H_e(P,S)$ is a sum of zero extensions of balanced arrays on hole sets of size $\le 2e$. Extension by zero preserves both marginal conditions.

Apply the same argument with pigeons and holes exchanged to each summand, keeping its hole set fixed. This splits each summand further into balanced arrays on at most $2e$ pigeons, without enlarging its hole set. □

4.4 Extending one balanced layer

Lemma 4.7 (square realisation)**.** *Let $|A|=|B|=2e$ and $f\in H_e(A,B)$. There is a function μ on the matchings of the $A\times B$ board (of all sizes) that satisfies every row and column relation, vanishes on matchings of size $<e$, and equals f on matchings of size e.*

Proof. The functional on F_e that is 0 below level e and f at level e vanishes on W_e: relations at bases of size $<e-1$ involve only zeros, and those at bases of size $e-1$ are the vanishing marginals of f. Lemma 4.5 and the remark after it give μ. □

Lemma 4.8 (tensor extension)**.** *Let $1\le e\le D-1$. Split the board into pigeon sets $A\sqcup A'$ and hole sets $B\sqcup B'$ with $|A|=|B|=2e$. Let μ be as in Lemma 4.7 on $A\times B$ and let ν be a unit degree-$(D-e)$ design on $A'\times B'$. For a matching M with $|M|\le D$ put*

$$G(M)=\begin{cases}0 & \text{if } M \text{ has an edge in } A\times B' \text{ or } A'\times B,\\ 0 & \text{if } |M\cap(A\times B)|<e,\\ \mu(M\cap(A\times B))\,\nu(M\cap(A'\times B')) & \text{otherwise.}\end{cases}$$

Then G, as a functional on F_D, vanishes on W_D and on all matchings of size $<e$, and $G(M)=f(M)$ for matchings $M\subseteq A\times B$ of size e (zero on other matchings of size e).

Proof. G is defined: in the third case the residual part has size at most $D-e$. At level e the residual part of a non-zero term is empty and $\nu(\emptyset)=1$.

Fix a base M with $|M|\le D-1$ and a free pigeon i; columns are symmetric. Write $a=|M\cap(A\times B)|$ and $b=|M\cap(A'\times B')|$. We check that the free row sum of G at (M,i) equals $G(M)$.

- M has a crossing edge: every extension has one too; both sides are 0.
- $i\in A$, $a\le e-2$: extensions into B have core size $\le e-1$ and extensions into B' cross; both sides are 0.
- $i\in A$, $a=e-1$: $G(M)=0$. Only extensions into B can be non-zero, and their sum is $\nu(M\cap(A'\times B'))\cdot\sum_{j\in B\text{ free}}\mu(M_{\text{core}}\cup\{(i,j)\})=\nu(\cdot)\,\mu(M_{\text{core}})=0$, by the row relation of μ and $\mu=0$ below e. Here $b\le D-1-a=D-e$, so ν is evaluated within its degree. This is the only place where the cutoff at level e matters.

- $i \in A$, $a \geq e$: the same row relation of μ gives the sum $G(M)$.
- $i \in A'$, $a < e$: every extension into B' still has core size $< e$, and extensions into B cross; both sides are 0.
- $i \in A'$, $a \geq e$: then $b \leq D-1-e$, so the row relation of ν at the base $M \cap (A' \times B')$ is one of its defining relations; extensions into B cross; the sum is $\mu(M_{\text{core}})\,\nu(M \cap (A' \times B')) = G(M)$.

So G satisfies every relation at bases of size $\leq D-1$, that is, $G(W_D) = 0$. □

Lemma 4.9 (padding a design)**.** *A unit degree-D design on the board $(n_0+k) \times n_0$ gives one on $(r+k) \times r$ for every $r \geq n_0$, and, by restriction, unit designs of every smaller degree.*

Proof. Add $r - n_0$ pigeons and $r - n_0$ holes and fix a bijection σ between them. Define the new design to be the old one times $[M_{\text{new}} \subseteq \sigma]$ on matchings without edges between old and new parts, and 0 on others. The proof of Lemma 4.8 applies with the old board as residual and σ as core; there is no cutoff, and the value at $\emptyset$ stays 1. □

4.5 Proof of Theorem 4.1

Let $n \geq n_0 + 2D - 2$, and let U be the unit degree-D design on $(n+k) \times n$ given by Lemma 4.9. Then U_D holds.

For ND_D we show that every functional λ on F_{D-1} vanishing on W_{D-1} extends to a functional on F_D vanishing on W_D (Section 2.7). Start with $\Lambda_0 = \lambda(1)\,U$; the remainder $\lambda - \Lambda_0|_{F_{D-1}}$ vanishes on W_{D-1} and at $\emptyset$. Suppose that for some $1 \leq e \leq D-1$ the remainder r_e, a functional on F_{D-1} vanishing on W_{D-1}, vanishes on all matchings of size $< e$. Its level-e layer f_e lies in H_e by the last paragraph of Section 2.7 with $t = D-1$. By Lemma 4.6, $f_e = \sum_\alpha f_\alpha$ with each f_α balanced on at most $2e$ pigeons and $2e$ holes. Since $n \geq 2D-1 > 2e$, enlarge each support to exactly $2e$ pigeons and $2e$ holes; the array stays balanced. The complement has $n - 2e \geq n_0$ holes and surplus k, so by Lemma 4.9 it has a unit degree-$(D-e)$ design. Lemmas 4.7 and 4.8 give G_α vanishing on W_D and below level e, with level-e layer f_α. Put $\Lambda_e = \Lambda_{e-1} + \sum_\alpha G_\alpha$. The new remainder $r_{e+1} = r_e - \sum_\alpha G_\alpha|_{F_{D-1}}$ vanishes below level $e+1$ and still vanishes on W_{D-1}. After the step $e = D-1$ the remainder is zero, so Λ_{D-1} extends λ and vanishes on W_D. (For $D = 1$ there are no correction steps.)

The PC conclusion is Proposition 2.19. □

4.6 Proof of Corollary 4.2

Take $D = 2^\ell - 1$, $n_0 = N_\ell$ and $k = p^\ell$. Beame and Riis's construction gives a unit degree-D design over $\mathbb{F}_p$ on $(n_0+k) \times n_0$; its defining conditions are linear with coefficients in $\mathbb{F}_p$, so it is also a design over every field F of characteristic p. For $\ell \geq 1$, $n_0 = N_\ell \geq N_1 = 1$. Theorem 4.1 excludes refutations of degree at most $2^\ell - 1$ whenever $n \geq N_\ell + 2(2^\ell - 1) - 2 = N_\ell + 2^{\ell+1} - 4$. □

Corollary 4.10 (exact degree at $p = 2$, without E's inputs)**.** *Let $\ell \geq 1$. Over every field of characteristic 2, the PC degree of* onto-FPHP$_n^{n+2^\ell}$ *is exactly 2^ℓ for $n \geq (4^\ell - 2^\ell)/2 + 2^{\ell+1} - 4$.*

Proof. At $p = 2$ the bound 2^ℓ of Corollary 4.2 equals the upper bound p^ℓ of Proposition 2.10. □

So at $p = 2$ the exact degree of Corollary 5.2 has another proof, at a larger threshold ($n \geq 40$ instead of 15 for $\ell = 3$), using neither of E's representation-theoretic inputs; both proofs use Riis's upper bound. This corollary was deduced while drafting the paper and checked in the model review

of the paper, not in the run that produced D. For odd p the two bounds differ, because Beame and Riis's designs stop at degree $2^\ell - 1$ while the Nullstellensatz degree may be as large as $p^\ell - 1$; designs of degree $p^\ell - 1$, if they exist, would give the exact degree for every p through this section alone (Section 10).

For $\ell = 1$ Corollary 4.2 gives degree at least 2 for $k = p$ and every $n \geq 1$, consistent with Theorem A. For $p = 2$, $\ell = 2$ it gives degree at least 4 for surplus 4 from $n \geq 10$; the Lean result gives exactly 4 from $n \geq 5$.

The Nullstellensatz route, for comparison. Theorem 8.1 of [2] states its bound in multiplier degree. It also yields the design used above: a certificate of product degree at most D, reduced modulo the Boolean axioms, has every non-zero multiplier of degree at most $D - 1$, since every axiom has positive degree; with $D = 2^\ell - 1$ this contradicts Theorem 8.1, and duality gives the unit design. The two routes give the same threshold; we cite the construction because it needs no convention conversion.

4.7 What was checked

The proof of Theorem 4.1 was written by a Codex agent (x1) in run 20261104-0759 (`03-FI-HOLE-AVOIDANCE.md`, `04-NS-PC-PADDING.md`). Two Claude agents reviewed it in the run (c1 re-did every case of Lemma 4.8 by hand); a second Codex agent adjudicated it against the primary sources of Church–Ellenberg and Rhoades; a separate model referee with no contact with the run re-read both sources and found the transfer consistent with its own small computations over $\mathbb{F}_2$ and $\mathbb{F}_3$. The adjudicator Astra checked the Beame–Riis construction in the author-hosted PDF (Definitions 7.1, 7.2, 8.2, Lemma 8.4, proof of Theorem 8.1) and the conversion above. The tensor construction of Lemma 4.8 was also checked by exact computation at $D = 3$ over $\mathbb{F}_2$, with supports of size 2 and 4, on the full 12×8 board (all 60,884 free-line equations per case). None of this is a Lean proof or a human review, and none of it re-proves Corollary B of [5], Theorem 3.12 of [18] or Lemma 8.4 of [2].

5 Periodicity and the exact degree

This section proves result E: whether onto-FPHP$_n^m$ has a PC refutation of degree at most D depends, for m large compared with D, only on m modulo the least power of p above D. Since the square board $m = n$ is satisfiable, this fixes the PC degree exactly.

Theorem 5.1 (periodicity)**.** *Let F be a field of characteristic p, $n \geq 1$, $D \geq 1$, and q a power of p with $q > D$. Let $m \equiv m' \pmod q$.*

(a) If $m, m' > 2D$, then onto-FPHP$_n^m$ *has a* PC$_F$ *refutation of degree at most D if and only if* onto-FPHP$_n^{m'}$ *has one.*

(b) If Theorem 8.4.5 of [11] holds (see below), the same holds for $m, m' \geq 2D$.

Here m is the number of pigeons and n the number of holes; n is arbitrary, and $m < n$ is allowed.

Corollary 5.2 (E, exact degree)**.** *Let p be prime, $k \geq 1$, $n \geq 1$ and $q = p^{v_p(k)}$. Over every field of characteristic p, the PC degree of* onto-FPHP$_n^{n+k}$ *is exactly q for every $n \geq 2q-1$. If Theorem 8.4.5 of [11] holds, the same holds for $n \geq 2q - 2$.*

Proof. If $q = 1$ the degree is 1 (Section 1.1). Let $q \geq p$ and $D = q - 1$. The board with n pigeons and n holes has a solution, a permutation matrix, so it has no refutation of any degree. Since $n \equiv n + k \pmod{q}$ and $n > 2D$ when $n \geq 2q - 1$, Theorem 5.1(a) gives no refutation of degree at most $q - 1$ for $n + k$ pigeons. Proposition 2.10 gives one of degree q. With (b), $n \geq 2D = 2q - 2$ suffices. □

In particular, for surplus p^ℓ the PC degree is exactly p^ℓ once $n \geq 2p^\ell - 1$: Riis's upper bound is attained, and the degree grows with ℓ as fast as it can. This is the strongest statement in this paper. It is also the one that rests on the deepest inputs, so we state its status with care.

Tier and inputs. T-WRITTEN. Part (a) is proved in Appendix A (outlined in Section 5.3), using Harman's equivalences [10] with the repairs of Section 7, and standard facts on tilting modules (tensor products and Levi restrictions of tilting modules are tilting; bounded tilting complexes compute the derived category). The input we consider most exposed there is the compatibility of Harman's equivalence with induction and restriction of arbitrary submodules (Lemma A.4), whose tilting-theoretic foundations we cite and do not re-prove.

Section 5.2 gives a shorter argument through the semisimplification functors of Hone, Klein, Pauwels, Sherman, Yacobi and Zhang [11], which covers (b) as well. It uses their Theorem 8.4.5. A model reconstruction of the proof of that theorem (Astra, November 2026) found that the published proof does not establish it: their Lemma 8.4.4 rests on Lemma 8.2.1, which is false as stated (Section 7). This is not a counterexample to Theorem 8.4.5, and the reconstruction found every other step of Section 5.2, and their Lemma 7.3.2, to hold. Until that step is repaired, (b) and the threshold $n \geq 2q - 2$ are conditional, and (a) rests on Appendix A.

Comparison. For $q = p$, Theorem A gives the exact degree p at the Lean tier from $n \geq 2p - 3$. For $q = p^2$, result F gives the exact degree p^2 from $n \geq 2p^2 - 3$ for every prime, on different inputs (Section 6). Result D gives 2^ℓ, which is exact when $p = 2$, on the inputs and thresholds of Section 4. Below the threshold the degree can be smaller: over $\mathbb{F}_2$ with $k = 4$, exact computation of the closure below gives degree 3 at $n = 3$ and $n = 4$, while Lean gives exactly 4 from $n = 5$ (`[1156] StdPCExact.std_f2_k4_exact`), below the threshold $2q - 1 = 7$ of Corollary 5.2.

5.1 The closure

We use the notation of Section 2.7 on the board with m pigeons and n holes, and write $\mathcal{B}_m$, $F_d(m)$. Let $C_0(m) \subseteq F_D(m)$ be the span of the reduced pigeon and hole axioms, and recursively

$$S_t = C_t \cap F_{D-1}, \qquad C_{t+1} = C_t + \sum_{i,j} \rho(x_{ij})\, S_t.$$

The chain is increasing in the finite-dimensional space $F_D(m)$, so it stabilises; let $C_\infty(m)$ be its limit. All C_t are invariant under permutations of the pigeons.

Lemma 5.3. onto-FPHP$_n^m$ *has a* PC$_F$ *refutation of degree at most* D *if and only if* $1 \in C_\infty(m)$.

Proof. ($\Rightarrow$) As in Proposition 2.19, every line f of a degree-D derivation has $\rho(f) \in C_\infty$: axioms reduce into C_0 or to 0, linear combinations stay in C_∞, and if xf is a legal multiplication then $\deg f \leq D - 1$, so $\rho(f) \in C_\infty \cap F_{D-1} = S_t$ for large t and $\rho(xf) \in C_{t+1}$.

($\Leftarrow$) We show by induction on t that every element of C_t, written as a multilinear matching polynomial, is a line of a degree-D derivation. For $t = 0$ this is clear. Let $s \in S_t$; by induction s is derivable, and $\deg s \leq D - 1$ as a polynomial, so $x_{ij}s$ is a legal multiplication. Its reduction

$\rho(x_{ij}s)$ differs from $x_{ij}s$ by multiples of Boolean, collision and functional axioms: reduce one repeated variable or forbidden pair at a time; each multiple has degree at most $\deg(x_{ij}s) \le D$, as in Lemma 2.6. Linear combinations give C_{t+1}. □

On the square board ($m = n$), every element of C_∞ vanishes at every permutation matrix (true for C_0 and preserved by the recursion), so $1 \notin C_\infty(n)$.

The recursion is a finite sequence of linear-algebra operations on matrices with entries in $\mathbb{F}_p$. Extending scalars to any field containing $\mathbb{F}_p$ commutes with sums, images and intersections, so $1 \in C_\infty$ over F if and only if $1 \in C_\infty$ over $\mathbb{F}_p$ if and only if over $\overline{\mathbb{F}}_p$. We may therefore work over $k = \overline{\mathbb{F}}_p$.

The closure as representation theory. The symmetric group S_m acts on pigeons, and all spaces above are kS_m-modules. For $e \ge 0$ and a fixed set of e holes listed in increasing order, the e-matchings onto those holes form the permutation module $E_e(m)$ of ordered e-tuples of distinct pigeons; so $F_d(m)$ is a direct sum of $\binom{n}{e}$ copies of $E_e(m)$, $e \le d$. For a hole j let $U_j = E_1(m)$, the natural permutation module on pigeons, and let $\mu_j \colon U_j \otimes F_{D-1}(m) \to F_D(m)$ send $u_i \otimes x_M$ to $\rho(x_{ij}x_M)$. Then

$$C_{t+1} = C_t + \textstyle\sum_j \mu_j(U_j \otimes S_t),$$

and the constant 1 is the image of the map $k \to F_D(m)$ onto the empty matching. So $C_\infty(m)$ is built from permutation modules by tensoring with U_j, images, intersections and sums, and the question is whether the composite $k \to F_D(m) \to F_D(m)/C_\infty(m)$ is zero.

5.2 A conditional proof through semisimplification

This argument is shorter than the one in Appendix A and covers part (b), but it depends on [11, Thm. 8.4.5], whose published proof is incomplete (Section 7). We give it because every other step has been checked, so a repair of that one step would make (b) unconditional.

It suffices to prove one step: $M = L + q$ pigeons against L pigeons, with $L \ge 2D$ (for (b)) or $L > 2D$ (for (a)). Applying the step repeatedly in either direction gives the theorem.

The functor. Let $H \subseteq S_q$ be an elementary abelian p-group of order q acting regularly on the last q pigeons. Let Φ be restriction from S_M to $S_L \times H$ followed by semisimplification in the H-variable: an H-module decomposes into indecomposables, and semisimplification discards the summands of dimension divisible by p (the *negligible* ones). This is the OTI functor of [11] for the semisimplification functor φ_H, which satisfies the hypotheses of their Theorem 8.4.5 (stated there). Φ is strong symmetric monoidal [11, §2.4]. It is *not* exact on all kS_M-modules.

External input. Call $\lambda \vdash M$ *q-stable* if $\sum_{i \ge 2} \lambda_i < q$ and, when $M < 3q$, $\lambda_1 \ge (M + q)/2$ [11, Def. 6.1.1]. Let $\mathrm{Pol}^{\mathrm{st}}_{M,q}$ be the Serre subcategory of polynomial functors of degree M whose simples are the L_{λ^t} with λ q-stable, and $\mathrm{Pol}^{\mathrm{tr}}_{L,q}$ its truncated counterpart in degree L, with simples $L_{(\lambda - qe_1)^t}$.

Theorem 5.4 ([11, Thm. 8.4.5])**.** *For φ satisfying the two hypotheses there (semisimplification does), the OTI functor corestricts to an equivalence of highest-weight categories* $\mathrm{Pol}^{\mathrm{st}}_{M,q} \to \mathrm{Pol}^{\mathrm{tr}}_{L,q}$.

By [11, Lemma 7.3.2], this functor commutes with Schur–Weyl duality: for the Schur functor $\mathcal{F}$, $(\mathcal{F} \boxtimes 1)\Phi \simeq \Phi\mathcal{F}$. We use the twisted Schur functor $\tilde{\mathcal{F}} = \mathcal{F}(-) \otimes \mathrm{sgn}$; twisting costs nothing on the H-side, since H is a p-group and its sign character is trivial in characteristic p.

Step 1: exactness on the bounded-tail range. Let $\mathcal{A}_D(M)$ be the Serre subcategory of polynomial functors whose simple labels are λ^t with $\lambda = (M-e, \tau)$, $|\tau| = e \le D$, and $\mathcal{A}_D(L)$ the same with L. These λ are q-stable: $e \le D < q$, and $M - e \ge (M+q)/2$ is equivalent to $L \ge 2e$. Theorem 5.4 restricts to an exact equivalence $\Theta\colon \mathcal{A}_D(M) \to \mathcal{A}_D(L)$ preserving e and τ. The kernel $K_D(M)$ of $\tilde{\mathcal{F}}$ on $\mathcal{A}_D(M)$ is generated by the simples whose λ is p-singular (some part repeated p times). Since $M - e = L + q - e > e \ge \tau_1$, a repetition in λ lies inside τ and remains one in $(L-e, \tau)$, where $L - e \ge e \ge \tau_1$. Hence $\Theta(K_D(M)) \subseteq K_D(L)$. (For $L > 2D$ this is an equality; at $L = 2D$ it can be strict.) The exact functor $\tilde{\mathcal{F}}\Theta$ therefore factors through the Serre quotient $\mathcal{A}_D(M)/K_D(M)$.

This quotient is the full image $\mathcal{R}_D(M)$ of $\tilde{\mathcal{F}}$ on $\mathcal{A}_D(M)$, a full subcategory of kS_M-modules whose kernels, cokernels and subobjects are the ordinary ones: for $X \in \mathcal{A}_D(M)$ and a submodule $T \le \tilde{\mathcal{F}}X$, the pullback $X_T = X \times_{G\tilde{\mathcal{F}}X} GT$ along the unit of the right adjoint G of $\tilde{\mathcal{F}}$ is a subobject of X with $\tilde{\mathcal{F}}X_T = T$; lifting the graph of a module map in the same way identifies morphisms. (Compare [11, Lemma 8.5.1].) By Lemma 7.3.2 the induced functor $\mathcal{R}_D(M) \to \mathcal{R}_D(L)$ is Φ. So Φ is exact on $\mathcal{R}_D(M)$. It need not be faithful there, and we never use faithfulness.

Step 2: every object used lies in the range. The module $E_e(M)$ is $\tilde{\mathcal{F}}$ of $\Lambda^{M-e}V \otimes V^{\otimes e}$, whose highest weight is $(M-e, 1^e)^t$; every composition factor has a label ν dominated by it, so ν^t dominates $(M-e, 1^e)$, has first part $\ge M - e$ and tail of size $\le e$. Hence $E_e(M)$, $e \le D$, and so $F_d(M)$, $d \le D$, lie in $\mathcal{R}_D(M)$. On a fixed hole set, $U_j \otimes E_e \cong E_{e+1} \oplus E_e^{\oplus e}$ (the extra pigeon is new, or equals the pigeon at one of the e holes), so $U_j \otimes F_{D-1}$ lies in the range too, and with it $U_j \otimes S$ for every submodule $S \le F_{D-1}$. All intersections, images and sums in the recursion therefore take place in $\mathcal{R}_D(M)$.

Step 3: the maps are the obvious ones. For a permutation module $k[X]$, restricted to H, the H-orbits of size > 1 give summands whose every H-endomorphism has constant diagonal on the orbit and hence trace 0; they are negligible and are killed. So $\Phi(k[X]) = k[X^H]$, and an equivariant map between permutation modules goes to its block from fixed basis vectors to fixed basis vectors. A matching fixed by H uses none of the last q pigeons (an edge at a moved pigeon would move), so $\Phi(F_d(M)) = F_d(L)$ and $\Phi(U_j(M)) = U_j(L)$, compatibly with inclusions and with the monoidal structure. On fixed basis vectors: a pigeon axiom of a remaining pigeon is unchanged; a hole axiom $\sum_i x_{ij} - 1$ loses exactly the terms at moved pigeons, which form a non-trivial orbit, and becomes the hole axiom of the smaller board; and μ_j sends $u_i \otimes x_N$ to the same matching, the same new matching, or 0, on both boards.

Step 4: the closure transfers. By induction on t, Φ carries $C_t(M) \le F_D(M)$ to $C_t(L) \le F_D(L)$: exactness transfers the intersection with F_{D-1}, the images and the sums; monoidality and Step 3 transfer $\mu_j(U_j \otimes S_t)$. Both chains stabilise after finitely many steps.

Step 5: detecting the constant. Let $c\colon k \to F_D(M)/C_\infty(M)$ be the image of the constant. If $c \ne 0$ it is injective, since k is simple; Φ is exact, so $\Phi(c)$ is injective, and $\Phi(k) = k \ne 0$, so $\Phi(c) \ne 0$. If $c = 0$ then $\Phi(c) = 0$. By Step 4, $\Phi(c)$ is the corresponding map for L pigeons. So $1 \in C_\infty(M)$ if and only if $1 \in C_\infty(L)$, and Lemma 5.3 finishes the step. □

5.3 The proof of part (a), through Harman's equivalences and tilting modules

For $m, m' > 2D$, Theorem 5.1(a) is proved without [11]; the proof is given in full in Appendix A. It transports the closure along Harman's equivalence between the bounded-tail image categories

of S_m and $S_{m'}$ in weight $\leq D$ [10, Thm. 2.9]. The multiplication step is written as induction and restriction of submodules, so the argument needs the equivalence to commute with induction and restriction of arbitrary submodules, not only of permutation modules. This is proved on the Schur-algebra side, where induction and restriction become tensoring with the natural module and a weight space of Levi restriction, both exact and preserving tilting modules; the compatibility holds on tilting modules because the twisted Schur functor is fully faithful there, and extends to all modules through $K^b(\mathrm{Tilt}) \simeq D^b$. The full faithfulness is proved directly in every characteristic, because the identification Harman's proof uses for it is false at $p = 2$; the period must be a power of p strictly greater than D; and the strict inequality $m, m' > 2D$ is needed (Section 7).

5.4 Checks

The statements were tested by exact computation of C_∞ (full row reduction over $\mathbb{F}_p$, no use of symmetry). For $D = 2$ the predicate "$1 \in C_\infty$" is periodic in m with period 4 for $p = 2$ ($m = 5, \ldots, 12$), 3 for $p = 3$ ($m = 5, \ldots, 13$) and 5 for $p = 5$ ($m = 5, \ldots, 14$), for $n = 3, 4$, and also for $n > m$. Period 2 fails for $p = 2$, $D = 2$ ($n = 3$: refutable at $m = 5$, not at $m = 7$), so the condition $q > D$ cannot be weakened to $q \geq D$. At the boundary $m = 2D$ over $\mathbb{F}_2$ ($D = 1, 2, 3$, $n \leq 7$) every congruent triple agreed. The mechanism itself was tested: for $q = p$ and H generated by a p-cycle, the Jordan type of the generator on $C_\infty(M)$ and on $F_D(M)/C_\infty(M)$ has no blocks of size strictly between 1 and p, and the number of blocks of size 1 equals the dimension of the corresponding space for $L = M - p$ pigeons (for example $(p, M, n, D) = (3, 9, 3, 2)$: $\dim C_\infty(9) = 199$, with 88 blocks of size 1, and $\dim C_\infty(6) = 88$). These are finite checks, not evidence for the general statement beyond the absence of a counterexample at these sizes.

Who did what. All in run 20261104-0759; all reviews are model reviews. The closure formulation (Lemma 5.3), the computations of Section 5.4 and the proof of Appendix A are by a Claude agent (c1). A Claude referee with no contact with the run reviewed that proof and found the characteristic-2 error in [10]; a Codex agent (x2) reviewed it and supplied the two-way scalar extension. The proof of Section 5.2 is by x2, who also found the counterexample to [11, Thm. 8.5.3] at the boundary and the exact-quotient argument that avoids it; c1 reviewed it, and a second Codex agent (x1) adjudicated it, including the boundary range. After the run, a model referee re-read [10] and [11] and re-ran the closure computations with its own code, and Astra adjudicated both proofs. Astra then reconstructed the proof of [11, Thm. 8.4.5] and Lemma 7.3.2 from the source: Lemma 7.3.2 holds, the proof of Theorem 8.4.5 has the gap described in Section 7, and Steps 1–5 hold conditional on that theorem. A second model re-checked the counterexample to Lemma 8.2.1 by hand.

6 Fixed-degree bands and the $\mathbb{F}_3$ target

This section gives a third mechanism, which works degree by degree. It reduces ND_d to a statement about the homogeneous ideals generated by row sums and by column sums, and proves that statement for every $d < p^2$. It cannot give growth in ℓ at fixed p, and it is not a third proof of it; what it gives is exactness at $v_p(k) \leq 2$ from a slightly lower threshold, and the $\mathbb{F}_3$ target, on inputs that differ from those of Sections 4 and 5.

Theorem 6.1 (F)**.** *Let p be prime, $k \geq 1$, $q = p^{v_p(k)}$, and $1 \leq D < \min(q, p^2)$. Over every field of characteristic p,* onto-FPHP$_n^{n+k}$ *has no PC refutation of degree at most D for any $n \geq 2D - 1$. Consequently, if $v_p(k) = 2$, its PC degree is exactly p^2 for every $n \geq 2p^2 - 3$.*

Corollary 6.2 (the $\mathbb{F}_3$ target)**.** *If* $9 \mid k$ *and* $n \geq 7$*,* onto-FPHP$_n^{n+k}$ *has no PC refutation of degree at most* 4 *over* $\mathbb{F}_3$*.*

The corollary is the case $p = 3$, $D = 4$. It has two proofs: through the identity (I) below in degrees $d \leq 4$, which for $p = 3$ needs only the first two bands and the connectivity theorem; and through exact computations on three square boards (Section 6.4).

Tier and inputs. T-WRITTEN. The ideal-identity proof uses the connectivity theorem of Björner, Lovász, Vrećica and Živaljević for chessboard complexes, as stated by Shareshian and Wachs [22, Thm. 1.1]. Degrees $d < 2p$ need nothing else. Degrees $2p \leq d < p^2$ (odd p) also use Yalçın's theorem on equivariant Moore spaces and the Dade group [24, Def. 1.1, Props. 3.8, 5.7], Proposition 3.3 of Miller [16], and uniqueness of Dade caps. The input we consider most exposed is Miller's Proposition 3.3 at its interface with the truncated chessboard cochain complex, where the output needed is an equality of integer dimensions of Brauer quotients (Section 6.3). The second proof of Corollary 6.2 replaces connectivity by Church–Ellenberg's Corollary B and exact computation.

6.1 From ND to a kernel statement

We use the notation of Section 2.7 on a board with m pigeons and n holes. Let V_d be the span of matchings of size exactly d. The *labels* at level d are the pairs $(N, \text{row } i)$ and $(N, \text{col } j)$ with $|N| = d - 1$ and i, j free in N. Let a_d send a label to the top part of its relation ($\sum_j x_{N \cup \{(i,j)\}}$, resp. $\sum_i x_{N \cup \{(i,j)\}}$), a map from the label space to V_d, and let β send a label to its base x_N.

Lemma 6.3. $W_d \cap F_{d-1} = W_{d-1} + \beta(\ker a_d)$*. Hence* ND$_d$ *holds if and only if* $\beta(\ker a_d) \subseteq W_{d-1}$*.*

Proof. An element of W_d is $w + \sum_g c_g \,\mathrm{rel}(g)$ with $w \in W_{d-1}$ and g over the level-d labels. Its degree-d part is $a_d(c)$, so it lies in F_{d-1} iff $c \in \ker a_d$, and then it equals $w - \beta(c)$. □

Fix total orders on pigeons and holes. The following elements of the label space lie in $\ker a_d$:

- $rr(N; a, b) = \sum_j (N \cup \{(b, j)\}, \text{row } a) - \sum_j (N \cup \{(a, j)\}, \text{row } b)$ for $|N| = d-2$ and free pigeons $a < b$; $cc(N; a, b)$ likewise for holes;
- $rc(N; i, j) = \sum_{i' \neq i} (N \cup \{(i', j)\}, \text{row } i) - \sum_{j' \neq j} (N \cup \{(i, j')\}, \text{col } j)$;
- the *anchored* elements $Z_N^{(t)}$, $1 \leq t \leq d$, $|N| = d - t$:

$$Z_N^{(t)} = \sum_E \Big(\sum_{\substack{i \text{ free} \\ i > \max \mathrm{pig}(E)}} (N \cup E, \text{row } i) - \sum_{\substack{j \text{ free} \\ j > \max \mathrm{hol}(E)}} (N \cup E, \text{col } j) \Big),$$

  the outer sum over $(t-1)$-matchings E on the pigeons and holes free in N ($\max \emptyset = -\infty$). Both halves map under a_d to the sum of all d-matchings containing N, each once.

Lemma 6.4. $\beta(rr)$*,* $\beta(cc)$*,* $\beta(rc)$ *lie in* W_{d-1}*, and with* $s = |N|$*,*

$$\beta(Z_N^{(t)}) \equiv \Big(\tbinom{m-s}{t} - \tbinom{n-s}{t} \Big) x_N \pmod{W_{d-1}}.$$

Proof. $\beta(rr(N;a,b))$ is the difference of the row relations at (N,b) and (N,a), and similarly for cc; $\beta(rc(N;i,j))$ is the column relation at (N,j) minus the row relation at (N,i). For Z, group the E by their pigeon set A, $|A| = t-1$; the row coefficient is the number of free pigeons above $\max A$. For fixed A, $\sum_{\mathrm{pig}(E)=A} x_{N\cup E} \equiv x_N$ modulo W_{d-1}, by applying one row relation at a time at bases of size $\le d-2$. Since t-subsets of the free pigeons correspond to pairs (A, a new largest element), the row half contributes $\binom{m-s}{t}x_N$; the column half contributes $\binom{n-s}{t}x_N$. □

If $m-n=k$ and $t<q=p^{v_p(k)}$, then $\binom{m-s}{t} \equiv \binom{n-s}{t} \pmod p$, by $(1+z)^k = (1+z^q)^{k/q} \equiv 1+O(z^q)$ over $\mathbb{F}_p$. So:

Proposition 6.5. *Let $D<q$ and suppose that for every $d \le D$*

$$(A)(m,n,d)\text{:}\quad \ker a_d \text{ is spanned by the elements } rr,\ cc,\ rc \text{ and } Z_N^{(t)}.$$

Then $\mathrm{ND}_1,\dots,\mathrm{ND}_D$ hold on the board, and since $W_0=0$ also U_D; by Proposition 2.19 there is no PC refutation of degree at most D.

(A) is homogeneous and does not see k; the surplus enters only through the binomial differences. (A) fails off the stable range, for example at $(p,m,n,d)=(3,7,4,3)$, where $\ker a_3$ has dimension 944 and the listed elements span 916 (computed twice, independently).

6.2 The ideal identity

Let $\mathcal{A}$ be the homogeneous matching ring: basis x_M over matchings, $x_M x_{M'} = x_{M\cup M'}$ if M, M' are edge-disjoint and $M\cup M'$ is a matching, and 0 otherwise. (So $x_{ij}^2=0$; this is the top-degree ring, not the Boolean algebra $\mathcal{B}$.) Let $R_i=\sum_j x_{ij}$, $C_j=\sum_i x_{ij}$, $I=(R_i)$, $J=(C_j)$, and σ_t the sum of all t-matchings. The identity in degree d is

$$\mathrm{(I)}(m,n,d)\text{:}\qquad (I\cap J)_d = (IJ)_d + \sigma_1\mathcal{A}_{d-1} + \sigma_p\mathcal{A}_{d-p},$$

with the last term absent for $d<p$.

Lemma 6.6. *If $\min(m,n)\ge 2d-1$, then (I)(m,n,d) implies (A)(m,n,d).*

Proof. To $s\in\ker a_d$ with row part α and column part γ associate $y(s)=\sum\alpha_i R_i = -\sum\gamma_j C_j \in (I\cap J)_d$ (the products are taken in $\mathcal{A}$, with the base of each label as coefficient). Then $y(rr)=y(cc)=0$, $y(rc(N;i,j))=x_N R_i C_j$ (because $R_i x_{ij}=0$), and $y(Z_N^{(t)})=x_N\sigma_t$. So every element on the right-hand side of (I) is y of a combination of rc's and Z's: $(IJ)_d$ is spanned by the $x_N R_i C_j$, which vanish unless i and j are free in N, and $\sigma_1\mathcal{A}_{d-1}$, $\sigma_p\mathcal{A}_{d-p}$ by the $x_N\sigma_1$, $x_N\sigma_p$. (The run's first form of (I) listed all $x_N\sigma_t$; the form above, with σ_1 and σ_p only, is the one proved.) By (I), subtracting suitable rc's and Z's leaves s' with $y(s')=0$, that is $\sum\alpha'_i R_i=0$ and $\sum\gamma'_j C_j=0$ separately. For a fixed set of d pigeons the row-only relation is a cocycle of the augmented cochain complex of the chessboard complex $\Delta(d,n)$ in dimension $d-2$, and the rr's are its coboundaries. For $n\ge 2d-1$ the connectivity theorem [22, Thm. 1.1] gives $\tilde H_{d-2}=\tilde H_{d-3}=0$ over $\mathbb{Z}$, hence $\tilde H^{d-2}=0$ over every field. So α' is a sum of rr's, and likewise γ' of cc's (using $m\ge 2d-1$). □

Degrees below p. Let $G=S_d$ act on ordered d-tuples by positions. The coefficient spaces of $(I\cap J)_d$ identify with V^G, where $V=P_m\otimes P_n$ and P_m is the span of functions on injective d-tuples in $[m]$ that are independent of at least one position; this identification needs no division. On the spanning tensors of deletion indicators, the norm $N_G=\sum_{g\in G} g$ gives $x_N\sigma_1$ when the row and column indicators omit the same position and $x_N R_i C_j$ when they omit different positions. Hence

$$N_G V = (IJ)_d + \sigma_1\mathcal{A}_{d-1},\qquad (I\cap J)_d/\big((IJ)_d+\sigma_1\mathcal{A}_{d-1}\big) \cong \hat H^0(G,V).$$

For $d<p$, $|G|=d!$ is invertible, the Tate group vanishes, and (I) holds.

6.3 The bands $p \le d < p^2$

We give the argument in these bands in outline. Each step was written out in full in the run, reviewed there, and reconstructed in Astra's audit of the local-to-global argument; the outline below follows that audit. It is the part of the paper whose written form is furthest from what is printed here. For $p \le d < p^2$ write $d = ap + r$ with $0 \le r < p$; then $a < p$, $v_p(d!) = a$, and $Q = (C_p)^a$ acting on a disjoint blocks of p positions is a Sylow p-subgroup of S_d. Since $[S_d : Q]$ is prime to p, restriction to Q followed by transfer is multiplication by a unit on $\hat{H}^0$, so it suffices to show that $\hat{H}^0(Q, V)$ is generated by transfers of classes that become $(p-1)!\, x_N \sigma_p$ in $\mathcal{A}$.

1. *One cycle* ($a = 1$, any number r of fixed positions). Projecting the augmented chessboard cochains of $\Delta(d, m)$ to the fixed-position cochains of $\Delta(r, m)$ has a kernel of free modules; connectivity gives exactness below the top; dimension shifting along the cyclic period shows that the actual map from functions on fixed-position r-tuples is onto on $\hat{H}^0$. This covers $p \le d < 2p$ for every prime, including $p = 2$, $d = 2, 3$.

2. *Full-vertex summands.* The fixed complexes of subgroups of Q are again chessboard complexes, tight Moore complexes by connectivity; Yalçın's theorem identifies P_m as capped endo-permutation with an explicit Dade class. For the tensor product $V = P_m \otimes P_n$ the relative terms cancel, giving $[V] = 2[\Omega_Q]$; its cap is the minimal second syzygy $C = \ker(R^a \to R)$, $R = \mathbb{F}_p[x_1, \ldots, x_a]/(x_i^p)$. Dade cap uniqueness makes every full-vertex indecomposable summand of V a copy of C, and $\hat{H}^0(Q, C)$ is generated by transfers from the cyclic factors.

3. *Proper-vertex summands.* Miller's Proposition 3.3, applied to the truncated chessboard cochain complexes, gives for every non-trivial $L \le Q$ the integer identity $\dim \mathrm{End}(V)(L) = (h_s(m)h_s(n))^2$ for the Brauer quotients, where h_s is the top cohomology dimension of the L-fixed chessboard complex. With Higman's criterion this forces every proper vertex into a coordinate hyperplane H_i, so those summands contribute only transfers from the H_i.

4. *Induction on a.* Restricted to H_i, V is the same kind of module with $a-1$ moving blocks and $r + p$ fixed positions; the one-cycle step allows any number of fixed positions, so induction on a reduces everything to single p-cycles, where the classes are fixed-label indicators whose transfer to S_d is $(p-1)!\, x_N \sigma_p$.

At $d \ge p^2$ the subgroup Q is no longer Sylow and the argument stops; nothing here concerns that range.

Proof of Theorem 6.1. For $d < \min(q, p^2)$ and $m, n \ge 2d-1$, (I) holds (above), hence (A) (Lemma 6.6), hence ND_d (Lemmas 6.3, 6.4); Proposition 6.5 excludes refutations of degree $\le D$ once $n \ge 2D-1$. All spaces are defined over $\mathbb{F}_p$, so the statement holds over every field of characteristic p. If $v_p(k) = 2$, take $D = p^2 - 1$; the upper bound is Proposition 2.10. □

6.4 The $\mathbb{F}_3$ target by finite certificates

The second proof of Corollary 6.2 avoids the ideal identity. Two lemmas move (A) from squares to rectangles.

Lemma 6.7 (hole avoidance; from Corollary B of [5])**.** *If $n \ge 2e + 1$, every array in H_e is a sum of arrays in H_e each of which avoids some hole. Dually, if the restriction of $v \in V_e$ to every board with one hole removed lies in the image of a_e there, then v lies in the image of a_e.*

This is the case $|S| > 2e$ of Lemma 4.6. The threshold is sharp: exact computation shows that the statement fails at $n = 2e$ on several small boards.

Lemma 6.8 (pigeon induction)**.** *Order the pigeons so that the removed pigeon ω is the largest. Then (A)$(m-1, n, d)$, (A)$(m-1, n-1, d-1)$ and Lemma 6.7 on $(m-1, n)$ at level $d-1$ imply (A)(m, n, d). The mirror statement removes a hole.*

Proof. Split the labels of the board (m, n) into those avoiding ω (G_1), the row labels of ω (G_2), and those whose base uses ω (G_3), and decompose $V_d(m) = V_d(m-1) \oplus \bigoplus_h x_{\omega h} V_{d-1}(m-1, n \setminus h)$. Let $s = (g_1, g_2, g_3) \in \ker a_d$. (a) The first component of $a_d(s)$ is $a_d(g_1)$ on the board $m-1$, so g_1 is in that kernel; each listed element of the smaller board is the G_1-part of the element of the same name on the larger board (the extra terms involve ω only in G_2, G_3, and ω largest preserves the order conditions of Z). Subtract. (b) If $g_1 = 0$, the component at h reads $g_2|_{\text{avoid } h} + a_{d-1}(g_{3,h}) = 0$, so by Lemma 6.7 g_2 lies in the image of a_{d-1} on $(m-1, n)$; the elements $rr(N; a, \omega)$ and $rc(N; \omega, j)$ have G_1-part 0 and G_2-parts exactly the needed extension sums. Subtract. (c) If $g_1 = g_2 = 0$, each $g_{3,h}$ lies in $\ker a_{d-1}$ on $(m-1, n \setminus h)$, and $x_{\omega h}$ times a listed element at base N' is the listed element of the same name at base $N' \cup \{(\omega, h)\}$. □

Starting from (A) at level 1 (which holds on every board), Lemma 6.8 and its mirror give (A) on all boards with $m, n \geq 2d-1$ from the single square statement $S(d) = \text{(A)}(2d-1, 2d-1, d)$. For $p = 3$ and $D = 4$ the squares are 3×3, 5×5 and 7×7. $S(2)$, $S(3)$ and $S(4)$ were verified over $\mathbb{F}_2$ and $\mathbb{F}_3$ by exact rank computations, by three implementations written by different agents; at $S(4)$ the top matrix has rank 27,068 on 58,800 labels, and the 47,972 listed elements span its 31,732-dimensional kernel. With Proposition 6.5 this proves Corollary 6.2 at tier T-WRITTEN with T-MEASURED certificates.

Lean. The library contains a conditional Lean theorem for the same target (published as finding K100, DOI 10.5281/zenodo.24056819): if $9 \mid k$, $n \geq 7$, and the spanning statement (A)$(n+k, n, r)$ holds for $r = 2, 3, 4$, then there is no standard PC refutation over $\mathbb{F}_3$ of degree at most 4; (A) at level 1 is proved on every board with positive numbers of pigeons and holes.[185] The Lean theorem does not formalise Lemma 6.8 or Lemma 6.7, so it needs (A) on every board $(n+k) \times n$ rather than on three squares. On the smallest board with positive surplus divisible by 9, 16×7, (A) is verified by exact computation at levels 2 (kernel dimension 343) and 3 (kernel dimension 24,780, two independent programs); level 4 was not decided within the time limit.

Who did what. All in run 20261104-0759 unless stated; all reviews are model reviews. The reduction (Lemmas 6.3–6.6, Proposition 6.5, Lemma 6.8) and the form of (I) are by a Claude agent (c2), with a Codex agent (x1), and were reviewed by a second Claude agent (c1). The hole-avoidance lemma is x1's, checked by c2 and adjudicated by a second Codex agent (x2). The identity (I) was proved band by band: in degree 2 and whenever $d!$ is invertible by x1 (reviewed by c2); at $d = p$ by x2 (reviewed by x1 and c2); for $p < d < 2p$ by x1 (reviewed by c2, accepted by x2); in the higher bands by x1 and x2 together, with c2 reviewing each step. The finite certificates $S(2)$–$S(4)$ were computed by c1 and, independently, by x1, and regenerated after the run by Astra with a third implementation. A model referee with no contact with the run re-derived Lemmas 1–5 and (I) for $d < 2p$, re-read the sources of the higher bands, and checked (I) and the spanning statement on ten boards with its own code. Astra's global audit reconstructed the local-to-global argument

[185] Lean: `Programme.«20261104-1627-proof-complexity»: RemainingSpanning.no_stdpc_le; level 1: MatchingLabels.spanning_one; the semantic form ConcreteDegreeFour.no_ppc`.

of Section 6.3. The Lean theorem is from run 20261104-1627 and was re-proved by a second agent there.

7 Errata in two sources

Writing the proofs of Section 5 required checking two recent sources closely. We record five points. For each we say whether it is a counterexample or an omitted justification, and we do not draw conclusions beyond the stated point; in particular none of them suggests that the other results of either paper fail. The first four points were sent on 8 November 2026 to Harman and to Yacobi, one of the authors of [11]. The fifth was found afterwards, in the reconstruction of [11, Thm. 8.4.5] described in Section 5.

Harman, Corollary 1.8(4), at $p = 2$ (counterexample). The corollary identifies the value of a right adjoint on a Young module with a tilting module, $\tilde{G}(Y(\lambda)) = T(\lambda')$ [10, Thm. 1.7(4), Cor. 1.8(4)]. At $p = 2$ in polynomial degree 2, $Y(2)$ is the trivial module k; the right-adjoint value has dimension $N(N+1)/2$, while $T(1,1) = \Lambda^2 V$ has dimension $N(N-1)/2$ (at $N = 2$: 3 against 1). Harman's proof of his Theorem 2.8 uses this identification to conclude that the twisted Schur functor is fully faithful on tilting modules. That conclusion is true in every characteristic: the tilting modules in question are the summands of the $\Lambda^\lambda V$; the ordinary Schur functor sends $\Lambda^\lambda V$ to $M(\lambda) \otimes \mathrm{sgn}$ and the twisted one to $M(\lambda)$; the Hom spaces on both sides have the dimension of the number of (λ, μ) double cosets; and the functor is faithful on them because $\Lambda^\mu V \subseteq V^{\otimes d}$ and $V^{\otimes d}$ is projective and injective with socle detected by the Schur functor. The second proof of Section 5.3 uses this argument instead of Corollary 1.8(4).

Harman, Theorem 2.5, the modulus (counterexample at the printed period). The theorem's printed modulus allows, at $r = p = 3$, the period 3. But the stable incidence composite through 3-subsets has scalar $\binom{7}{3} \equiv 2$ at rank 7 and $\binom{10}{3} \equiv 0$ at rank 10 in $\mathbb{F}_3$; both ranks exceed $2r$, so no functor identifying these stable maps can respect this composite at period 3. A period that is a power of p strictly greater than r avoids the problem, and that is what we use. We do not know whether this has been noted before; we record it because our statement depends on the correct modulus.

Harman, Theorem 2.9 (omitted justification). The proof applies Lemma 2.10 to a restricted right adjoint without showing that it lands in the Serre subcategory where the lemma applies. The conclusion needed, that the Serre quotient by the kernel of the twisted Schur functor is its full image with ambient kernels and subobjects, follows from the pullback-and-graph argument of Section 5.2 (compare [11, Lemma 8.5.1]). This is not a counterexample to Theorem 2.9.

Hone et al., Theorem 8.5.3, at the boundary of Definition 6.1.1 (counterexample). Take $p = q = 2$ and $d = 4$; the label $(3,1)$ is 2-stable, since $\lambda_1 = 3 \geq (4+2)/2$, and it truncates to the 2-singular label $(1,1)$. Let W be the sum-zero subspace of $\mathbb{F}_2^4$ (a submodule of the permutation module of S_4) and C the line of the all-ones vector. In W/C with basis $a = e_1 + e_2$, $b = e_1 + e_3$, the transposition (34) acts by $a \mapsto a$, $b \mapsto b + a$. So W/C is free over $\mathbb{F}_2[\langle(34)\rangle]$ and is killed by semisimplification, although it is a non-zero object of the stable image category: it is a subquotient of $E_1(4)$, which belongs to that category, and the category is closed under ambient subquotients. The equivalence asserted in [11, Thm. 8.5.3] is therefore not faithful at that boundary. The example does not refute Theorem 8.4.5 of the same paper: the descended functor can be exact while killing

this object. This is why the proof of Section 5.2 detects the constant through a map out of the simple unit and never uses faithfulness. With strict inequality in Definition 6.1.1(2b) the boundary case does not arise.

Hone et al., Lemma 8.2.1 (counterexample), and the proof of Theorem 8.4.5. Lemma 8.2.1 of [11] states: for a highest-weight category $(\mathcal{C}, \Lambda)$, a maximal λ, and an additive functor φ to an abelian category that is exact on the Serre subcategory of $\Lambda \setminus \{\lambda\}$ and whose comparison maps $\varphi(\Delta_\lambda) \to \tilde{\varphi}(\Delta_\lambda)$ and $\tilde{\varphi}(\nabla_\lambda) \to \varphi(\nabla_\lambda)$ are isomorphisms ($\tilde{\varphi}$ the tilting derived functor), the functors $L\varphi$, $\tilde{\varphi}$, $R\varphi$ agree and $\tilde{\varphi}$ is a direct summand of φ. This is false. Let $\mathcal{C}$ be the representations of the Kronecker quiver (vertices $1, 2$, two arrows $1 \to 2$) over any field, ordered $1 < 2$. Then $\Delta_1 = \nabla_1 = S_1$, $\Delta_2 = S_2$ is projective, $\nabla_2 = I_2$ is the injective of dimension vector $(2, 1)$, and the tilting objects are the sums of S_1 and I_2. Let $\psi(X)$ be the sum of the images of all maps from $S_1 \oplus S_2 \oplus I_2$ to X. It is additive, exact on the Serre subcategory generated by S_1, and the identity on S_1, S_2, I_2 and the maps between them, so both hypotheses hold and $\tilde{\psi}$ is the identity. For the projective cover P_1 of S_1, of dimension vector $(1, 2)$, $\mathrm{Hom}(S_1, P_1) = \mathrm{Hom}(I_2, P_1) = 0$ and $\mathrm{Hom}(S_2, P_1) = k^2$, so $\psi(P_1) = S_2^2$, of which $P_1 = \tilde{\psi}(P_1)$ is not a direct summand; and $L\psi(S_1) = 0 \neq S_1$. The error in the proof is the claim that $P_\bullet \to i_* i^* P_\bullet$ is a homotopy equivalence: a short exact sequence of complexes with contractible kernel gives a quasi-isomorphism, which an additive functor need not preserve. Lemma 8.4.4 uses Lemma 8.2.1 to identify the OTI functor with its tilting derived functor on all stable objects, and Theorem 8.4.5 rests on Lemma 8.4.4. So the published proof of Theorem 8.4.5 is incomplete. This is not a counterexample to Theorem 8.4.5, which may hold for the OTI functor by an argument specific to it. The point was found by Astra in a model reconstruction of the proof and checked by hand by a second model.

Further minor points found in the same reading (a misprint in [11, §7.3], where $M^{\lambda^t} \otimes \mathrm{sgn}$ should read $M^\lambda \otimes \mathrm{sgn}$; the justification of [10, Prop. 2.7] at $\ell = 1$) and, in the same reconstruction, two points in [11, §8.3, §7.3] (the genericity condition before Definition 8.3.1 is not sufficient for Lemma 8.3.2 in characteristic 2, while coefficients linearly independent over $\mathbb{F}_p$, which the paper gives as an example, suffice; and in the proof of Lemma 7.3.4 the dominance inequality should read $\geq$, with target degree $M - q$) do not affect anything here and were not included in those notes.

8 Relation to the literature

The same family. Three sources treat the PC or Nullstellensatz degree of onto-FPHP$_n^{n+k}$ with $p \mid k$.

- Riis's criterion [19] (Theorem 8 of [17]) gives the upper bound $p^{v_p(k)}$ for every n (Proposition 2.10). Our results show that it is attained: at the Lean tier for $v_p(k) = 1$ from $n = 2p-3$ (A), and at the written tier for every k from $n = 2p^{v_p(k)} - 1$ (E).

- Beame and Riis [2] prove Nullstellensatz degree at least $2^\ell - 1$ (multiplier degree) for surplus p^ℓ and $n \geq ((p+2)^\ell - p^\ell)/2$, and at most $p^\ell - 1$ for $n \geq p^\ell$, and list PC lower bounds for this family as open. Their designs are the input of D.

- Riis and Sitharam [20, Thm. 33] state PC degree at least $2^\ell - 1$ for surplus p^ℓ once $N \geq (21+3p^2)(2^\ell-2)-p$. We have read the statement and the preceding lemmas (pp. 36–38, same linear encoding) but not checked the proof, which delegates to a tree translation (Lemmas 31, 32) that we have not reconstructed. We draw no conclusion about its correctness, and none from the fact that Razborov's later survey asks Question 7 without citing it. For comparison:

at $\ell = 2$ it gives 3 for $N \geq 2(21 + 3p^2) - p$, against $p + 1$ from $n = 2p - 1$ (B, Lean) and p^2 from $n = 2p^2 - 3$ (F, written); at $p = 2$, $\ell = 3$ it gives 7 for $N \geq 196$, against 5 from $n = 7$ (C, Lean) and 8 from $n = 15$ (E, written).

Razborov's Question 7. The mathematical statements here concern the exact eventual degree and, for surplus p^ℓ, growth as ℓ varies; under E's named inputs that degree is p^ℓ. We make no claim of resolving the historical formulation of Question 7 or of priority over Theorem 33 of [20]. None of the Lean results bears on growth.

Other characteristics and other principles. Buss, Grigoriev, Impagliazzo and Pitassi [3] prove linear PC degree for Tseitin formulas mod p in characteristic $q \neq p$ (when the field contains a root of unity of order p) and for counting principles in other characteristics; Alekhnovich and Razborov [1] extend linear bounds to non-binomial cases; Buss, Impagliazzo, Krajíček, Pudlák, Razborov and Sgall [4] prove Nullstellensatz lower bounds for counting principles mod q over $\mathbb{Z}_m$ with $q \nmid m$. All concern the regime in which the characteristic does not divide the relevant modulus, and none treats the onto principle with $p \mid k$. Impagliazzo, Pudlák and Sgall [12] give Razborov's bound for the injective principle, where the onto and functional axioms are absent and the bound does not depend on p. The applications of general degree methods [15] that we examined do not yield bounds for the linear onto functional system when p divides k, and in the sources we examined we found no suitable low-degree reduction from the cited counting or Tseitin principles to this system.

The algebraic inputs. The proofs of D, E and F use published results from representation stability and modular representation theory: FI-modules [5], matrix loci [18], stable periodicity [10], semisimplification [11], chessboard complexes [22], Moore spaces and the Dade group [24], and endosplit resolutions [16]. We make no claim about whether they have been used in proof complexity before.

Search. The searches behind this section (web, ECCC, and the citations of the sources; October and November 2026), with the list of sources opened and of those read only as abstracts, are published with each of the findings behind A–C, under *prior art*.

9 How these results were produced, and what was checked

Who did what. All results in this paper were found and proved by AI research agents (Claude models by Anthropic and Codex models by OpenAI) working in an automated research harness in October and November 2026. The author built and ran the harness, chose the problems, directed the runs, read the outputs and decided what to publish. He is not a specialist in this area and does not follow every proof in detail himself. He publishes the results as they stand, with the evidence for each stated by its tier, and takes responsibility for publishing them. The harness records every claim together with its evidence, its reviews and the files behind it; those records are the basis of the roles stated below. Results A–C come from runs that produce Lean proofs; D–G from one exploratory run (20261104-0759) with two Claude and two Codex agents, followed by adjudication.

The Lean results. The formal statements are supported by kernel-checked proof terms. Their interpretation as the claims of this paper also depends on the definitions and the correspondence established in Section 2. The library is public, with the exact toolchain and a manifest; each finding

records the commit it was checked at. Its checks rebuild the proof modules with the recorded toolchain and then check the published declarations' types and axiom dependencies with `#check` and `#print axioms`. Every theorem uses only `propext`, `Classical.choice` and `Quot.sound`. On 6 November 2026 a separate model reproduction rebuilt the named proof targets from the public repository and checked the published declarations; hardware, timings and memory are recorded in the "How to verify" section of each finding. For A and B, a second agent in the same run supplied Lean re-proofs, reusing between one and nine of the first agent's lemmas for A and six for each statement of B. For C, a second agent independently checked the level-3 certificate with a program written from the Lean definitions; this was not a second Lean re-proof.

The written results. Each proof of D–G was written by one agent and reconstructed by at least one other, of the other vendor in most cases; a separate model referee with no contact with the run re-read the primary sources for each of the three mechanisms; and the adjudicator Astra (a Codex model outside the run) checked each against the primary sources and wrote the scoped verdicts we report. Section 4.7 and the corresponding paragraphs of Sections 5–6 give the roles per result. Where reviewers co-developed an argument before reviewing it, the run records say so. The paper's later prose reconstructions (Section 3, Appendix A) and added corollaries were checked in a separate model review of the paper, which led to corrections. This is a demanding process, but it is model review, and we do not present it as more.

What was not done. No human expert has reviewed any proof in this paper. The external theorems named as inputs were checked at the level of their statements and the interfaces we use, not re-proved; the exception is [11, Thm. 8.4.5], whose proof was reconstructed, with the result described in Sections 5 and 7. D–G are not formalised. The finite computations are exact, but are evidence only for the cases computed. We would welcome corrections, and will record them with the published findings.

10 Open problems

1. *Formalise E.* A first step is planned for a revision of this paper: a Lean proof that periodicity, and hence the exact degree $p^{v_p(k)}$, follows from the existence of a functor with the properties used in Section 5.2 or Appendix A, so that the remaining trust is isolated in explicit functor properties and the listed external inputs. The inputs are deep; a formalisation of F in the bands $d < 2p$, which needs only the connectivity theorem, is closer.

2. *Complete the proof of [11, Thm. 8.4.5]* for the semisimplification functor, without their Lemma 8.2.1. This would make Theorem 5.1(b) and the threshold $n \geq 2q - 2$ unconditional.

3. *The uniform spanning statement.* Prove the spanning statements (A)$(n + k, n, r)$, $r = 2, 3, 4$, uniformly over $\mathbb{F}_3$ for $n \geq 7$ and $9 \mid k$ ((A) fails off the stable range, Section 6.1). Alternatively, formalise Lemmas 6.7 and 6.8 and import kernel-checked square certificates $S(2)$, $S(3)$, $S(4)$ in the formal spanning definitions; this would discharge the hypotheses of the Lean theorem behind Corollary 6.2.

4. *Thresholds.* The degree is $p^{v_p(k)}$ from $n = 2p^{v_p(k)} - 1$ (E) and, for $v_p(k) \leq 2$, from $2p^{v_p(k)} - 3$ (A, F). What is it below? Over $\mathbb{F}_2$ with $k = 4$ it is 3 at $n = 3, 4$ and 4 from $n = 5$, so the threshold of E is not sharp.

5. *The ideal identity beyond p^2.* Investigate the full identity $(I \cap J)_d = (IJ)_d + \sum_{t=1}^{d} \sigma_t \mathcal{A}_{d-t}$, the form tested by exact computation in the run, for $d \geq p^2$. For $d < p^2$ the proof of Section 6.3 establishes the stronger form with σ_1 and σ_p only; at $d = p^2$ the elementary abelian subgroup is no longer Sylow and the transfer argument stops.

6. *Designs of degree $p^\ell - 1$.* Beame and Riis leave the Nullstellensatz degree for surplus p^ℓ between $2^\ell - 1$ and $p^\ell - 1$ for odd p. Unit designs of degree $p^\ell - 1$ would close that gap and, through Theorem 4.1, give the exact PC degree p^ℓ on the inputs of Section 4 alone.

7. *Riis–Sitharam, Theorem 33.* Check its proof.

A The second proof of periodicity

This appendix proves Theorem 5.1(a), for $m, m' > 2D$. It is the proof on which part (a), and hence the exact degree for $n \geq 2q-1$, rests; it does not use [11]. It extends natural isomorphisms only between exact functors, so the flaw in [11, Lemma 8.2.1] (Section 7), which concerns additive functors, does not affect it. It transports the closure of Section 5.1 along Harman's stable equivalences [10]. Those equivalences are compatible with induction and restriction of permutation modules; the closure needs compatibility for *arbitrary* submodules, which we prove on the Schur-algebra side with standard facts on tilting modules (Lemma A.4).

Status. T-WRITTEN. The proof was written by a Claude agent (c1) in run 20261104-0759, reviewed by a model referee with no contact with the run (who found the characteristic-2 error of Section 7 and supplied the replacement argument of Lemma A.2) and by a Codex agent (x2, who supplied the two-way scalar extension), and adjudicated by Astra, who accepted the coherence argument of Section A.3 conditional on the named inputs and without re-proving their foundations. The inputs are: Harman's Theorem 2.5 with the corrected modulus and his Corollary 1.6 and Proposition 2.7 (for $\ell = 1$, which the referee checked directly); Ringel duality and the equivalence $K^b(\mathrm{Tilt}) \simeq D^b$ for highest-weight categories with finitely many weights [21, 9]; the facts that tensor products of tilting modules for GL_N are tilting and that restriction to a Levi subgroup preserves tilting modules [23, 6, 14]; the Hom-dimension formula for permutation modules [13, 13.19]; and standard properties of the Schur functor [8].

Throughout, $k = \overline{\mathbb{F}}_p$ (Section 5.1 reduces to this case), q is a power of p with $q > D$, and $m \equiv m' \pmod q$ with $m, m' > 2D$. For a partition λ and $m \geq |\lambda| + \lambda_1$ put $\lambda(m) = (m - |\lambda|, \lambda)$.

A.1 Harman's equivalences, corrected

Permutation and Young modules. Let $\mathrm{Perm}_m^{\leq r}$ be the category of direct sums of permutation modules $M(\lambda(m))$ of S_m with $|\lambda| \leq r$, and $\mathrm{Young}_m^{\leq r}$ its Karoubi envelope, the direct sums of Young modules $Y(\lambda(m))$. Homomorphisms between permutation modules have the stable Carter–Lusztig bases of [10, §2]; a *stable map* is one whose coefficients in these bases do not depend on m.

Theorem A.1 (Harman [10, Thm. 2.5, Cor. 2.6], modulus corrected)**.** *Let $r \geq 0$ and let q be a power of p with $q > r$. (For $r = 0$ the categories consist of sums of the trivial module and the equivalence is the evident one.) For $m, m' > 2r$ with $m \equiv m' \pmod q$, sending $M(\lambda(m))$ to $M(\lambda(m'))$ and each stable map to the stable map with the same coefficients defines an equivalence $E^{\mathrm{P}} \colon \mathrm{Perm}_m^{\leq r} \to \mathrm{Perm}_{m'}^{\leq r}$, and hence an equivalence E^{Y} of Young categories.*

The structure constants of composition are binomial coefficients in m of degree at most r, and $\binom{m}{d}$ mod p depends on m only modulo the least power of p above d (Lucas). The printed modulus $p^{\lceil \log_p r \rceil}$ is too small when r is a power of p (Section 7); with $q > r$ the proof of [10] applies unchanged. For $\ell = 1$, [10, Prop. 2.7] states that E^{P} commutes with restriction to S_{m-1} and induction from S_{m-1}, up to natural isomorphism, on permutation modules. The relevant maps are explicit: restriction sends the basis element h_τ to $\sum h_{\tau - e_{ij}}$ and induction adjoins a singleton block, with coefficients 0 and 1 that do not depend on m; so this can be checked directly.

Schur algebras. For $N \geq m$ let $S(N,m)$ be the Schur algebra, so that $S(N,m)$-modules are the polynomial representations of GL_N of degree m, and let $\mathcal{S}^{\leq r}(N,m)$ be the Serre subcategory generated by the simples $L(\lambda(m)')$, $|\lambda| \leq r$. It is a highest-weight category whose tilting objects are the $T(\lambda(m)')$ [10, Prop. 1.3 and proof of Thm. 2.8]. Let $\mathcal{F}$ be the Schur functor and $\tilde{\mathcal{F}} = \mathcal{F}(-) \otimes \mathrm{sgn}$; it is exact and $\tilde{\mathcal{F}}(T(\lambda')) = Y(\lambda)$ [10, Cor. 1.8(1),(2)].

Lemma A.2 (full faithfulness on tilting modules)**.** *In every characteristic, $\tilde{\mathcal{F}}$ restricts to an equivalence* $\mathrm{Tilt}(\mathcal{S}^{\leq r}(N,m)) \to \mathrm{Young}_m^{\leq r}$.

Proof. Harman deduces this from [10, Cor. 1.8(4)], which is false at $p = 2$ (Section 7); here is a proof that does not use it. The exterior powers $\Lambda^\mu V = \Lambda^{\mu_1} V \otimes \Lambda^{\mu_2} V \otimes \cdots$ are tilting, and $T(\mu')$ is a summand of $\Lambda^\mu V$ with multiplicity one; so the tilting objects of $\mathcal{S}^{\leq r}(N,m)$ are the summands of the $\Lambda^{\lambda(m)} V$, $|\lambda| \leq r$, and $\tilde{\mathcal{F}}(\Lambda^\mu V) = M(\mu)$.

Dimensions. Since $\Lambda^\lambda V$ has a Weyl filtration and $\Lambda^\mu V$ a good filtration, $\dim \mathrm{Hom}(\Lambda^\lambda V, \Lambda^\mu V) = \sum_\nu (\Lambda^\lambda V : \Delta(\nu))(\Lambda^\mu V : \nabla(\nu)) = \sum_\rho K_{\rho\lambda} K_{\rho\mu}$, the number of (S_λ, S_μ) double cosets in S_m, which is $\dim \mathrm{Hom}_{S_m}(M(\lambda), M(\mu))$ in every characteristic [13, 13.19].

Faithfulness. $\Lambda^\mu V$ embeds in $V^{\otimes m}$ (antisymmetrisation is injective in every characteristic). For $N \geq m$, $V^{\otimes m}$ is projective and injective, and every simple submodule L of it has $\mathcal{F}(L) \neq 0$ [8, §6]. If $f \colon \Lambda^\lambda V \to \Lambda^\mu V$ is non-zero, its image is a non-zero submodule of $V^{\otimes m}$, so it contains such an L, and since $\mathcal{F}$ is exact, $\mathcal{F}(\mathrm{im}\, f) \neq 0$, so $\tilde{\mathcal{F}}(f) \neq 0$.

A faithful functor between Hom spaces of equal finite dimension is bijective on them. So $\tilde{\mathcal{F}}$ is fully faithful on the $\Lambda^{\lambda(m)} V$, hence on their summands; its essential image is $\mathrm{Young}_m^{\leq r}$. □

Proposition A.3 (Harman's Theorems 2.8 and 2.9, with the repairs)**.** *Let $N \geq m$, $N' \geq m'$. There is an equivalence of highest-weight categories $\Theta \colon \mathcal{S}^{\leq r}(N,m) \to \mathcal{S}^{\leq r}(N',m')$ with $\Theta(T(\lambda(m)')) \cong T(\lambda(m')')$, together with a natural isomorphism $\zeta \colon \tilde{\mathcal{F}}\Theta \cong E^{\mathrm{Y}} \tilde{\mathcal{F}}$ on tilting objects. Θ descends to an equivalence of abelian categories $E \colon \mathcal{R}_m^{\leq r} \to \mathcal{R}_{m'}^{\leq r}$ between the essential images $\mathcal{R}_m^{\leq r} = \tilde{\mathcal{F}}(\mathcal{S}^{\leq r}(N,m)) \subseteq \mathrm{Rep}(S_m)$, whose kernels, cokernels and subobjects are the ordinary ones, and E restricts to E^{P} on permutation modules up to the isomorphisms ζ.*

Proof. By Lemma A.2 and Theorem A.1, composing $\tilde{\mathcal{F}}$, then E^{Y}, then a quasi-inverse of $\tilde{\mathcal{F}}$ at (N',m') gives an equivalence $\mathrm{Tilt}(\mathcal{S}^{\leq r}(N,m)) \to \mathrm{Tilt}(\mathcal{S}^{\leq r}(N',m'))$ that preserves labels. Ringel duality [10, Cor. 1.6] extends it to Θ; on tilting objects Θ agrees with the given equivalence up to natural isomorphism, since both are determined by the image of a full tilting object and its endomorphism ring. (This is a property of the construction, not its definition; we use only this property.)

For $|\lambda| \leq r < m/2$ the first part $m - |\lambda|$ of $\lambda(m)$ exceeds every other part, so $\lambda(m)$ is p-singular if and only if $\lambda(m')$ is. Hence Θ maps the Serre subcategory generated by the p-singular simples, which is the kernel of $\tilde{\mathcal{F}}$, onto its counterpart. It remains to identify the Serre quotient $\mathcal{S}^{\leq r}/\ker \tilde{\mathcal{F}}$ with $\mathcal{R}_m^{\leq r}$. Harman applies Gabriel's lemma to $\mathcal{S}^{\leq r}$, which needs a right adjoint landing in $\mathcal{S}^{\leq r}$; that is not shown (Section 7). Instead: the right adjoint $\tilde{\mathcal{G}}$ of $\tilde{\mathcal{F}}$ on all of $S(N,m)$-mod has

invertible counit, so $S(N,m)$-mod$/\ker\tilde{\mathcal{F}} \simeq \mathrm{Rep}(S_m)$ by Gabriel's lemma [7]; and since $\mathcal{S}^{\le r}$ is a Serre subcategory, for $X \in \mathcal{S}^{\le r}$ and $T \le \tilde{\mathcal{F}}X$ the pullback $X_T = X \times_{\tilde{\mathcal{G}}\tilde{\mathcal{F}}X} \tilde{\mathcal{G}}T$ is a subobject of X in $\mathcal{S}^{\le r}$ with $\tilde{\mathcal{F}}X_T = T$, and lifting graphs the same way gives every morphism as a roof inside $\mathcal{S}^{\le r}$. So the quotient is the full subcategory $\mathcal{R}_m^{\le r}$, closed under submodules and quotients in $\mathrm{Rep}(S_m)$. The exact equivalence Θ descends to an exact equivalence E. □

A.2 Compatibility with induction and restriction

Write E_m for the equivalence of Proposition A.3 at weight $\le D$ between S_m and $S_{m'}$, and E'_{m-1} for the one at weight $\le D-1$ between S_{m-1} and $S_{m'-1}$ (note $m-1 > 2(D-1)$ and $m-1 \equiv m'-1$).

Lemma A.4 (B)**.** *There are natural isomorphisms*

$$E_m \circ \mathrm{Ind} \cong \mathrm{Ind} \circ E'_{m-1} \quad \text{on } \mathcal{R}_{m-1}^{\le D-1}, \qquad E'_{m-1} \circ \mathrm{Res} \cong \mathrm{Res} \circ E_m \quad \text{on } \mathcal{R}_m^{\le D-1},$$

which on permutation modules are the isomorphisms of [10, Prop. 2.7] with $\ell = 1$.

Proof. Fix $N \ge \max(m, m') + 1$. On the Schur side take $\Phi_{\mathrm{Ind}} = - \otimes V \colon S(N, m-1)\text{-mod} \to S(N,m)\text{-mod}$ and let $\Phi_{\mathrm{Res}} \colon S(N,m)\text{-mod} \to S(N-1, m-1)\text{-mod}$ be the part of weight 1 for the last factor of the Levi subgroup $GL_{N-1} \times GL_1$.

(i) $\tilde{\mathcal{F}}(M \otimes V) \cong \mathrm{Ind}\,\tilde{\mathcal{F}}(M)$ and $\tilde{\mathcal{F}}(\Phi_{\mathrm{Res}} M) \cong \mathrm{Res}\,\tilde{\mathcal{F}}(M)$, naturally: the (1^m)-weight space of $M \otimes V$ is $\bigoplus_i M_{(1^m)-\varepsilon_i}$, which is the induced module, and the transposition $(m\ N)$ carries the (1^m)-weight space of M to the weight $(1^{m-1}, 0, \dots, 0, 1)$, S_{m-1}-equivariantly; the sign twist commutes with both. *(ii)* Both functors are exact and send tilting modules to tilting modules [23, 6, 14]; a weight summand of a tilting module for the Levi subgroup is tilting. *(iii)* By branching, $- \otimes V$ maps $\mathcal{S}^{\le D-1}(N, m-1)$ into $\mathcal{S}^{\le D}(N,m)$ (composition factors of $\Delta(\mu) \otimes V$ lie below $\mu + \varepsilon_i$), and Φ_{Res} maps $\mathcal{S}^{\le D-1}(N,m)$ into $\mathcal{S}^{\le D-1}(N-1, m-1)$. *(iv)* By (i), both functors preserve the kernels of $\tilde{\mathcal{F}}$.

Step 1 (tilting level). Let $\phi \in \{\mathrm{Ind}, \mathrm{Res}\}$ with Schur-side partner Φ, and T a tilting object. Then, naturally in T,

$$\tilde{\mathcal{F}}\Theta\Phi T \cong E^{\mathrm{Y}}\tilde{\mathcal{F}}\Phi T \cong E^{\mathrm{Y}}\phi\tilde{\mathcal{F}}T \cong \phi E^{\mathrm{Y}}\tilde{\mathcal{F}}T \cong \phi\tilde{\mathcal{F}}\Theta T \cong \tilde{\mathcal{F}}\Phi\Theta T,$$

using ζ, (i), [10, Prop. 2.7] (on permutation modules, hence on their summands), ζ again, and (i). Both $\Theta\Phi T$ and $\Phi\Theta T$ are tilting by (ii), and $\tilde{\mathcal{F}}$ is fully faithful on tilting objects (Lemma A.2), so $\Theta\Phi \cong \Phi\Theta$ naturally on tilting objects.

Step 2 (all objects). The categories involved are highest-weight with finitely many weights, so $K^b(\mathrm{Tilt}) \to D^b$ is an equivalence [21, 9]. An exact functor induces a functor on D^b computed termwise on tilting complexes, and the natural isomorphism of Step 1 applied termwise gives one on $K^b(\mathrm{Tilt})$, hence on D^b. Taking H^0 on the heart gives $\Theta\Phi \cong \Phi\Theta$ on all objects, restricting to the isomorphism of Step 1 on tilting objects.

Step 2′ (restriction in weight). The same argument shows that Θ at weight $\le D$, restricted to weight $\le D-1$, is isomorphic to Θ at weight $\le D-1$: both are exact and agree with the same Young-module equivalence on tilting objects. This is needed because $C_t \cap F_{D-1}$ is transported at weight D and then restricted.

Step 3 (descent). By (iv) and the universal property of Serre quotients for exact functors and natural transformations, the isomorphism descends to $E \circ \phi \cong \phi \circ E$ on the image categories. On permutation modules it is the isomorphism of Step 1, that is, Harman's. □

A.3 Coherence

Two points make the isomorphisms usable for subobjects; Astra accepted both, conditional on the inputs above.

Independence of N. Induction uses $S(N, m-1)$ and restriction lands in $S(N-1, m-1)$. Truncation to the weights with last coordinate 0 is an exact Morita equivalence $S(N, m-1)$-mod $\to$ $S(N-1, m-1)$-mod for $N-1 \geq m-1$; it preserves tilting objects and labels and commutes with $\tilde{\mathcal{F}}$. Applying Step 2 to $\Theta^{(N)}$ and to $\Theta^{(N-1)}$ conjugated by truncation (both exact, both isomorphic to the same Young functor on tilting objects) gives a natural isomorphism between them that is the identity on Young modules. So E'_{m-1} does not depend on N, and the output of restriction can be fed into induction.

Fixed identifications. Fix, on each side, the descent isomorphism $\delta\colon E\tilde{\mathcal{F}} \to \tilde{\mathcal{F}}\Theta$ and the tilting comparison $\alpha\colon \tilde{\mathcal{F}}\Theta \to E^{\mathrm{Y}}\tilde{\mathcal{F}}$ of Proposition A.3. On tilting objects their composite descends, by full faithfulness of $\tilde{\mathcal{F}}$ there (Lemma A.2), to a natural isomorphism $\eta\colon E|_{\mathrm{Young}} \to E^{\mathrm{Y}}$; write η' for the same at $m-1$. The same δ and α are used in constructing the isomorphisms of Lemma A.4. Let $c\colon E^{\mathrm{P}}_m\mathrm{Ind} \to \mathrm{Ind}\, E^{\mathrm{P}}_{m-1}$ and $c'\colon E^{\mathrm{P}}_{m-1}\mathrm{Res} \to \mathrm{Res}\, E^{\mathrm{P}}_m$ be Harman's isomorphisms on permutation modules. Unwinding the descended isomorphisms β (induction) and γ (restriction), the Schur-functor isomorphisms of (i) cancel by naturality of δ and α, and for permutation objects P, Q

$$\beta_P = (\mathrm{Ind}\,\eta'_P)^{-1} \circ c_P \circ \eta_{\mathrm{Ind}\,P}, \qquad \gamma_Q = (\mathrm{Res}\,\eta_Q)^{-1} \circ c'_Q \circ \eta'_{\mathrm{Res}\,Q}.$$

All maps used below (the multiplication pieces g_j, ε_j, the inclusions, the axiom maps and the constant) are stable maps defined from the same stable decompositions as c. For each inclusion $T \to P$ of a submodule into a permutation object, naturality gives a commuting square whose ambient map is the fixed component above; taking images identifies the transported subobject. So in the induction below every subobject is carried to the corresponding subobject on the smaller board, and no unknown automorphism enters.

A.4 Transport of the closure

The multiplication step as induction. Fix a hole j and let S_{m-1} fix pigeon m. Multiplication by x_{mj} maps $\mathrm{Res}\, F_{D-1}(m)$ into the summand $P_j \subseteq \mathrm{Res}\, F_D(m)$ spanned by matchings containing the edge (m, j); call this map g_j. P_j is a permutation module of S_{m-1} of weight $\leq D-1$ (matchings of size $\leq D-1$ on pigeons $[m-1]$ and holes other than j). Let $\varepsilon_j\colon \mathrm{Ind}\, P_j \to F_D(m)$ be adjoint to the inclusion. Both are stable maps with coefficients 0 and 1. For an S_m-submodule $S \subseteq F_{D-1}(m)$,

$$\textstyle\sum_i \rho(x_{ij})S = \varepsilon_j\big(\mathrm{Ind}\, g_j(\mathrm{Res}\, S)\big),$$

because $\sum_i \rho(x_{ij})S = \sum_{\sigma\in S_m} \sigma\,\rho(x_{mj})\,S$ and the image of $\mathrm{Ind}\, T \to F_D$ is $S_m \cdot T$. So every $C_t(m)$ is obtained from permutation modules of weight $\leq D$ (for S_m) or $\leq D-1$ (for S_{m-1}) and stable maps by images, preimages ($C_t \cap F_{D-1} = \iota^{-1}(C_t)$ for the inclusion ι), finite sums, and induction and restriction of submodules. The constant is the stable map $k \to F_D(m)$.

Proof of Theorem 5.1(a). By induction on t, $\eta\big(E_m(C_t(m))\big) = C_t(m')$ as subobjects of $F_D(m')$. For $t = 0$, C_0 is the sum of the images of the axiom maps, which are stable. For the step: E_m is exact, so it preserves images, preimages and sums of subobjects, and the subobjects are the ordinary ones (Proposition A.3); stable maps go to the same stable maps up to η (Section A.3); induction and restriction of the subobjects $C_t \cap F_{D-1}$ and $g_j(\mathrm{Res}\, S)$ are transported by Lemma A.4, with Step 2′ for the change of weight. Both chains stabilise. Since E_m is an equivalence, the composite $k \to F_D(m) \to F_D(m)/C_\infty(m)$ is zero if and only if its image $k \to F_D(m') \to F_D(m')/C_\infty(m')$ is. Lemma 5.3 and the reduction to $\overline{\mathbb{F}}_p$ (Section 5.1) finish the proof. □

The strict inequality $m, m' > 2D$ is used twice: in Theorem A.1 and in the comparison of p-singular labels in Proposition A.3. At $m = 2D$ this proof does not apply, and Theorem 5.1(b) rests on the conditional proof of Section 5.2.

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